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Essays on environmental change and inequality

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Abstract

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This dissertation is comprised of three essays each of which may be described as empirical investigations relating environmental change to a distinct concept of inequality: economic inequality as measured by the global distribution of income; political inequality as reflected in the demand for democratic institutions; and gendered cultural inequality as expressed through domestic violence.

Chapter 1 explores the relationship between anthropogenic climate change and global inequality, a subject bridging two of the defining challenges of the 21st century but that nonetheless remains virtually unstudied at subnational levels. To make overdue progress addressing this research gap, I adapt recently developed methods for estimating climate impacts to overcome long-standing empirical limitations and apply these refinements to fine-scaled data on globally representative income distributions.

I document new evidence that temperature shocks significantly and persistently impact distributions of income within countries, an effect driven by concentrations of harm onto the lowest income-earners in warm climates as well as a surprising vulnerability of the top 1% in these countries to environmental shocks. Integrating these inequality effects over observed distributions of

income and the spatial incidence of global warming between 1981 and 2016, I find that climate change has regressively redistributed global shares in a reduced stock of income both between and within countries largely by depriving the world's poorest of economic opportunity that would otherwise have been available. For the poorest percentile of positive income-earners in our data, we find that income levels absent asymmetrical warming since 1981 would be 29% [18, 41] higher than historically realized. Altogether, these results constitute the most comprehensive evidence yet of the regressive impact of climate change on economic inequalities between and within countries.

Despite widespread regard of climate change as a threat to global stability, the nature of the relationship between environmental change, political preferences, and regime change is not well characterized in environmental political economy. In Chapter 2, I consider two possible frameworks by repurposing influential existing theories of regime change. Under a "mercury uprising" hypothesis, adverse environmental shocks diminish the opportunity cost of contesting autocratic rule thereby improving prospects for democratic reform. On the other hand, a "state of exception" hypothesis would suggest that the public responds to these shocks with a willingness to compromise civil liberties for authoritarian features of governance which may more credibly ensure security thereby leading to a diminution of democratic institutions.

I empirically test the relative merits of these theories first by measuring the impacts of identified temperature shocks on survey-based proxies for the demand for democracy. Next, I use multi-valued and dichotomous measures of democracy to investigate the dynamic effects of these shocks on the democratic quality of institutions. The results from these analyses consistently support the latter theory.

A particularly striking finding is that a positive 1°C temperature shock increases the propensity for autocracies in temperate geographies to democratize by between 0.030-0.075 percentage points depending on the operating definition of democracy but reduces this propensity for their tropical counterparts by 0.032-0.048 percentage points. The same shocks also increase the propensity for warm-climate democracies to break down by 0.011-0.040 percentage points. In these cases, effects are substantial and significant but exhibit very limited persistence beyond the year of the shock after

accounting for autocorrelation in the shock itself. Finally, we find that the same shocks have null effects on the propensity for temperate democracies to undergo reversal, suggesting these institutions have historically proven impervious to environmental disruption.

In Chapter 3, I explore how explore how drought-induced weather shocks impact spousal violence in a previously never-violent relationship. Here, I combine high-resolution weather data with comprehensive duration data on the incidence of physical and emotional violence in India to estimate how drought-induced negative income shocks may affect these health outcomes.

My findings suggest that negative income shocks substantially reduce the hazard into marital violence. Specifically, a negative precipitation shock of a magnitude one might ordinarily expect once every 4-5 years on average reduces the risk of a never-violent relationship becoming violent by 0.037 percentage points, representing a 70% reduction in the baseline hazard of this transition during the formative years of a marriage.

I also replicate a recently published study relating these weather shocks to the risk of child marriage using updated methods and much-improved data. I find that these effects are milder but more persistent than characterized in the original study. In addition, I find that this result is reproduced in Sub-Saharan African countries, contradicting the headline result of the original study which had reported opposite-signed effects in the two settings and attributed the difference to opposite consumption smoothing incentives motivated by opposite consumption-smoothing incentives imposed by the cultural practices of dowry and bride price payments. Instead, secondary results suggest that child marriage seem to be even less economically motivated than marriages occurring later in life.

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Chapter 1: Climate inequalities

> "There is no progress trying to do welfare economics without [going] beyond the Pareto principle and taking on the real problems of distribution, inequality, and poverty.'
>
> Amartya Sen (Sen et al., 2020)

1.1 Introduction

Mitigating anthropogenic climate change and narrowing economic inequalities are widely considered among the most pressing global challenges of the 21st century. For example, the Sustainable Development Goals adopted by all United Nations members explicitly resolve to "take urgent action to combat climate change and its impacts" and to "reduce inequality within and among countries" (United Nations, 2015). These global priorities are also understood to be intimately related, often framed as twinned crises (Pande, 2023): the Intergovernmental Panel on Climate Change identifies the "widening opportunities between and within countries" (IPCC, 2018) as first-order considerations in any sufficient policy response to climate change. Likewise, the World Inequality Report remarks that "climate mitigation is largely a distributional issue, not only between countries but also within them" (Chancel et al., 2022). When surveyed in 2021, an overwhelming majority of published environmental economists indicated a belief that climate change will "likely" or "very likely" exacerbate inequalities between (89%) and within (71%) countries (Howard & Sylvan, 2021).

These frames exemplify how the distributional implications of climate change are increasingly operationalized along its international and intranational dimensions. The conventional wisdom that the social consequences of climate change tend to concentrate in less developed countries with

tropical geographies is supported by a robust empirical literature and reinforced by the visibility of calamitous extreme weather events. In contrast, within-country 'climate inequality' remains remarkably undercharacterized and largely inactive as a subject of academic inquiry even as public and institutional demand for these equity considerations has surged (Office of Management and Budget, 2023) and intranational disparities increasingly dominate total global inequality[1]. As a result, our understanding of how climate change affects the economic well-being of the Earth's eight billion human inhabitants is largely limited to aggregate comparisons across fewer than 200 polities, necessarily agnostic to urgent considerations such as social stratification and mass immiseration. Addressing this ignorance is necessary for adequately specifying the problem of climate change, let alone formulating an efficient and equitable response to it. This chapter aims to initiate and stimulate long overdue progress in this area by providing the most comprehensive characterization yet of the dynamic and distributional impact of climate change on economic inequality.

Credible progress in this area first requires overcoming long-standing challenges in the tradition of estimating the causal effect of weather variation on economic outcomes of interest (Pindyck, 2013). In Section 1.2, I briefly summarize specific technical issues in this niche literature to contextualize the methodological choices made in the present analysis. First, estimates of cumulative impacts implied by these "damage functions" are known to diverge by orders of magnitude depending on the modeler's assumptions about the persistence of these effects. While 76% of recently surveyed climate economists indicated a belief in highly persistent or permanent effects of weather on macroeconomic aggregates like GDP, statistical tests of their presence have often been inconclusive (M. Burke et al., 2015; Newell et al., 2021). Second, recent work has challenged the validity of raw weather variables conventionally used for as explanatory variables in these regressions on the grounds of identificaiton issues. Finally, I discuss the use of these reduced-form damage functions to make inferences about systematic climate change.

My implementation is briefly described in Section 1.3 with more detailed exposition provided in Section A.2. I adapt recent innovations in empirical macroeconomics to construct more plau-

[1]Within-country income inequality now accounts for almost 70% of total income inequality as measured by the entropy-based Theil measure of inequality, an increase from just over 40% in 1980. (Chancel et al., 2022)

sibly exogenous shocks and estimate dynamic impulse response functions by the method of local projections. To credibly measure global income distributions, the analysis uses new data from the most recent update to the World Inequality Database, which provides well-balanced percentile-year panels of pre-tax income for a set of countries now representing almost the entire global population. By applying a refined econometric apparatus to unprecedentedly rich distributional data, my empirical analysis accomplishes two objectives. First, it substantially improves the frontier of an analytic tradeoff between subnational disaggregation and global representativeness which has historically hindered the joint study of global inequality and climate change. Second, the dynamic estimates which result are entirely invariant to hotly contested assumptions about the persistence structure of these economic impacts, thus obviating the dominant source of extreme estimation instability observed in the climate impacts literature.

The results of this empirical analysis are documented in Section 1.3.2. Here, we present dynamic distributional income response surfaces which depict how the impact of a temperature shock may vary across income groups and climates and accumulate over time. We find evidence of substantial, significant, and persistent impacts on within-country inequality, particularly in warmer countries. This distributional effect is driven by substantial transfers of national income shares among countries in warm climates. More specifically, the distributional impacts in these settings are marked by severe reductions in income among the lowest income-earning decile and to a lesser extent, a perhaps surprising precipitous fall in income among the top 1%. These losses are shown to compound for at least three years following a shock even when accounting for persistence in the shock itself. As impacts to warmer countries are shown to be significantly negative across the entire income distribution, national income in sufficiently warm countries are expected to uniformly decrease in response to a transitory temperature shock. Thus, positive temperature shocks in already-hot countries result in regressive redistribution of shares in a persistently diminished national income.

In Section 1.4, we present a counterfactual analysis which produces the most disaggregated evidence yet of the impact of historically observed climate change on the global distribution of

income. This entails integrating the distributional response surfaces estimated in the previous section over the observed histories of country-percentile-year income and the incidence of temperature shocks between 1981 and 2016. My results suggest that systematic changes to the distribution of temperature shocks over this 36 years has unambiguously worsened total global inequality, both between and within countries. This effect mainly stems from foregone income growth for the poorest 20% of global income-earners. For the lowest positive income-earning global percentile in our data, the analysis implies that cumulative income growth over this 36-year period would be 29% higher than historically observed had the climate remained stable at 1980 levels. The evidence is also consistent with the standard finding in the literature on aggregate impacts that climate change is extremely likely to have reduced total global income over this period though we cannot reject a null of no impact for segments of the global middle class, income groups largely comprised of individuals resident in cool climates where impacts are expected to be relatively mild.

Section 1.5 concludes by summarizing the contributions and limitations of the present analysis and suggesting promising extensions of this work. In particular, I briefly consider how the subject of climate inequalities poses urgent challenges the economic analysis of climate change and the design of climate policy.

1.1.1 Related literature

Spurred by methodological advances, accessibility of computing power, and political urgency, the past decade has seen a proliferation of economic research dedicated to estimating the social impacts of climate change. To contextualize the relative sparsity of within-country distributional impact studies in climate economics, it is instructive to familiarize the reader with the subfield's taxonomy of empirical designs into complementary "top-down" and "bottom-up" approaches each premised on an underlying utilitarian cost-benefit logic.

Top-down approaches restrict attention to a widely available outcome of interest. This outcome is taken as a comprehensive enough proxy for social welfare that climatically attributable impacts to the stock of this outcome may represent the socially relevant climate "damage". The pioneering

works in this category include Dell et al. (2012) and M. Burke et al. (2015), which relate annual observations of country-level temperature to GDP growth. Alternatively, bottom-up approaches consider impacts to distinct "sectors" of an economy; mortality (T. Carleton et al., 2022), energy consumption (Rode et al., 2021), and labor (Graff Zivin & Neidell, 2014) are common areas of focus. Results using this approach can be presented as partial costs or combined with other sectoral damages in an integrated model which accounts for their interdependencies. The advantage of the former approach is in its convenience and interpretability: the wide availability and cross-country comparability of economic aggregates such as those provided by national accounts substantially simplifies the task of calculating global climate impacts. The benefit of the latter approach is in its ability to account for disaggregated and non-market impacts to which single-dimensional aggregates are largely indifferent.

Conversely, the simplicity of the top-down approach and comprehesiveness of the bottom-up approach pose a critical tradeoff between geographic disaggregation and global coverage. In perhaps the most prominent existing work directly concerned with characterizing global 'climate inequality', Diffenbaugh and Burke (2019) estimate that the ratio between the top and bottom deciles of the global income distribution was 25% larger in 2010 than it would have been had anthropogenic climate forcings been constrained to 1960 levels, all else held equal. But this is necessarily calculated by assigning all individuals their country's mean GDP and crudely defining global deciles accordingly, an unavoidable restriction when limited to 160 or so distinct economic units. The authors remark that "documenting the impact of global warming on within-country inequality remains an important challenge."

S. Hsiang et al. (2017) is a relatively rare study which makes the opposite compromise between specificity and breadth in order to make progress in this area. Integrating fine-scaled agricultural, crime, energy, mortality, and labor data, the authors determine that climate risks in the continental United States are disproportionately borne by counties in the US South and Midwest, regions generally already poorer than their coastal counterparts; climate change then is likely to imply worsening inequality within the United States. Yet to make more generalizable claims about the implications

of global warming for global inequality, analyses must extend to settings which lack comparably rich or granular data.

The distributional focus of this paper complements a much more active area of research measuring unequal impacts across countries. Like Diffenbaugh and Burke (2019), a subset of these studies explore the implications of this type of inequality directly, for example by highlighting the inverse relationship between country-level contributions to cumulative global greenhouse gas emissions and exposure to their social consequences (Chancel & Piketty, 2015; Kotz et al., 2024; Ricke et al., 2018). There is by now general agreement that non-linearities in the response of aggregate variables to marginal warming concentrate damages and risk in relatively warm and poor countries which have contributed proportionally little to anthropogenic warming (Howard & Sylvan, 2021). However, estimated magnitudes of these impacts are notoriously divergent, largely because of unsettled disagreement over the persistence of these non-linear effects (Newell et al., 2021). We draw from recent work by Nath et al. (2024) and Bilal and Känzig (2024) which use less restrictive time series methods to circumvent these statistical bugbears.

Similar non-linearities emerge at the micro level across a range of outcomes as varied as labor supply (Graff Zivin & Neidell, 2014), agricultural yields (Schlenker & Roberts, 2009), and standardized test performance (J. Goodman et al., 2020). T. A. Carleton and Hsiang (2016) overviews the proliferation of multidisciplinary studies in this category whose results describe patterns of setting-specific inequality suggestive of a broader micro-level environmental inequality similar to those observed in the aggregate. The present study bridges this micro evidence with the aforementioned macro evidence on inequalities between countries, estimating distributional responses consistent with the general features of both categories of impacts. We also document evidence of negative impacts to an elite minority, which I have not seen examined in existing work.

The literature on climate impacts on economic inequality *within* countries has remained comparatively sparse in large part due to an absence of adequate research infrastructure. Estimation of distributional equivalents to the aggregate damage functions standard in the literature requires data comparable in credibility and coverage to domestic production data whose construction across

countries follows rigorous standards set forth by the international System of National Accounts (European Commission et al., 2008). Historically, distributional measurement of welfare has lacked and arguably been crowded out by the wider interest and investment in measuring aggregate production (Jorgenson, 2018). While secondary inequality data products compiling country-year inequality variables from a wide array of official and survey sources date at least as far back as Deininger and Squire (1996), the validity of these products for cross-country causal analysis is limited by inconsistencies across primary sources in construction and availability (Atkinson & Brandolini, 2001). For example, the most recent rigorous evaluation I could find of these secondary datasets concluded that results derived from the conveniently packaged Standardized World Income Inequality Database should be regarded as "not sufficiently credible" and that analyses using the United Nations' long-running World Income Inequality Database should at best be accompanied by justifications for their sample and variable selection criteria (Jenkins, 2015). Nonetheless, these inequality products have been used in at least two recent studies (Cevik & Jalles, 2023; Gilli et al., 2024) relating the inequality measures provided in these products to variations in weather.

In the absence of credible global inequality measures, a handful of works have managed to contribute to the study of within-country inequality impacts by using idiosyncratic data available for individual countries. S. Hsiang et al. (2017), described earlier, is a particularly impressive example. Elsewhere, Marx (2024) uses rich tax data from France to infer that years with marginally more days above 30°C are associated with wider income disparities between the richest and poorest cantons (administrative units described as roughly equivalent to 10 municipalities). Dasgupta et al. (2023) uses data from perhaps the most unequal country in the world to project a 3-6 Gini point increase in South Africa by 2100 under a moderate warming scenario.

The research programs of Thomas Piketty, Emmanuel Saez, and their co-authors since the turn of the century are widely credited with revitalizing public and academic interest in the measurement of within-country inequality. Beginning with work tracing the long-run evolution of income and wealth inequality in France (Piketty, 2003) and the United States (Piketty & Saez, 2003), their mobilization of extensive tax returns data has enabled credible measurement of top incomes over

long time horizons, stimulating new programs of research and equipping the general public with new political language around which to mobilize (Jones, 2015). This paper takes advantage of the cross-country research apparatus that has rapidly developed and been made accessible in recent years with the intention of maturing the economic analysis of climate change in a similar manner. Palagi et al. (2022) is the only other climate-economic study I could locate which makes use of the same "distributional national accounts" (World Inequality Lab, 2024) used in this study. The authors find that extreme precipitation negatively impacts low-income individuals much more severely in countries more dependent on agriculture.

Finally, I locate this paper in the context of a growing demand for distributional considerations in economic analyses of climate change and the design of climate policy. For example, Chancel et al. (2023) estimates that within-country inequality of carbon emissions have recently exceeded the equivalent "carbon inequality" between countries and advocates for targeted carbon taxation programs based on individual footprints to more efficiently internalize the climate change externality. The modern SCC framework accommodates equity considerations by concavifying the underlying utility function so that the welfare weights assigned to individuals are inversely related to their relative levels of consumption; a logarithmic specification is commonly chosen, for example. Variations of this more progressive utilitarianism have been institutionalized to some degree in a handful of countries including recently in the United States (Office of Management and Budget, 2023). Theoretically, the way equity weighting is implemented should result in substantially higher SCC estimates depending on the extent to which the distributive effect of climate change is modeled as regressive. This paper provides the first comprehensive estimate of the global regressivity of climate change at subnational resolution.

1.2 Data

1.2.1 Distributional income data

We make use of the recently published grouped income series from the World Inequality Database (World Inequality Lab, 2024). Its predecessor, the World Top Income Database launched in 2011,

drew from comprehensive tax records from over 30 mostly Western countries to capture the distribution of income among their top income-earners with unprecedented precision. A critical innovation used to develop the new product is the systematic integration of this tax data with national accounts and household survey data in order to impute the full income distribution within those countries (Piketty et al., 2019) in a manner patterned after the System of National Accounts (European Commission et al., 2008). The newest generation of WID data extends coverage within those countries—now claiming to capture "100% of income" in the United States (Piketty et al., 2018)—and to countries with relatively limited available data. By 2019, historical data for over 120 countries had been incorporated into the database. As of this writing, their equal-split adult pre-tax income series was available for 215 countries and territories collectively representing almost the entire global population with distributional data imputed every year since 1980. All income estimates are converted to 2023 US dollars using the provided PPP-adjusted exchange rates. Accompanying metadata scores the availability and reliability of each category of input source by country, unsurprisingly indicating that the input mix of these sources varies substantially by country with less developed countries being more reliant on household surveys and wealthier countries on tax records.

Notably for the percentile-level panels, the income data exhibits substantial bottom-coding with almost every country-year reporting exactly zero income for at least the bottom four percentiles of the pre-tax income distribution. Even if an accurate measure of adult unemployment, this evokes well-known deficiencies of income as an identifier and measure of poverty compared to alternatives like consumption expenditures (Meyer & Sullivan, 2003). For example, while the unemployed subpopulation of a given economy may well be composed of its most impoverished, it is likely to also include unemployed individuals of all levels of precarity and social status and does not account for security provided by communal and social safety nets. It also positively biases estimated changes in welfare among the poor since zero incomes mechanically cannot decrease.

Given these inherent disadvantages of income measures as a proxy for relevant welfare, we intentionally do not make precise claims about the poverty implications of our findings. Instead, we

consider the incomes of the bottom 10 percentiles as a group in all analyses and consider estimated impacts for this quantile to be a conservative bound on true impacts on the very poorest because it implicitly assumes perfect equality among its constituent population (Cowell, 2011).

1.2.2 Historical weather data

We merge all income data with an annual country-level panel of population-weighted near-surface air temperature data derived from the Global Meteorological Forcing Dataset for Land Surface Modeling (GMFD). which corrects model-based biases in NCEP/NCAR reanalysis product using observational meteorological station data (Dept. of Civil and Environmental Engineering, Princeton University, 2006). The GMFD reports historical weather data for the entire global surface at 0.25-degree resolution[2] for every three hours from the start of January 1, 1948 to the end of December 31, 2015. We use the product which reports daily averages of near-surface air temperature.

While other temperature series of similarly high resolution can provide a larger estimation sample by beginning before 1948 or updating more recently than 2015, results reported in the broad climate impacts literature have been generally consistent across data products. We favor the GMFD product as it has the unique feature of being the only reference product used to bias-correct and downscale the CMIP6 projection data series produced by the NASA Earth Exchange Global Daily Downscaled Projections; this allows the results of the present analysis to seamlessly integrate with future work exploring its implications under different future warming scenarios or attribution studies which attempt to decompose estimated climate change impacts into their anthropogenic and natural contributions, as introduced in Section A.4.

1.2.3 Temperature as an explanatory variable

It has been customary in the climate impacts literature to use functions of observed weather variables such as temperature or precipitation as explanatory variables in causal analyses. Histori-

[2]At this resolution, cells near the equator are approximately 27 km^2 (17 mi^2).

cally, their use in panel settings has been justified by an intuition that annual fluctuations in these geophysical conditions are plausibly exogenous with respect to most economic activity. Among several challenges to this intuition, Nath et al. (2024) objects to their validiity on the grounds that because levels of temperature do not satisfy conditions of an identified shock[3] and exhibit substantial autocorrelation, they are ill-suited for dynamic causal inference.

I follow Kahn et al. (2021) in defining temperature shocks $\widehat{\tau}_{it}$ as deviations in temperature relative to a climate 'state' variable $\overline{T}_{it}$ in turn defined as the M-period moving average of local temperature $\overline{T}_{it}$:

$$\overline{T}_{it} := \frac{1}{M}\sum_{m=1}^{M} T_{i,t-m} - 18 \tag{1.1a}$$

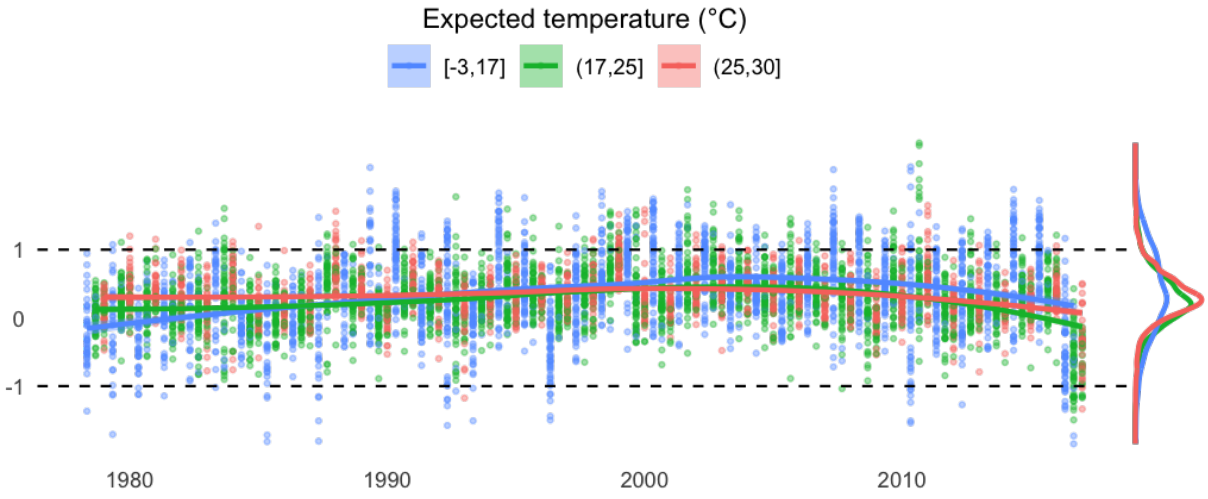
$$\widehat{\tau}_{it} := T_{it} - \overline{T}_{it} \tag{1.1b}$$

We subtract 18 in Equation (1.1a) so that our main effects in state-dependent regressions correspond to an intermediate representative climate of 18°C. For sufficiently large M, this definition of the state variable $\overline{T}_{it}$ has an appealing coherence with the convention of using long-run moving averages of weather realizations to define the slow-moving state of the climate at a fixed period[4]Most commonly, these "climate normals" are defined as 30-year averages to accommodate natural variation which would occur even in a stable climate so we set $M = 30$ in our main results. The most economically relevant example of this natural variability is the El Niño Southern Oscillation, an ocean-warming phenomenon which can vary average global surface temperatures by as much as 0.4°C across its three phase cycles which may span 2-7 years. Other minor oscillations can span multiple decades.

[3]To crudely summarize the conditions as outlined in the handbook chapter of Ramey (2016), a valid shock in a time series setting must i) measure unanticipated movements in relevant exogenous variables, ii) be exogenous with respect to current and lagged endogenous variables, and iii) be uncorrelated with other exogenous shocks included in the model. Temperature specified in levels do not satisfy the first condition and the third condition is violated in commonly used non-linear specifications where temperature enters as a polynomial.

[4]Nath et al., 2024 use a state variable that is constant for a given country. They suggest this is justified by the moderate warming over their observation period but I find that observed warming in my sample is non-trivial: the median country observes, on average, a 0.26°C (IQR: 0.18-0.36) increase in these 30-year temperature normals every 15 years.

Figure 1.1: Country-level time series of identified temperature shocks



Points correspond to shocks observed in individual country-years. The three colors correspond to an equal-sized binning of country-years by their moving-average temperature. Flexible local regression curves describe time trends split by this binning. Marginal distribution plots on the right show that colder country-years observe greater variance in shocks.

Figure 1.1 depicts the implied history of temperature shocks $\widehat{\tau}_{it}$ constructed in this manner with individual points corresponding to distinct country-year observations. These observations are divided into three equally sized groups according to their corresponding climate states $\overline{T}_{it}$; points and local regression curves are colored accordingly. The marginal distribution plots on the right demonstrate that colder seasonal countries experience more positive and more volatile temperature shocks than their warmer counterparts, a reflection of both greater natural weather variability in these settings and the fact that absolute anthropogenic warming is greater in regions closer to the Earth's poles[5].

The positive values of the curves in Figure 1.1 over almost the entire observation period captures the idea that systematic warming is characterized by the increased frequency and greater magnitude of positive (hot) shocks relative to negative (cold) shocks. Intervals where the curves are closer to 0 thus reflect relative balance in the incidence of hot and cold shocks, corresponding to periods of relative climate stability and slower warming.

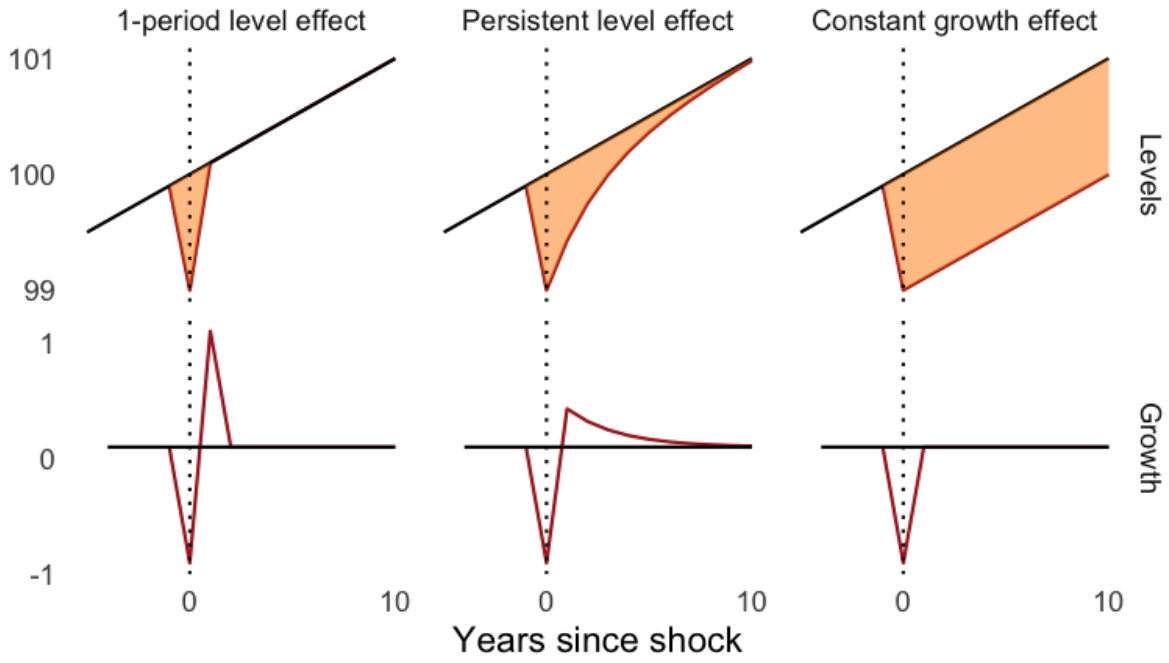
1.2.4 Modeling the persistence of social impacts: levels vs. growth effects

Figure 1.2 illustrates the conceptual distinction between transitory level effects and permanent growth effects. Each column corresponds to a different persistence structure for the dynamic effect of an isolated temperature shock in period 0 on an outcome measured in log levels (top panels) and first differences in these log levels (bottom), a convenient approximation of a growth rate. Black and red lines depict the evolution of the outcome variables in the absence and presence of a unit shock in period 0 respectively under the different described dynamics. Filled areas in orange measure the cumulative loss (or gain) in the outcome which may be attributed to the shock.

Under a one-period level effect, the level of, say, GDP falls by an estimated level $\widehat{\beta}_0$ in the year of the shock but fully recovers to its pre-shock trend (black) in the next period. In this scenario, the production lost in the year of the shock is exactly offset by an above-trend rate of growth which exactly offsets the negative growth effect in the year of the shock. The shock has no additional effect

[5]Figure A.6 provides a more direct illustration of this phenomenon for different global climate models

Figure 1.2: Levels and growth under different persistence structures



on future production and the area in orange represents the one-off impact from the shock.

Under the constant growth effect depicted in the rightmost panels, the same shock incurs the same effect on the levels but this effect is permanent so that GDP in all future periods is reduced by the same amount relative to the no-shock counterfactual. The difference between the two trajectories thus compound over time as observed by the orange area which grows linearly with the projection horizon. The corresponding growth effect depicted in the bottom panel is entirely transitory with no compensatory positive growth relative to trend.

The middle column describes an intermediate case where the economy requires several periods to return to its pre-shock trend. Cumulative differences attributable to the one-off shock continue to rise over time but to a diminishing degree until eventually compensatory positive growth exactly offsets the initial negative growth, i.e., the integral of the growth curve equals zero in the long run.

Now consider the following two models, which we may refer to as the levels and growth models respectively.

$$\log Y_{it} = \beta^L X_{it} + v_{it} \tag{1.2a}$$

$$\Delta \log Y_{it} = \beta^G X_{it} + u_{it} \tag{1.2b}$$

For simplicity, imagine the outcome variable Y_{it} represents a conventional economic aggregate like GDP and X_{it} a perfectly identified temperature shock. In the levels model, β^L represents an estimate of the change in log GDP attributable to a unit shock. Graphically, this impact is represented by the vertical distance between the black and red lines in the top-left plot. Since this is an entirely static model, there is no mechansim for X_{it} to affect future values of Y_{it} and so the model assumes a perfect return to trend in all ensuing periods. In the growth plot directly below, this is represented by a $\beta^L - 0 = \beta^L$ effect in period 0, an exactly offsetting $0 - \beta^L = -\beta^L$ effect on growth in period 1, and no effect on growth in all ensuing periods. In models specified in levels, level effects which persist for p periods can be accommodated by including p additional lags of X in the regression. A permanent growth effect can only be accommodated with the inclusion of infinite lags.

In the growth model, $\beta^G \neq 0$ represents a permanent change in the difference between consecutive values of Y_{it}. This is represented by the gap between the black pre-shock trend line and the observed GDP series in the top-right plot staying constant at β^G for all periods following the shock and is never offset. Level effects which persist for p periods can be accommodated by including p lags of X and finding that the sum of the $p + 1$ coefficients is exactly 0 so that the initial negative growth effect is eventually offset by sufficient compensatory growth.

The differing restrictions imposed by these stylized regression models is a simplification of perhaps the most influential unsettled question in climate economics: the persistence of climate impacts. For example, two studies projecting the proportional loss in cumulative global production by 2100 arrive at point estimates of -20% (M. Burke et al., 2015) and -3.4% (Casey et al., 2023) under the same worst-case warming scenario. Methodologically, the two studies use similar methods and data except for their choice of outcome variable.

Intermediate persistence structures where level effects are diminishingly persistent can be accommodated by either model by including sufficient lags of the explanatory variables. Of course, identifying variation is reduced with the addition of additional explanatory variables, a problem exacerbated in specifications where shocks enter non-linearly and settings where shocks exhibit significant autocorrelation, both general features of modern climate models using absolute temperature as their explanatory variables. In addition, models specified in levels are notoriously vulnerable to unit root concerns, further undermining credible identificaiton.

Indeed, the appendix to M. Burke et al. (2015) concludes that the identifying variation which remains after lag augmentation to test for persistence "cannot reject the hypothesis that this effect is a true growth effect nor the hypothesis that it is a temporary level effect", rendering these extremely influential assumptions effectively unfalsifiable. Absent substantially improved data, resolving this issue may require richer data or an alternative methodological approach relatively robust to misspecification.

1.3 Empirical analysis

1.3.1 Methods: cumulative temperature multipliers

To characterize the dynamic effects of an identified temperature shock on national income distributions, we adapt the method of local projections first introduced in Jordà (2005). As Jordà (2023) notes, this semi-parametric method of estimating impulse response functions has found growing appeal outside of applied macroeconomics in recent years as recent econometric literature has demonstrated its flexibility in incorporating non-linearities (Cloyne et al., 2023), its coherence with the potential outcomes framework (Dube et al., 2024), and its robustness to misspecification and non-stationarity (Montiel Olea & Plagborg-Møller, 2021; Montiel Olea et al., 2024; Piger & Stockwell, 2023). Berg et al. (2024), Bilal and Känzig (2024), and Nath et al. (2024) are the first works I am aware of to apply this method to estimate dynamic economic responses to climate shocks, demonstrating how these features can help address the identification issues described in Section 1.2.

Our application of this method to our inequality setting entails the following steps:

1. Estimating the state-dependent impulse response of income to an identified temperature shock
2. Estimating the state-dependent impulse response of temperature shocks to an identified temperature shock
3. Using both to estimate a state-dependent multiplier to isolate the dynamic income response net of the shock's effect on future shocks
4. Statistically evaluating the significance and persistence of this multiplier
5. Repeating the above steps for each income quantile of interest

We briefly describe this process below and provide additional exposition in Section A.2 using an application to country-level GDP data.

Estimating state-dependent impulse responses

We use country-year panel datasets to estimate dynamic responses to temperature for an outcome of interest represented by the outcome variable y_{it}. For example, y may represent the average logarithm of income of the bottom 10% of individuals in country i in year t and we may repeat the estimation process for each income decile and use the results to infer a total effect on national income distributions.

The method of local projections entails estimating $H + 1$ iterations of the following single-equation model each corresponding to different horizons $h \in \{0, 1, ..., H\}$ periods after an identified temperature shock:

$$\begin{aligned} \Delta_h y_{i,t+h} &:= y_{i,t+h} - y_{i,t-1} \\ &= \beta_{1h}\widehat{\tau}_{it} + \beta_{2h}\widehat{\tau}_{it} \cdot \overline{T}_{it} + \lambda_h \overline{T}_{it} + \boldsymbol{\gamma}_h^\top \mathbf{Z}_{it} + \mu_i + \eta_t + u_{i,t+h} \end{aligned} \tag{1.3}$$

In our main specification, the matrix $\mathbf{Z}_{it}$ includes lagged controls $\{\widehat{\tau}_{i,t-j}, \widehat{\tau}_{i,t-j} \cdot \overline{T}_{it}, \Delta y_{i,t-j}\}_{j=1}^{p}$. The inclusion of p lags of the temperature shock $\widehat{\tau}$ is included to address potential autocorrelation in the explanatory variable. Interactions of these lagged shocks with the state variable capture

potential state-dependencies in these autocovariance structures motivated partially by the finding in Figure 1.1 that shocks in cooler climates tend to be more positive on average and feature greater year-to-year variability. Lags of the outcome are included as a guard against trend non-stationarity. Finally, the sets μ_i and η_t of fixed effects are included to absorb time-invariant unit-specific effects and common year-specific effects. In all, this model differs from the state-dependent model of Nath et al. (2024) only in how temperature shocks are constructed and in permitting the state variable to vary over time as described in Section 1.2.3.

For each projection horizon h, the main regression coefficients of interest attribute to a unit temperature shock a main effect β_{1h} representing a change in the "long difference" between y observed h periods after the shock and y observed the period before the shock. Equivalently, this can be interpreted as the change in y h periods after the shock relative to a counterfactual absent the shock. In most of our results, y represents the logarithm of income for some quantile of the national income distribution; in these cases, coefficients are interpreted as percentage-point impacts on income.

An interaction term β_{2h} allows the effect at each period to scale with an observation's time-varying expected temperature $\overline{T}_{it}$, defined in Section 1.2.3. This 'state dependence' is motivated by the standard result in the climate impacts literature that the marginal effect of a given temperature shock on aggregate economic outcomes is increasingly negative for warmer climates.

The inclusion of a main effect λ_h associated with the time-varying state variable allows for the possibility that expected impacts may depend on changes in expected temperatures, net of the impact of climate change on the distribution of temperature shocks. For example, if a hot country stabilizes at a climate that is 1°C warmer than some reference period, the productive potential of the country is altered even in the absence of temperature shocks as the new climate will be differently conducive to some economic activity. Neglecting this possibility amounts to assuming perfect climate adaptation in the long run and that climate change is only socially impactful by biasing the trend in temperature shocks.

It is worthwhile to take stock of the assumptions underlying the sequence of regressions spec-

ified in Equation (1.3). First, because coefficients for each horizon are estimated through distinct regressions using differently constructed variables from different subsamples of the data, there are in principle no restrictions imposed by the model on the shape of the resulting impulse response function for a given value of the state variable; it is in this sense that the method is partially non-parametric.

This flexibility is particularly appealing in the context of estimating responses to weather shocks. As summarized in Section 1.2.4, fully parametric methods have necessarily required the modeler to assume either of the extreme persistence patterns depicted in Figure 1.2 with limited ability to accommodate intermediate persistence dynamics. By instead allowing the modeler to remain mostly agnostic to the data generating process, this method circumvents a notorious source of extreme estimation instability and can accommodate any persistence structure in principle without restriction[6]. Specifically, the sequence of coefficient estimates $\{\beta_{1h}\}_{h=0}^{H}$ in Equation (1.3) maps directly to the logged differences between the black and red time series in the first "levels" row or equivalently the sequence of its first differences correspond to the red time series depicted in the second "growth" row. Thus, a one-period level effect would be represented by a sequence where $\beta_{1h} = 0$ for all horizons $h > 0$ while a constant growth effect would be reflected in a sequence converging to a non-zero constant[7].

Another relevant benefit of the method is that local projections specified in long differences have recently been shown to be particularly robust to biases associated with highly persistent and even non-stationary processes Jordà (2024) and Piger and Stockwell (2023). This is a long-standing concern with regressions specified in levels such as those which use logged GDP as an outcome (Campbell & Mankiw, 1987; Nelson & Plosser, 1982). Altogether, the method is relatively robust to different forms of misspecification

These minimal restrictions naturally come with costs in estimation efficiency; IRFs estimated by local projection regressions are fundamentally noisier than their parametric and structural coun-

[7]Residuals across the projection horizon are generally autocorrelated but do not affect consistency of point estimates.
[7]Of course, under a conservative null hypothesis of non-persistence at all horizons, a permanent growth effect cannot be affirmatively evidenced and the modeler can only produce evidence of a lower bound on persistence

terparts such as vector autoregressions[8]. For this reason, we expect resulting estimates to be less precise than more restrictive methods, particularly for increasingly distant projection horizons.

At the same time, the model is not entirely free of restrictions. Most critically, the model specified in Equation (1.3) imposes within-horizon linearity assumptions on the coefficients of interest. Since the shock enters additively into Equation (1.3), responses to the shock are assumed to scale proportionally so that a 1°C shock has exactly twice the impact of a 0.5° shock and the exact negative of the effect of a -1°C shock.

By the same principle, the linear interaction term imposes a symmetry in the state dependence such that holding all other factors fixed, the impact of a shock is assumed to be $k\beta_{2h}$ percentage points larger in a climate some quantity k degrees warmer than baseline and $-k\beta_{2h}$ in a climate k degrees cooler than baseline. While appreciating that unpredictable non-linearities abound in the interactions between social and climate systems (Dietz et al., 2021), I consider these necessary and palatable restrictions.

Collecting estimates of the coefficients of interest $\{\widehat{\beta}_{1h}, \widehat{\beta}_{2h}\}_{h=0}^{H}$ from Equation (1.3) and adopting the notation of Jordà (2023), our set of local projection estimates imply a family $\mathcal{R}_{\tau \to y}(h; \overline{T})$ of state-dependent impulse responses as a function of the state variable $\overline{T}$:

$$\mathcal{R}_{\tau \to y}(h; \overline{T}) := \widehat{\beta}_{1h} + \widehat{\beta}_{2h} \cdot \overline{T} \tag{1.4}$$

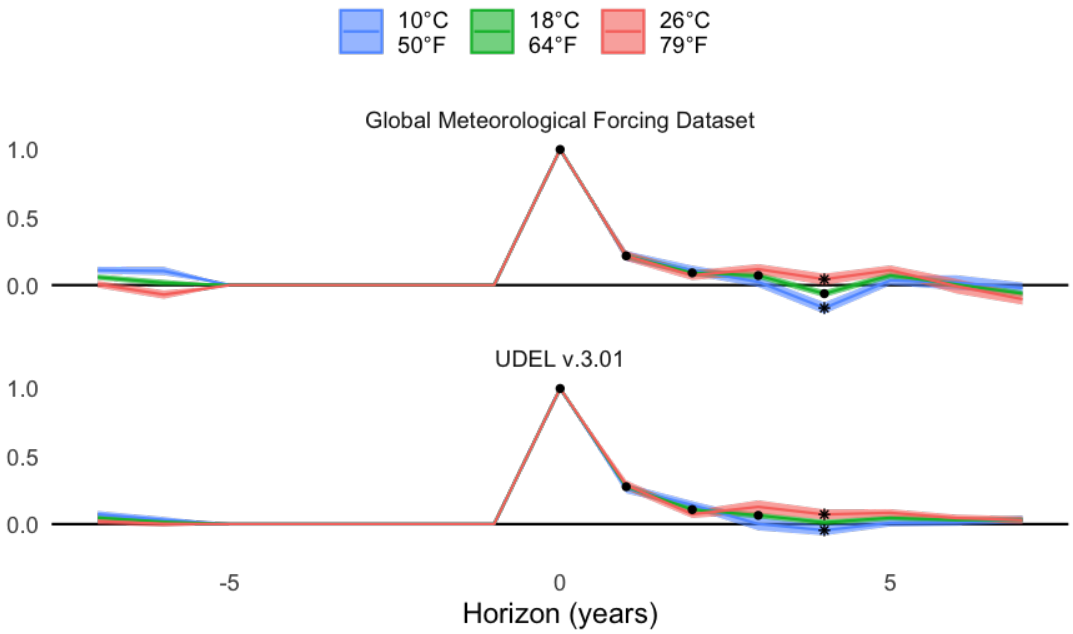
Estimating state-dependent cumulative multipliers

To make progress accounting for the persistence of temperature shocks themselves (see: Section 1.2.3), Figure 1.3 depicts impulse responses which result from using temperature shocks themselves as an outcome variable y as constructed by two commonly used datasets for gridded historically observed weather. We plot separate impulse responses for three representative states cor-

[8]Different compromises in this bias-variance tradeoff can be obtained by imposing additional restrictive parametric assumptions such as the functional approximation approach of Barnichon and Matthes (2018) which can substantially diminish estimation error.

responding to the 20th and 80th percentiles of country-year expected temperatures in our sample (10°C colored blue and 26°C colored red) and their average (18°C, roughly the 40th percentile and colored green). This coloring scheme and these reference temperatures will be used throughout the rest of this paper. [9]

Figure 1.3: State-dependent impulse response of temperature shocks to a unit temperature shock



State-dependent impulse responses of identified temperature shocks estimated using two widely used temperature datasets. The points in black denote significant evidence of autocorrelation. In both cases, persistence is large in magnitude for the first 1-2 periods following a shock before flattening. Negative horizons are included as a check for significant pre-shock trends.

Points along the IRFs depicted in black correspond to point estimates which are deemed significantly persistent by a joint hypothesis test with $\alpha = 0.05$. The underlying test is based on the significance-band test recently introduced in Inoue et al. (2024) and adapted for our state-dependent

[9]Note that for expositional convenience, we have transformed our state variable to center on the intermediate reference temperature of 18°C so that the responses in green correspond to the sequence of estimated main effects $\widehat{\beta}_{1h}$ while the responses in blue and red incorporate equivalent but opposite interaction effects corresponding to ±8°C deviations from this reference state

setting as described in Section A.2.2. In short, we have adopted a convention where filled circles correspond to significant main effects and asterisks correspond to significant interaction effects under a joint null hypothesis of non-persistence at each horizon. The presence of a significant effect up to horizon h is interpreted as evidence of a lower bound of h-period persistence. Thus, the IRFs presented in Figure 1.3 constitute significant evidence that temperature shocks positively influence successive temperature shocks for at least four periods although the magnitude of this persistence is mostly diminished within 1-2 years.

As emphasized by Nath et al., we will want to directly account for this persistence when attributing social impacts to a unit temperature shock. An analogy can be drawn to the estimation of multipliers in empirical macroeconomics: when estimating the impulse response of, for example, GDP to a fiscal policy shock, one must account for the tendency of stimulus programs to be implemented in stages or to be followed by additional, if typically smaller, rounds of stimulus. Otherwise, the effect sizes implied by simple impulse response functions misattribute cumulative changes in the outcome only to the initial unit shock; the calculation of multipliers scale these effects at each horizon by the corresponding accumulation of shocks. As such, we will want to present our results in the form of these cumulative temperature multipliers rather than as raw impulse response functions.

In Section A.2.3, I describe how we modify methods for efficiently estimating cumulative temperature multipliers in order to implement this transformation while maintaining the ability to statistically test for persistence. Ultimately, the process of estimating cumulative temperature multipliers for some outcome y amounts to estimating a cumulative variant of the original single-equation model in Equation (1.3). That is, for every discrete income quantile q of interest and projection horizon $h \in \{0, 1, ..., H\}$, we estimate the following state-dependent model:

$$\Delta_h^c y_{i,t+h}^q = m_{1,h}^q \widehat{\tau}_{i,t+h}^c + m_{2,h}^q \widehat{\tau}_{i,t+h}^c \overline{T}_{it} + \lambda_h \overline{T}_{it} + \mathbf{Z}_{it}^q \boldsymbol{\gamma}_h^q + \mu_i + \eta_t + \varepsilon_{i,t+h}^q \tag{1.5}$$

Here, variables denoted by a superscript c are cumulative versions of the original variables constructed using the one-step process summarized by Algorithm 1 in Section A.2.3. The quantile-specific matrix of controls $\mathbf{Z}_{it}^{q}$ include lags $\ell \in \{1, ..., L\}$ of the shocks, the interaction of the shock with the state variable, the outcome $\Delta y_{i,t-\ell}^{q}$, and country-level aggregate growth $\Delta y_{i,t-\ell}$.

State-dependent cumulative temperature multiplier estimates $\{\widehat{m}_{1,h} + \widehat{m}_{2,h}\overline{T}_{it}\}_{h=0}^{H}$ describe the effect of an identified temperature shock on the cumulative stock of outcome y^q after h periods as a function of the expected temperature.

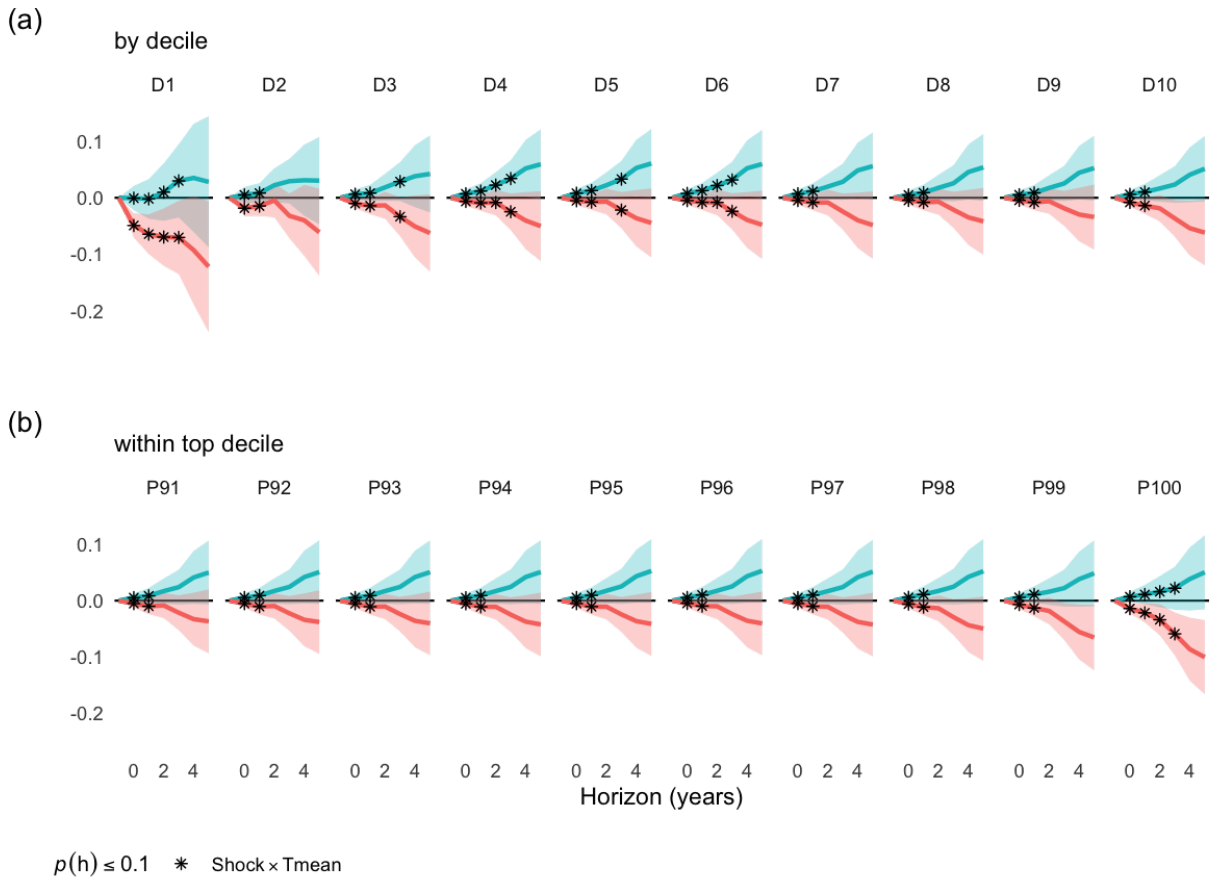
1.3.2 Empirical results: evidence for a temperature effect on within-country inequality

Panel (a) of Figure 1.4 summarizes the results of estimating separate sets of state-dependent umulative temperature multipliers for all 10 within-country income deciles. Here, color-coding follows the same intuitive scheme as previous plots with blue and red corresponding to representative expected temperatures of 10 and 26°C respectively[10]. The points in asterisks indicate significant interactions which impose differentiated effects based on expected climate. These effects persist for between 1-3 years, concentrating negative impacts in hot countries while keeping cooler countries relatively insulated, consistent with the aggregate results standard in the climate literature and the dynamic GDP results reported in Section A.2.3. Estimation error is larger than when estimating GDP impacts in Figure A.2b, unsurprising given the difference in temporal coverage and data quality.

Importantly, we are now able to compare impacts across quantiles within countries. Focusing on hot-climate countries, the estimates reported in Panel (a) imply an incidence pattern wherein negative effects of a transitory temperature shock to hot countries are found at all quantiles but are particularly dire at the bottom. For 26°C countries, the 90% confidence intervals for the impact of a temperature shock on cumulative income a year after the shock is $[-10.0, -2.9]$ percent for the bottom decile compared to $[-3.2, 0.3]$% for the second decile. This generally improves as one moves up the income distribution, stabilizing at approximately $[-2.4, 0.6]$ percent by the fourth

[10]We omit the green 18°C main effect to avoid over-busy plots, but it is of course implied as the average of the two depicted curves

Figure 1.4: Cumulative income response to a 1°C shock



decile through to the 99th percentile. Interestingly, the estimated impacts then discontinuously drops to $[-3.9, -0.5]$ for the richest percentile. Panel (b) shows this discontinuity within the top decile and also indicates this effect is significantly persistent for two additional years. In contrast, cumulative growth incidence is much flatter across the income distribution with point estimates suggesting positive impacts are somewhat larger for the middle classes between the third and sixth deciles.

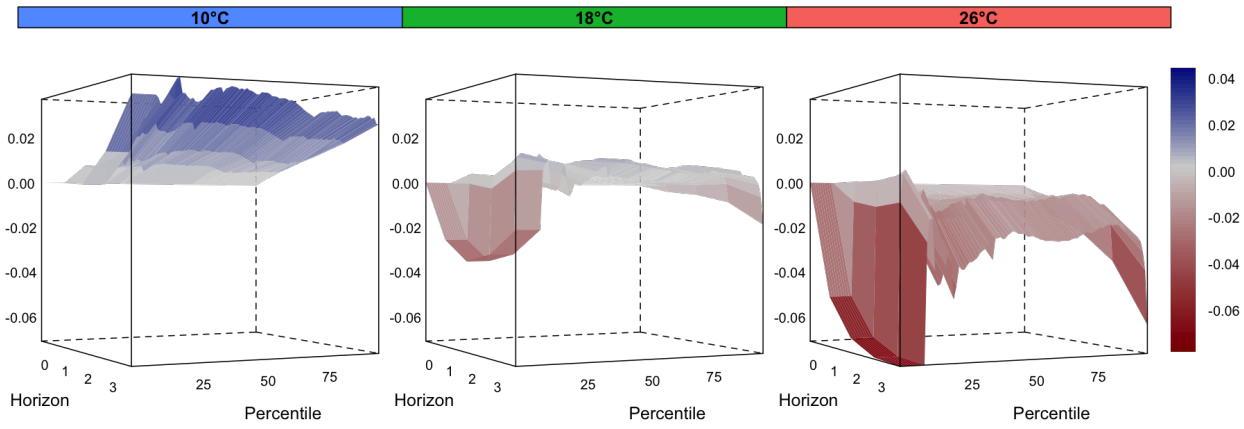
Figure 1.5a and the bottom panel of Figure 1.5b capture these discontinuities at the extremes more clearly by plotting point estimates of percentile-level impacts over time for our three representative climates. These figures imply that negative expected impacts to hot countries are indeed concentrated on both the bottom 10% and the top 1% of the distribution[11]. To the best of my knowledge, this particular vulnerability of the highest incomes in hot and generally poor countries to transitory environmental shocks has not been previously observed or theorized although it is consistent with recent work outside environmental economics demonstrating this group's disproportionate sensitivity to aggregate conditions even compared to other members of the top income decile (Alvaredo et al., 2018; Roine et al., 2009). In ongoing work, I investigate the extent to which this may be driven by weather-induced capital depreciation, capital shares in agricultural income, and the relative unavailability of weather derivatives in these settings.

This first set of new results describe inequalities in the growth incidence of impacts across a national income distribution as a function of the local climate. The disproportionate impacts to the bottom decile strongly imply a regressive effect: In the aftermath of an isolated temperature shock, a representative member of the the bottom decile in a 26°C country can expect to earn 7% less in total over the next three years than they would had the shock never occurred. This compares to losses of 2.2% for the median adult in their country and a *gain* of 3.0% for their counterpart in a country with a 10°C climate. At the same time, a representative member of the 100th percentile in the same 26°C country, represented by the thinnest sliver at the top of the rightmost distributional

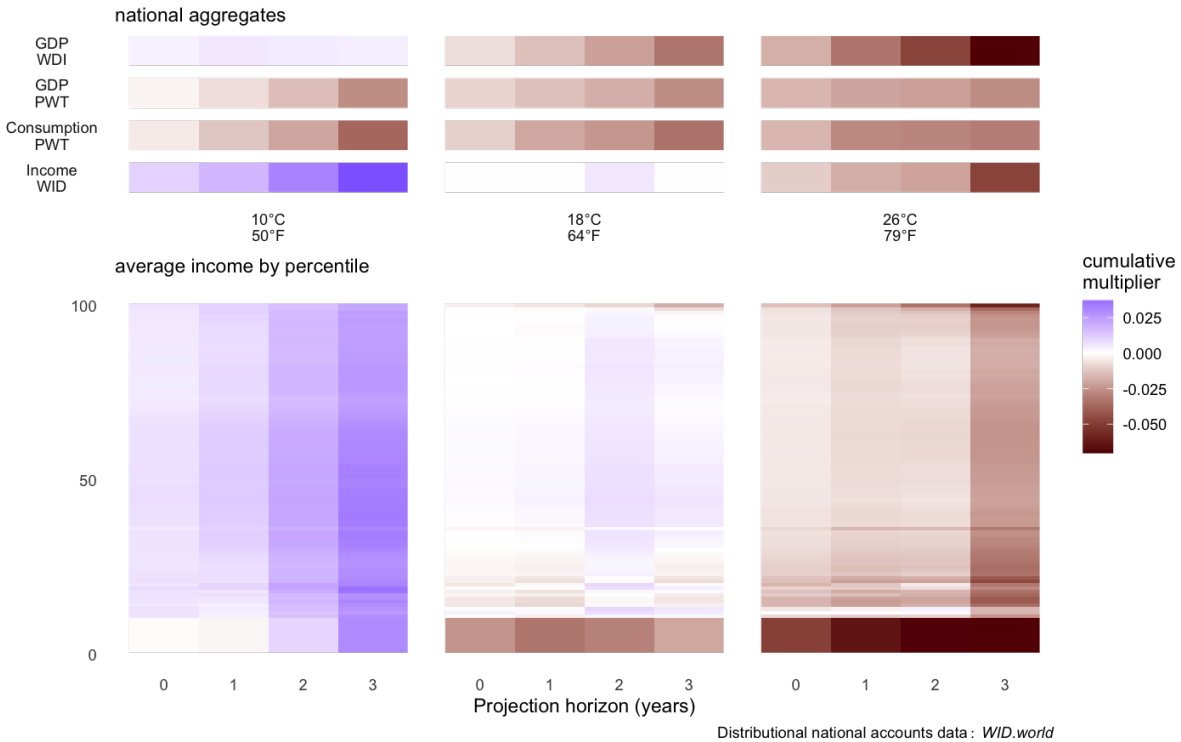
[11]As mentioned in Section 1.2, the income data for the bottom 10% is not well suited for decomposition into percentile effects but one would expect an ideal decomposition would reveal a similarly steep within-decile gradient in which negative impacts are driven by the poorest percentiles within the bottom 10%.

Figure 1.5: Point estimates of cumulative income growth incidence responses to a 1°C shock

(a) Incidence patterns for three representative climates



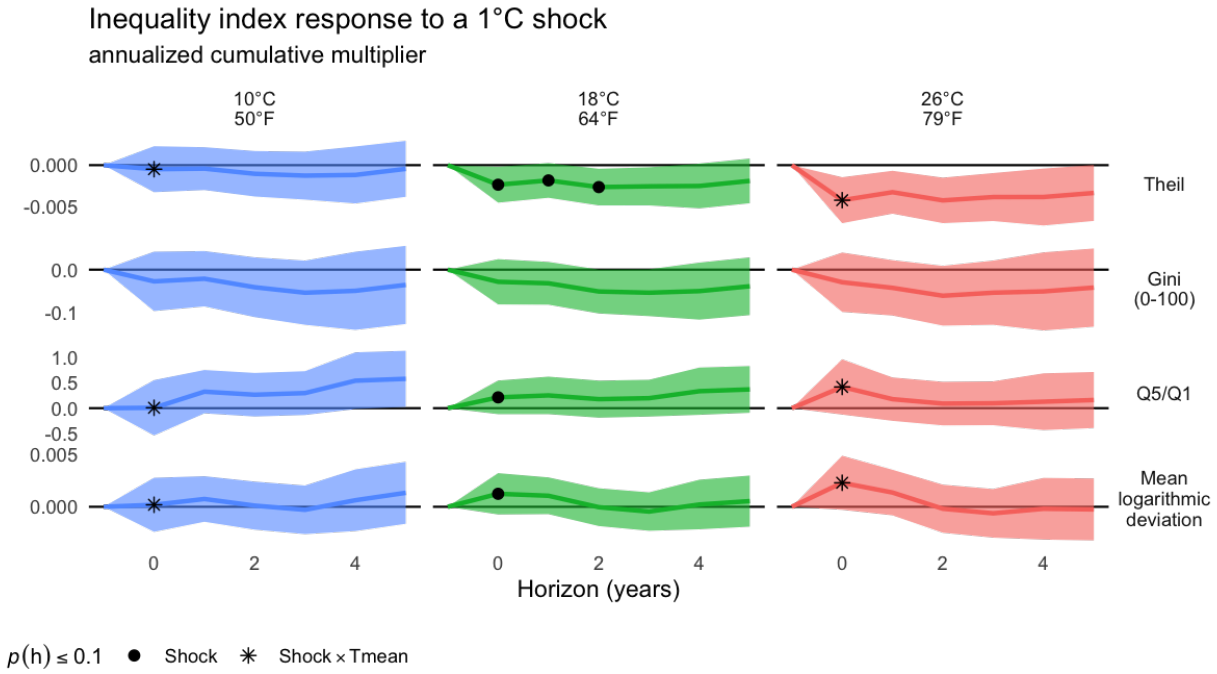
(b) Aggregate vs. distributional responses



contour plot in (b), can expect to earn 5.9% less over those three years than they would otherwise compared to 3.6% for the 99th percentile.

While these results strongly imply substantial redistributive effects from a temperature shock, their implied effect on inequality depends both on the initial distribution of income shares across quantiles and the measure of choice, especially given that growth effects are non-monotonic in income quantiles. Figure 1.6 depicts the annualized effect of a temperature shock on various measures of inequality. Units are not directly comparable across measures but all measures are increasing in (their respective conception of) inequality. Measures are provided in order from the most sensitive to redistribution of top incomes (the Theil index) to the most sensitive to redistribution of bottom incomes (the mean logarithmic deviation index).

Figure 1.6: Inequality index response to a 1°C shock



The filled circles in the middle column drawn with green error bands correspond to significant main effects of the temperature shock on the given measure of inequality. The asterisks on the plots on either side correspond to significant interactions, measuring linear heterogeneity by expected

temperature. In light of the growth incidence results just discussed, the near-zero impacts to cold-country inequality and substantial effects to hot-country inequality are unsurprising. Null effects on the Gini index, known to be relatively insensitive to changes at either end of the income distribution, are consistent with the disproportionate impacts to the bottom 10% and top 1%.

By the Theil index[12], a transitory temperature shock appears to actually *improve* inequality while inequality as measured by the ratio of incomes of the top 20% to the bottom 20% and mean logarithmic deviation[13] both suggest the opposite. From the previous analysis, we can surmise that the progressive effect on the Theil index is being driven by the transfer away from top 1% while for the other two measures, the regressive transfer of income shares away from the bottom 10% outweighs this 'improvement' in inequality. While the impacts to the bottom 10% are substantially larger proportionally and represent a population segment 10 times as large, for the Theil measure, the disproportionately large share of income originally held by the top 1% constitutes a large enough share of total national income that these measures disagree on the net effect on inequality. Of course, since the overall effect on income is negative at all percentiles for hot countries, these inconsistencies amount to differently equitable shares of a strictly smaller national income.

1.4 Counterfactual analysis

1.4.1 From identified temperature shocks to systematic climate change

The results presented in Section 1.3.2 are expressed as responses to a 1°C shock, which notably is much larger in magnitude than shocks we would expect to observe in a typical year in any country. Referring back to the time series of shocks illustrated in Figure 1.1, the median magnitude of a shock varies by the rate of climate change which will in turn vary across geographies and time.

[13]One interpretation of the Theil measure is as a geometric measure of inequality. For any given income distribution, an individual with income \$100,000 giving \$10 to someone with income \$10,000 would improve the Theil index measure of inequality by the exact same amount as someone with a \$1,000 income giving the same \$10 to someone with income \$100 because the ratio of incomes between the donor and recipient are the same (Cowell, 2011).
By the same principle, a \$100 income earner regressively transferring 10% of their income to a \$1,000 income earner would offset the progressivity of a \$100,000 income-earner transferring a ten-thousandth of their income to a \$10,000 income earner. If that is still not intuitive, then you are in good company (Sen, 1973).

[13]One interpretation of the mean logarithmic deviation measure is as the average disutility of a given income distribution compared to an equal distribution of the same total income, assuming logarithmic utility.

For the warmest 33% of country-years in our data, the median shock was approximately 0.19°C in absolute value in the 1970s and then increased to between 0.30-0.37° in the ensuing decades. Thus, hot-country effects can be understood as responses to a shock 3-5 times larger than is typical. For the coldest 33% of country-years, the median magnitude of a shock was approximately 0.35°C in the 1970s and 1980s, 0.51-0.53°C in the 1990s and 2000s, and 0.59°C in the 2010s. Their effect sizes then can be interpreted as the response to a shock just 2-3 times larger in magnitude than the median.

So while those results consistently demonstrated much larger distributional responses to the same shock for residents in warmer geographies, this disparity is at least partially offset by the relatively mild incidence of shocks in these settings. The counterfactual analysis we document in this section is valuable as a means of interpreting these reduced-form results practically, accounting for the spatial incidence of temperature shocks as well as other factors such as the absolute distribution of global incomes across country-quantiles and the spatial distribution of the global population across different geographies as observed in the historical record.

In the absence of a structural climate-economy model, it is standard practice to extrapolate reduced-form effects estimated using short-term weather variation to infer long-term impacts of sustained climate change. Broadly, this involves applying the estimated impact model to a counterfactual climate scenario to produce a counterfactual trajectory for the social outcome of interest. The difference between the social outcome of interest under this counterfactual relative to its trajectory under a reference scenario (e.g., for historical attribution studies like the present analysis, the actual historically observed trajectory of this outcome) are then attributed to the differences in the climate under the two scenarios.

In our case, we are estimating the effect of changes to the global climate system between 1980 and 2016 on the global distribution of income. Using the observed history of country-year temperatures, we calculate the implied history of country-year termperature shocks as defined in Equations (1.1a) and (1.1b). For the bottom national income decile and each of the 11th to 100th national income percentiles, we collect $B = 500$ bootstrap estimates of the state-dependent cumulative tem-

perature multipliers of income (Equation (1.5)). Then integrating these percentile-bootstrap specific estimates to the observed histories of temperature shocks implies a counterfactual history of country-percentile-year income absent any temperature shocks between 1980 and 2016. The differences between this simulated history of global income distributions and the observed history of global income distributions are then attributed to changes in the climate over this 36-year period, a long-enough horizon to dominantly attribute to anthropogenic causes by the conventional definition of a climate normal.

This process (and its analogous implementation for projecting future climate change impacts) offers several significant advantages over prior approaches to this type of counterfactual attribution exercise. First, since estimates of cumulative damages result from local projection regressions, dynamic impacts are identified directly from medium-term long differences in income and disciplined by a statistical test of the persistence of these impacts rather than imposed by modeling choices, circumventing the contentious levels vs. growth criticism described in Section 1.2.4.

Second, models which use levels of temperature as an explanatory variable have been challenged on the grounds that these counterfactual exercises require them to extrapolate effects of treatments that have never been observed in the estimation sample. For example, the model of M. Burke et al. (2015) specifies a globally generalized damage function which attributes to a given level of temperature an expected impact on GDP growth. This means that studying scenarios where annual country temperature levels exceed any observed in the estimation sample, much less the history of the human species, requires increasingly heroic extrapolations. This is much less of a concern in our model where the treatment variable is an identified temperature shock which is dynamically anchored by a time-varying state variable defined as a moving average of temperature levels, which can accommodate a capacity for adaptation in the sense of S. M. Hsiang (2016) wherein the "direct" harm of exposure to a given temperature level can be partially offset by Bayesian "belief effects" wherein adaptive expectations about the distribution from which local weather realizations are drawn.[14]

[14]More extreme adaptation can be accommodated by lower values of M, the number of lags used to construct the moving average defining the state variable. However, a review of the empirical literature on climate adaptation,

Finally, I argue that our approach to attribution is more intuitive than those deployed by comparable studies which use identified temperature shocks as their treatment variable. Bilal and Känzig (2024) and Nath et al. (2024) are other recent papers which use local projection regressions to attribute economic impacts to identified temperature shocks. Curiously, both define their shocks as residuals from slightly different autoregressive distributed lag models of temperature levels. As I understand it, doing so seems to assume a model of nearly immediate and costless climate adaptability such that if a country is exposed to, for example, 40°C weather five years in a row, the fifth year will impose no economic impact because it will register as a 0°C shock. Because of this hyper-adaptability, the time series of historically observed temperature shocks is largely homoskedastic around a constant mean of zero, a stark contrast to the positively biased time series we observed in Figure 1.1[15]. Because of this absolute symmetry in positive and negative shocks and the within-horizon linearity assumption discussed in Section 1.3.1, were they to adopt the approach I had outlined for attributing damages to long-term climate change, the cumulative impacts of historically observed temperature shocks would net out to essentially nil[16].

1.4.2 Results

In constructing these counterfactual series, each bootstrap b assigns each country-quantile-year to a global percentile corresponding to their population-weighted income ranking across the global income distribution for that year. Finally, we calculate the long-term growth between 1981 and 2016 implied by each series for each bootstrap series. The resulting long-term growth incidence curve is depicted in Figure 1.7a.

however, concludes "limited evidence" of adaptive behavior moderating the direct impacts of weather shocks in the context of climate change. (M. Burke & Emerick, 2016)

[16]In order to project the trajectory of GDP under different warming scenarios, Nath et al. instead seem to modify a delta approach commonly used in studies which use damage functions specified in temperature levels. This entails linearizing the difference between country-level temperatures at the end of the 21st century and a baseline period ending in 2010. Under the most extreme warming scenario, the population-weighted mean temperature change over this period is 4.3°C with the largest differences in countries furthest from the equator. Nath et al. then multiply these projected long differences in *levels* of temperature by their estimated cumulative responses to an identified temperature *shock*, seemingly interpreting the result as the cumulative loss in GDP attributable to climate change over these 90 years. The motivation for this approach is unclear to me especially in combination with the hyperadaptability implied by the construction of the temperature shock.

These results may be interpreted as impacts of observed climate change insofar as it the influence of climate change is sufficiently well-captured by asymmetries in the incidence of temperature shocks and changes to expected temperatures over the 36-year period of analysis. Section A.4 presents additional results decomposing these effects into their anthropogenic and natural contributions by applying methods from climate attribution science to output from state-of-the-art climate models, arriving at largely the same conclusions.

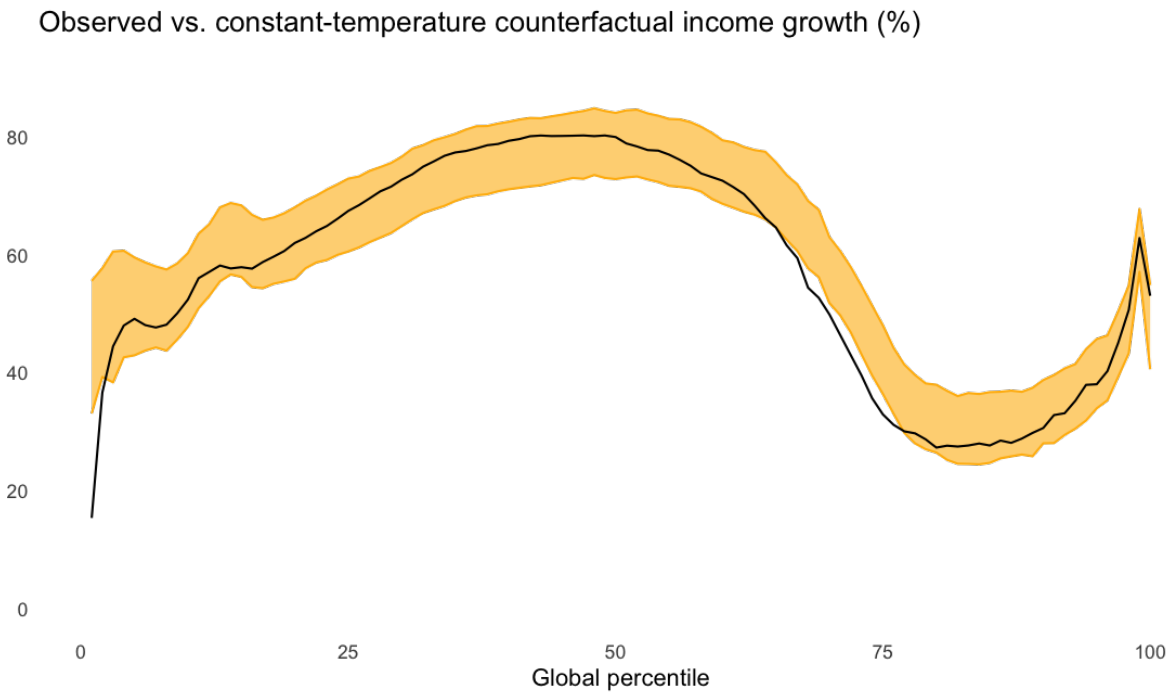
Here, the historically observed level of growth over this period is represented by a black filled line, reproducing a version of the "elephant curve" famously documented in Lakner and Milanovic (2016). The blue band depict the 90% confidence interval for the counterfactual removing the influence of temperature shocks on income. Regions where this band is above the observed incidence curve imply deprivations of income attributable to the incidence of temperature shocks over this period. Point estimates imply probable net harm for around 65% of the global sample.

Despite proportionally milder absolute warming in the poorest countries (see Figure A.6), this negative incidence still concentrates on the world's poorest 20%, representing 1.2 billion people in our 2016 sample and mostly located in warm developing countries. Importantly, as in Alvaredo et al. (2018), calculations underpinning this figure omit the bottom 10% of each country's income distribution because of the inclusion of unemployed adults and compositional instability effects which arise from assigning deciles of very large populations to percentiles representing smaller populations. For the poorest percentile in our data, we find that incomes absent these disproportionate temperature shocks would be 29% $[18, 41]$ higher than they are today. As these effects exclude precisely the subpopulation we have already found to be the most economically vulnerable to temperature shocks, this growth incidence curve should be interpreted as a likely very conservative estimate of deprivations to the global poor.

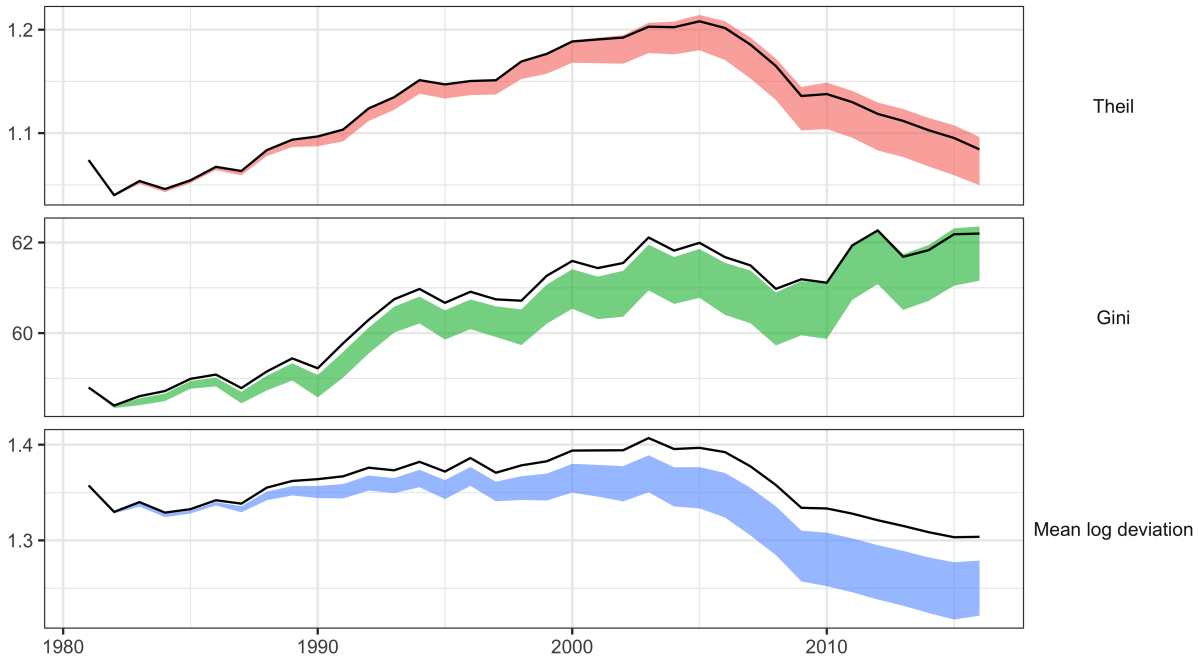
Elsewhere, estimates also imply likely harm from warming for the 52nd-97th percentiles. That the top 3% are relatively unaffected despite the negative effects for the top 1% in Section 1.3 is because these groups are dominated by extreme incomes in relatively cold countries. Effects on the global middle class are relatively tempered with slightly positive point estimates estimated for es-

Figure 1.7: Global inequality, observed vs. zero-warming counterfactual

(a) Global incidence of growth, 1981-2016



(b) Global inequality indices, 1981-2016



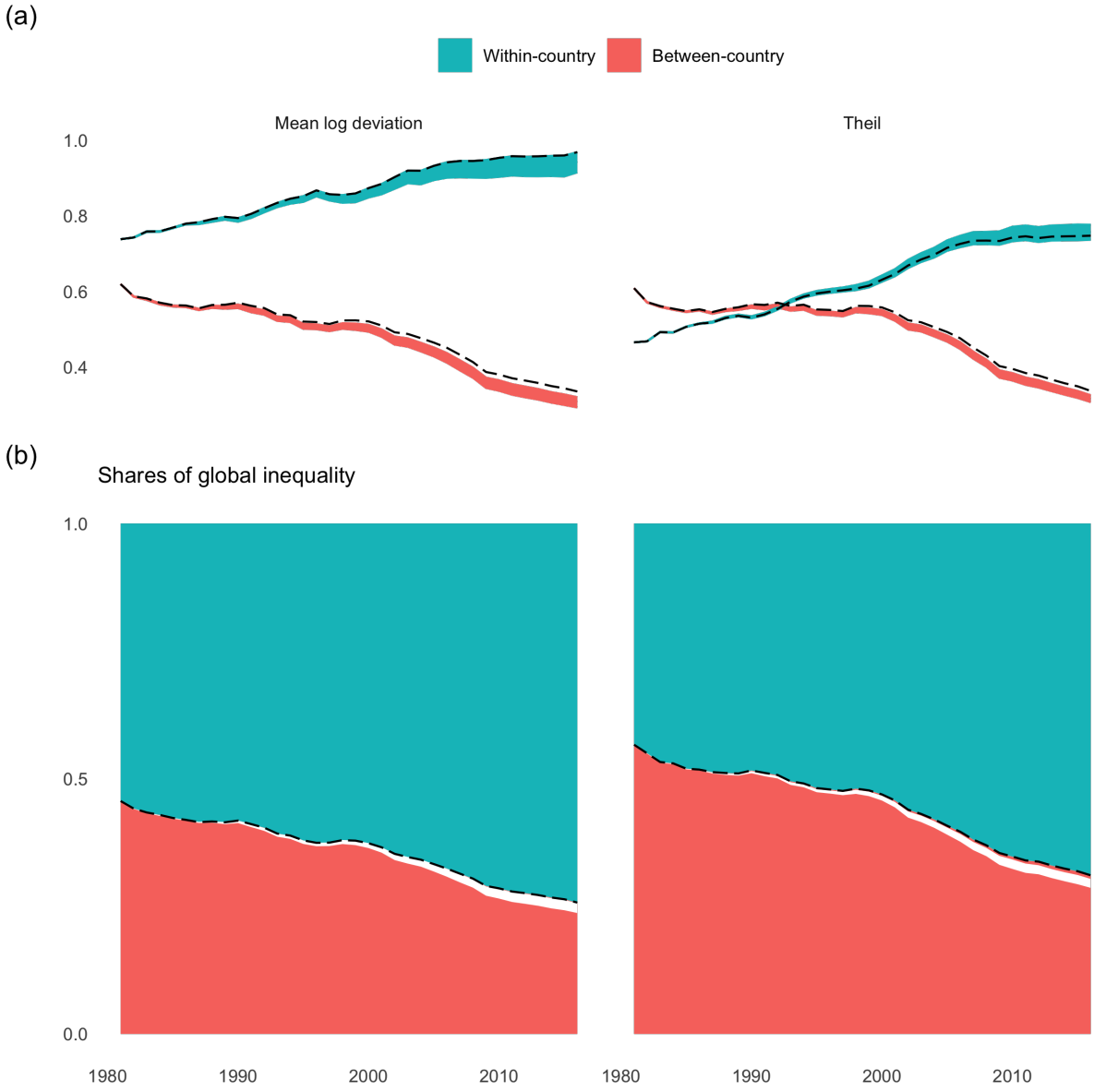
sentially the entire second quartile, also consistent with the within-country incidence curves which found the middle class experienced the mildest economic impacts across all climates.

Despite these different-signed effects, the counterfactual growth incidence band is noticeably flatter than the elephant curve, implying that historical climate change has been globally regressive in addition to being a likely drag on total growth. Figure 1.7b plots the time series for the global equivalents of the within-country inequality measures introduced earlier. In our within-country analysis, we identified opposing regressive transfers from the bottom decile and progressive transfers from the top decile which opposed one another to produce contradictory effects on within-country inequality depending on the measure of choice. In comparison, the historical effects of these shocks on global inequality are large and regressive across all measures, increasing the global Theil index by 0.9% $[-1.8, 2.8]$, the Gini index by 0.5 points $[-0.2, 1.0]$, and the mean logarithmic deviation index by 3.9% $[1.6, 6.4]$.

Figure 1.8 decomposes these effects on global inequality into their within-country and between-country dimensions. Panel (a) compares the observed (black dashed line) and counterfactual (filled colored bands) evolution of these entropy-based decomposable measures in levels. As measured by mean logarithmic deviation, which is relatively sensitive to the welfare of low-income quantiles, observed temperature shocks have unambiguously exacerbated both varieties of inequality decreasing within-country inequality by 2.6% [0.0, 5.6] and between-country inequality by 8.7% [4.9, 13.3]. For the Theil index, the relative sensitivity to negative impacts to top incomes observed in Section 1.3 re-appear, resulting in an estimated 1.3% [-1.3, 3.8] *improvement* in within-country inequality as a result of temperature shocks. These same shocks still worsen between-country inequality by 6.1% [3.6, 9.0].

Panel (b) adopts the same color scheme to visualize their relative contributions to global inequality. The dashed black lines trace the stark observed trend of within-country inequality (teal) increasingly dominating between-country inequality (red) since 1981. The region in white separating the two colors corresponds to the counterfactual shares had average temperatures remained fixed since 1981. That these white regions are entirely under the black dashed lines imply that warming

Figure 1.8: Global inequality between and within countries, 1981-2016



since 1981 has very likely slowed this trend by reducing the within-country share of global inequality by between 1.1 percentage points $[0.1, 2.0]$ by the MLD measure and 1.6 percentage points $[0.7, 2.5]$ by the Theil index. Thus, the incidence of temperature shocks primarily influence the global distribution income by slowing convergence between countries.

1.5 Conclusion

The results presented in this paper constitute the first comprehensive evidence of regressivity in the global incidence of climate change, complementing an expansive and active economic literature which has remained effectively agnostic on global distributional impacts at subnational levels. By importing methods recently developed in empirical macroeconomics to distributional data thus far largely unused in environmental economics, we were able to overcome several long-standing statistical and practical constraints which have long obstructed progress on this urgent research question.

Given the sparsity of other work on this broad topic, several limitations of the present study suggeset avenues for potentially fruitful future research. Firstly, even the impressive coverage of the World Inequality Database is fundamentally limited in its ability to speak to the most important of distributional considerations, namely the impacts of climate change on global poverty. Our evidence affirms and quantifies the conventional understanding that the world's poorest have been and will continue to be by far the most severely impacted by climate change. But due to the limitations of pretax income as a measure of poverty and welfare, even these impacts are likely to be a conservative lower bound on true impacts.

In revisions in progress, the distributional analysis reported here will be supplemented by an analysis of compositional effects identifying which income groups from which countries have been transitioning up and down the global income distribution as a result of climate change. If income effects are as severe and state-dependent as described here, then it becomes plausible that an important distributional consequence of climate change is that inhospitable environments becomes increasingly deterministic of economic development, severely limiting the upward mobility of pop-

ulations in hotter climates.

To the best of my knowledge, the vulnerability of the highest incomes in hot and generally poor countries to transitory environmental shocks has not been previously documented or theorized. Weather-induced capital depreciation, capital shares in agricultural income, and the relative unavailability of insurance instruments such as weather derivatives in developing countries are example plausible mechanisms that call for empirical testing.

From a policy perspective, the results presented in this paper suggest that incorporating equity considerations as through a concave utility function would indeed increase the social weighting of the most impacted by climate change. But it is may still be worth taking seriously criticisms that a "profound indifference to inequality" (Sen et al., 2020) remains embedded even in these progressive varieties of cost-benefit analyses predominant in the climate economic literature. For example, Prest et al. (2024) show that one proposed implementation would increase the US government's official estimate of the social cost of carbon by a factor of eight. But somewhat paradoxically, these greater cost estimates are primarily used to prescribe an increased price on carbon which may then disproportionately burden these same segments of the population whose increased welfare weight induced the inequality-exacerbating policy. Equity weighting should of course factor into both the estimation of costs and the design of policy internalizing them.

To this end, it is intuitive that democratic pressures and a richer portfolio of policy instruments should be more effectively mobilized domestically than coordinated across a Westphalian international system (Barrett, 2003; Keohane & Victor, 2011; Levin et al., 2009). If so, levers available domestically but not internationally—such as compensatory fiscal transfers—may yet emerge as viable instruments of climate policy.

Chapter 2: Temperature, institutions, and the political climate

> "A change in the weather is sufficient to recreate the world and ourselves."
>
> Marcel Proust, *In Search of Lost Time*

2.1 Introduction

In 2007, a panel of recently retired United States military officers published an assessment of the national security risks presented by climate change, the result of consultations with intelligence officers, climate scientists, and high-level figures in government and business. The report concluded that projected environmental pressures would lead to "widespread political instability and the likelihood of failed states" and "add to tensions even in stable regions of the world." (Sullivan et al., 2007) In a period defined by military-industrial expansion and the mainstreaming of climate advocacy, the report found influence by popularizing the term "threat multiplier" to describe how environmental outcomes associated with climate change may amplify social risks interpreted as falling within the interest of national security (S. Goodman & Baudu, 2023). By the following year, the government of the United Kingdom had revised its national security strategy, naming climate change "potentially the greatest challenge to global stability" (Her Majesty's Government, 2008).

This chapter investigates the degree to which environmental change may influence political preferences and facilitate institutional transformation. In a first step, we measure the impacts of identified temperature shocks on proxies for the demand for democracy. Using nationally representative survey data from over 300,000 respondents from up to 98 countries, we find evidence that exclusively for respondents in warmer climates, experiencing an additional positive tempera-

ture shock in the three years prior to being surveyed significantly diminished respondent attitudes towards democracy and increased support for features of government associated with authoritarianism. The same shocks were also associated with more inclusive conceptualizations of democracy in cooler countries with the opposite effect for warm countries.

We complement these findings with an analysis of aggregate effects on institutional change. To measure effects on the propensity for democratic "intensity", we use various data products which score country-year regimes along different aspects of democracy. We find relatively imprecise but suggestive evidence that a marginal temperature shock facilitates democratic backsliding in warm countries but that this effect does not persist beyond the year of the shock and is null for other geographies.

Finally, we look at the effects of these same shocks on the propensity for discrete regime change. Here, we report significant evidence that a transitory temperature shock greatly decreases the likelihood of warm-climate autocracies to transition to democracies while increasing the propensity for warm-climate democracies to regress to autocracies. For temperate countries, the same shock greatly increases the probability of democratic transition but has null effects on the propensity for democratic reversal. These effects are also largely concentrated in the period of the shock but remain mildly elevated for an additional 1-4 years.

2.1.1 The demand for democracy

Summarizing the institutional risk posed by climate change, the aforementioned Military Advisory Board report concluded that the consequences of climate change "will likely foster political instability where societal demands exceed the capacity of government to cope." While such a direct relationship between the incidence of climate change and the demand for institutional change is undertheorized by my reading, two provisional hypotheses can be constructed from the broader literature on the economic determinants of democratic deepening, transition, and stability (as well as their negations).

Under a "mercury uprising" theory, adverse climate outcomes affect the latent demand for in-

clusive institutions and in turn the propensity for concessionary reform. This is motivated by invoking the influential economic model of Acemoglu and Robinson (2001) wherein diminutions in the perceived opportunity cost of challenging autocratic rule result in a more democratic political equilibrium; in other words, citizens have less to lose by taking costly political action to demand institutional reform of elites. For example, P. J. Burke and Leigh (2010) reports evidence that economic downturns worldwide are associated with democratic improvements while Acemoglu et al. (2019) highlight that democratic transitions are correlated with temporary reductions in GDP growth rates. Brückner and Ciccone (2011) provides evidence associating negative income shocks with a transitory increase in the propensity for select sub-Saharan autocracies to democratize. Aidt and Leon (2015) reports complementary evidence linking drought-induced rioting to improvements in democratic institutions.

Perhaps notably, the assessment of the Military Advisory Board concluded the opposite effect, suggesting that for particularly vulnerable tropical and subtropical countries, these demands will manifest in "movement toward increased authoritarianism." For the United States and Europe, it alludes to intergroup fracturing and resentments associated with the arrival of environmental refugees and migrants.

These potential consequences are more consistent with an alternative"state of exception" theory wherein citizens respond to threats associated with climate change with an increased willingness to trade features of liberal democratic institutions for security perceived as more credibly ensured by authoritarian and exclusionary forms of governance. The COVID-19 pandemic provides a particularly resonant intuition for this dynamic, wherein emergency restrictions on movement and privacy were generally deployed with wide approval and successful health outcomes. Alsan et al. (2023) reported that experimentally induced increases in perceived health insecurity at the outset of the pandemic led to significant increases in respondents' willingness to trade off civil liberties. Amat et al. (2020) found similar effects on the preference for technocratic and authoritarian government in the Spanish experience of the pandemic. Mittiga (2022) is a political theory paper which directly considers how these pandemic experiences may legitimize authoritarian approaches to responding

to climate change in the future. Fritsche et al. (2012) finds that simply reminding respondents of local adverse consequences of climate change increased discriminatory attitudes towards out-groups such as prisoners and accentuated authoritarian preferences.

All in all, our results are highly consistent with this latter "state of exception" interpretation.

2.1.2 Other related literature

In the nearly two decades since the Military Advisory Board report operationalized climate change as a "threat multiplier", a prolific interdisciplinary social scientific literature has emerged dedicated to empirically examining climate change as a phenomenon of social disruption (T. A. Carleton & Hsiang, 2016; Intergovernmental Panel on Climate Change, 2023). The present analysis is closely related to an active subset of this work relating various environmental outcomes to the onset of conflict; Buhaug and von Uexkull (2021) and Koubi (2019) provide reviews of this research, construed broadly enough to include concepts of ranging from psychological aggression to self-harm to crime to warfare. Obradovich (2017) and a tertiary result in Dell et al. (2012) each find that higher temperatures in developing countries are associated with irregular changes in political leadership. Bellemare (2015) links food price increases to increased incidence in various forms of social unrest.

2.2 Data

2.2.1 Surveys of political preferences

Data comes from the Integrated Values Survey, which harmonizes the World Values Survey (Haerpfer et al., 2022) and the European Values Study (European Values Study, 2022). In total, the final dataset summarizes the responses of 663,965 total respondents to 506 nationally representative surveys conducted in 116 countries between 1981 and 2022. We retain data on the respondent's demographic information, including their birth country, immigration status, income range, and political preferences.

Our primary outcomes of interest are derived from responses to questions identified as indicative

Table 2.1: Summary statistics for IVS respondent data

	Unique	Missing Pct.	Mean	SD	Min	Median	Max
Year	36	0	2005.2	10.3	1981.0	2007.0	2022.0
Male	2	1	0.47	0.53	0.0	0.0	1.0
Citizen	2	74	0.98	0.1	0.0	1.0	1.0
Support index	41	55	0.0	0.8	-3.4	0.1	0.8
Skepticism index	65	72	0.0	0.8	-1.7	0.0	2.0
Authoritarian index	65	30	0.0	0.7	-1.3	-0.1	1.9
Inclusiveness index	140902	69	0.0	0.5	-2.0	-0.1	1.2
Democracy	2	38	0.74	0.36	0.0	1.0	1.0
Polity score	19	41	6.3	5.3	-10.0	9.0	10.0

of attitudes towards democracy or its negation. We assign these questions to one of four categories as summarized in Table 2.2. For each category, we then construct an inverse-covariance weighted index of responses to the questions using the method of Anderson (2008), which assigns greater weight to more informative component questions.

2.2.2 Ordinal measures of democracy

We use two sources of democracy 'scores' which purport to measure how democratic a given country-year regime is. The first is a measure provided by the Polity Project (Marshall & Gurr, 2020) which ranges discretely between -10 and 10 in increments of 0.5 with more positive values corresponding to higher perceived levels of democracy. The Project recommends a classification where country-years scoring below -5 are labeled autocracies, those above +5 are democracies, and those in between are 'anocracies'.

The second source of democracy scores comes from the Varieties of Democracy (V-Dem) Project (Coppedge et al., 2025). These measures are produced b combining primary sourcing with expert elicitation to assign a panel of country-year observations a score ranging continuously between 0 and 1. There are up to 400 variables by which observations are scored and these are in turn aggregated to produce five indices which operationalize democracy according to its deliberative, egalitarian, electoral, liberal, and participatory connotations respectively. Differences between these concepts of democracy can be pertinent if, for example, the extension of civil liberties is a

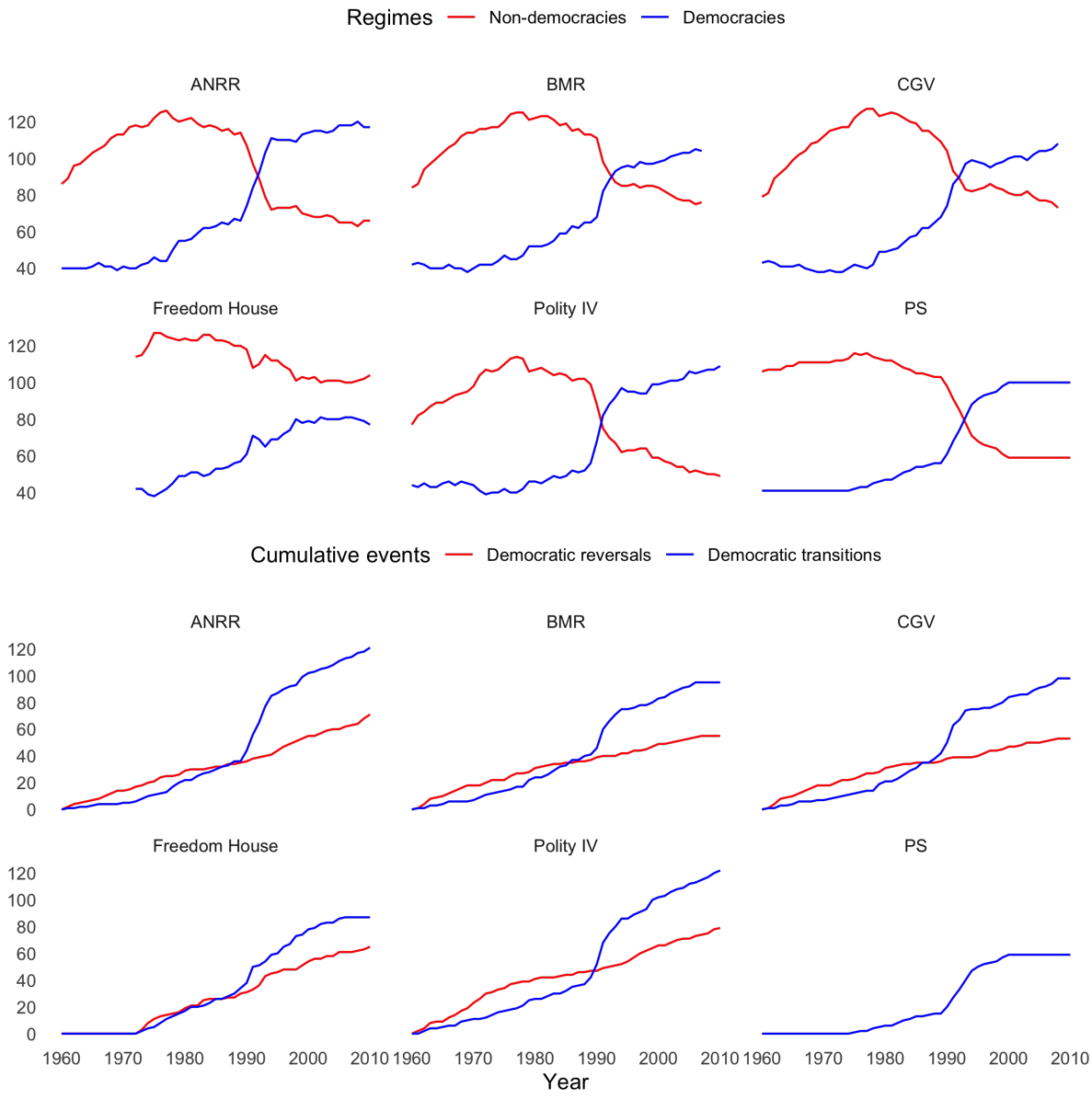
systematically slower-moving process or is considered a much greater democratic concession than the administration of a nominally competitive election. A contemporary example is Chile's ongoing constitutional reform process which began in 2021 in response to civil uprising in 2019. That the drawn-out process has seen the rejection of multiple proposed constitutions by referendum may be interpreted as an exercise of improved democratic deliberation and participation. But until a replacement is ratified, democracy as measured by formalized rights and protections are still constrained by the 1980 Constitution being challenged.

The distribution of scores are summarized in Chapter B, showing a bimodal distribution with peaks at -6 and +10 for the Polity IV scores and heavily right-skewed distributions for each of the V-Dem scores. Correlations between these indexes are summarized in Table B.1. The wide use of these ambitious intensive measurements of democracy seems likely to reflect their usefulness more so than it does their credibility, especially for capturing intra-regime variation; Knutsen et al. (2024) and Little and Meng (2024) provide thoughtful critiques of these data products along these lines.

2.2.3 Dichotomous indicators of democracy

To study systematic transitions in and out of democracy, we require data which crudely labels country-year regimes a label of 'free' or 'democratic' or their negation. Table B.2 summarizes differences between notable products which serve this purpose and have been commonly used in the political economy literature. For each data product, Section 2.2.3 plots the time series of the number of democracies and non-democracies worldwide as well as the cumulative number of transition events. In our main results, we will restrict attention to the aforementioned Polity index (made dichotomous by using a score of +5 as a threshold for democratic status) as well as the panel datasets from Acemoglu et al. (2019) and Cheibub et al. (2010).

Figure 2.1: Time series of regimes and regime changes by data source



2.2.4 Weather

We merge all data with population-weighted near-surface air temperature data derived from the Global Meteorological Forcing Dataset for Land Surface Modeling (GMFD). This data corrects model-based biases in NCEP/NCAR reanalysis product using observational meteorological station data (Dept. of Civil and Environmental Engineering, Princeton University, 2006). The GMFD reports historical weather data for the entire global surface at 0.25-degree resolution[1] for every three hours from the start of January 1, 1948 to the end of December 31, 2015. For interpretive convenience, we center population-weighted average temperatures $\overline{T}$ at 18°C, roughly in the 40th percentile of country-year temperatures in our dataset.

As in Chapter 1, we define temperature shocks $\widehat{\tau}_{it}$ in relation to an M-period moving average of local temperature $\overline{T}_{it}$:

$$\overline{T}_{it} := \frac{1}{M} \sum_{m=1}^{M} T_{i,t-m} - 18 \tag{2.1a}$$

$$\widehat{\tau}_{it} := T_{it} - \overline{T}_{it} \tag{2.1b}$$

To avoid redundancy, I refer the reader to Section 1.2 in Chapter 1 for a graphical depiction of the resulting history of temperature shocks constructed in this manner (Figure 1.1) as well as additional elaboration on this construction.

We next define a variable τ_{ct}^{N} representing the recent exposure to temperature shocks of a respondent in country c surveyed in year t:

$$\tau_{ct}^{N} := \sum_{j=1}^{N} \mathbb{1}(\widehat{\tau}_{c,t-j} > \underline{\tau}) \tag{2.2}$$

This shock will be used when estimating impacts on the demand for democracy as measured by cross-sectional surveys from the IVS. Because we cannot infer the month of the year when

[1]At this resolution, cells near the equator are approximately 27 km^2 (17 mi^2).

respondents were surveyed, shocks defined using temperature observed only in the calendar year the respondent was surveyed would be strongly influenced by weather only respondents surveyed later in the year would have experienced. Using exposure over a longer period before the calendar year helps mitigate the effect of this non-equivalence and also accounts for persistence in temperature shocks such as that caused by natural periodicity in the climate system (e.g., the El Niño Southern Oscillation which spans between 2-7 years). In our preferred construction, we set $N = 3$ to span the persistence in temperature shocks documented in Chapter 1.

This variable also depends on the value $\underline{\tau}$, the threshold value defining a temperature shock. As a benchmark, we use a common value of 1°C (corresponding to the 75th percentile in our full estimation sample) for all settings though there is also a reasonable case to use relative thresholds, for example based on country-specific variability in temperature.

Table 2.2: Categorization of relevant survey questions from the Integrated Values Survey

1. Support for democratic institutions
• Agree: Democracies are good for governance
• Agree: Democracy has its problems but is better than all alternatives
2. Democratic skepticism/ineffectiveness
• Agree: Democracies cause disorder
• Agree: Democracies are too bureaucratic
• Agree: Democracies are bad for the economy
3. Support for authoritarian leadership
• Preference for a strong leader who does not bother with parliament and elections
• Preference for army rule
• Preference for governance by experts over elected officials
4. Inclusiveness of definition of democracy
• Agree: Holds open elections
• Agree: Extends civil rights
• Agree: Promotes equality of women
• Disagree: Army takes over when government is incompetent
• Agree: Redistributes from the rich to the poor
• Agree: Provides unemployment insurance
• Disagree: Government by religious leaders

2.3 Methods

2.3.1 The demand for democracy

We modify the specification of Acemoglu et al. (2024), which uses the same IVS dataset to estimate the effect of exposure to democracy on attitudes towards democracy.

$$Y_{i,w,c,t,a} = \beta_1 \tau^N_{ct} + \beta_2 (\tau^N_{ct} \times \overline{T}_{ct}) + \boldsymbol{\gamma}^\top \mathbf{W}_{i,w,c,t,a} + \varepsilon_{i,w,c,t,a} \tag{2.3}$$

Indexes refer to respondent i, a resident of country c who in year t was surveyed as part of IVS wave w at age a. The matrix $\mathbf{W}_{i,w,c,t,a}$ of controls include the expected temperature $\overline{T}_{ct}$ in country c at year t, respondent gender, respondent town size, and fixed effects for country, year of survey, age of respondent, respondent's birth cohort , and IVS wave. Standard errors are clustered at the country and year levels.

2.3.2 Democratic deepening and backsliding

To estimate dynamic effects of temperature shocks on an outcome y measuring the degree of democracy, we estimate a series of state-dependent local projection regressions on a panel of country-year regime observations for projection horizons $h \in \{0, ...H\}$:

$$y_{c,t+h} = \beta_{1h} \widehat{\tau}_{ct} + \beta_{2h} (\widehat{\tau}_{ct} \times \overline{T}_{ct}) + \lambda_h \overline{T}_{ct} + \boldsymbol{\gamma}_h^\top \mathbf{Z}_{ct} + \mu_c + \eta_t + u_{c,t+h} \tag{2.4}$$

The primary quantity of interest for each projection horizon is $\beta_{1h} + \beta_{2h} \overline{T}_{it}$, the first derivative of the equation with respect to the temperature shock $\widehat{\tau}$. The first term represents the expected effect of a unit temperature shock on the outcome of interest for a country-year with an expected temperature of 18°C. The second term represents the linear dependence of this effect for country-years with different expected temperatuers. For example, the model would imply that a unit shock in a country with an expected annual temperature of 26°C would be expected to increase outcome y for the country by $\widehat{\beta_{1h}} + \widehat{\beta_{2h}} \cdot (26 - 18)$ units h periods after the shock relative to the counterfactual

where no shock occurred.

Controls $\mathbf{Z}_{ct}$ include lags $\ell \in \{1, ..., L\}$ lags each of the shock $\widehat{\tau}_{c,t-\ell}$, the shock $\widehat{\tau}_{c,t-\ell}$ interacted with expected temperature $\overline{T}_{c,t}$, logged precipitation, GDP growth, and the outcome variable $y_{c,t-\ell}$.

The reader may refer back to Section 1.3 and Section A.2 for exposition of this method and interpretation of its results.

2.3.3 Discrete regime change

To measure impacts on the propensity for autocracies to transition to democracies and vice versa, we estimate Equation (2.4) using a binary dependent variable indicating the country-year's regime type as the outcome variable. The corresponding coefficient estimates are interpreted as the state-dependent effect of an exogenous 1°C shock on the absolute propensity to transition to a democracy (or regress to an autocracy) h years after the shock. For interpretation as a proportional hazard, the magnitude of this effect can be compared to the unconditional annual propensity to undergo a regime change.

2.4 Results

2.4.1 The demand for democracy

Table 2.3 reports the main results of our demand analysis. To minimize susceptibility to inferential error from multiple testing, we restrict attention here to results for the four survey category indices but provide the equivalent results for their component survey questions in Tables B.3 to B.6. To aid interpretation of our non-linear state-dependent model, Figure 2.2 depicts these coefficients graphically as a function of the expected temperature in the respondent's country of residence in the year of their survey.

Our favored specification, depicted as a green filled line with a corresponding 90% confidence interval, uses an explanatory variable defined as the number of temperature shocks above 1°C the respondent was exposed to in the three years preceding the year of the interview. Additional dashed lines correspond to alternative constructions which vary the number of years considered.

The coefficients on the main effect imply that recent exposure to an additional positive temperature shock for respondents in a represenentative 18°C decreases support for democracy and leads respondents to favor more restrictive definitions of democracy while increasing skepticism of democracy and preferences for authoritarian features of government. These effects are large in magnitude, ranging between 0.12-0.13 standard deviations for the attitudinal indexes and 0.46 standard deviations for the inclusiveness index.

The state-dependency in this effect is significant at the 5% level for all outcomes except the democracy-support index. Among cold countries, effects are positive but nell with the exception of the large and highly significant impacts to the inclusiveness index. That these impacts change sign for all outcomes at some intermediate temperature well within the range of human habitation is consistent with the convention that marginal temperature shocks are increasingly harmful for countries above a economically optimal temperature of approximately 13°C and increasingly beneficial for increasingly cool countries (M. Burke et al., 2015; Nath et al., 2024). That the impacts are larger in magnitude and much more precisely estimated for the inclusiveness index despite being estimated for by far the smallest sample is most likely a reflection of the diversity and informativeness of questions from which it is constructed compared to the two highly correlated variables which comprise the democratic support index (see Table 2.2).

These estimates are generally consistent as we vary the period of exposure defining the temperature shock with a general pattern that increasingly restricting attention to environmental exposure in the most recent years tends to increase the magnitude of the impact. While the repeated cross-sectional design and other limitations of the survey data limits our ability to capture dynamic impacts, this provides some suggestive evidence that these demand effects are concentrated either contemporaneous to the period of the shock or with a one-period lag and unlikely to be highly persistent.

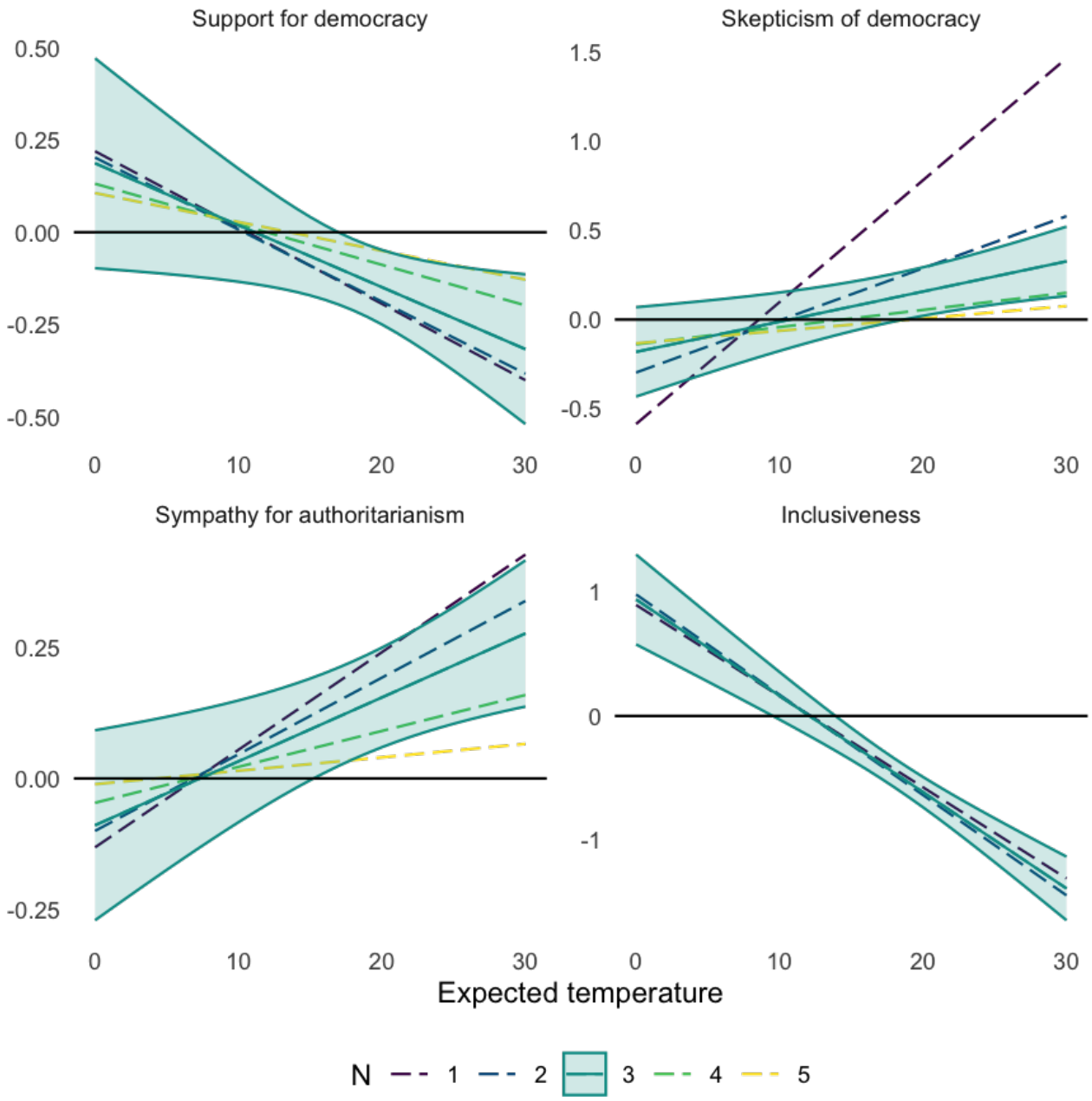
All in all, these results are strongly consistent with a ‘state of exception’ demand story.

Table 2.3: Impact of an additional 1°C shock in prior three years on attitudes towards democracy

Index:	Support	Skepticism	Authoritarian	Inclusiveness
Model:	(1)	(2)	(3)	(4)
Variables				
1°C Shock	-0.1152$^{\cdot}$	0.1235	0.1303*	-0.4582**
	(0.0585)	(0.0815)	(0.0568)	(0.0704)
1°C Shock $\times \overline{T}$	-0.0168$^{\cdot}$	0.0169*	0.0122*	-0.0775**
	(0.0090)	(0.0072)	(0.0052)	(0.0116)
$\overline{T}$	-0.6651$^{\cdot}$	-0.2732	0.0832	-0.3258*
	(0.3382)	(0.3732)	(0.1460)	(0.0970)
Gender	Yes	Yes	Yes	Yes
Precipitation	Yes	Yes	Yes	Yes
Fixed effects, # unique values				
Country	75	76	98	58
Year	11	13	21	6
Age	84	87	87	83
Birth year	92	93	101	86
Survey wave	4	4	8	3
Observations	150,292	170,798	317,655	77,392
Outcome mean	0.02250	-0.00387	-0.03017	-0.05107
R^2	0.08924	0.10248	0.18527	0.17156

Clustered (Country & Year) standard-errors in parentheses
*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, .: 0.1*

Figure 2.2: Effect on political preferences of an additional 1°C shock in last *N* years



2.4.2 Institutional change

While our prior results provided strong evidence of transitory temperature effects on the latent demand for democracy, they are not directly informative of the mobilization of these new preferences to effect institutional change. To this end, Figures 2.3 and 2.4 depict the dynamic state-dependent cumulative responses of institutional democracy as measured along intensive and extensive margins respectively.

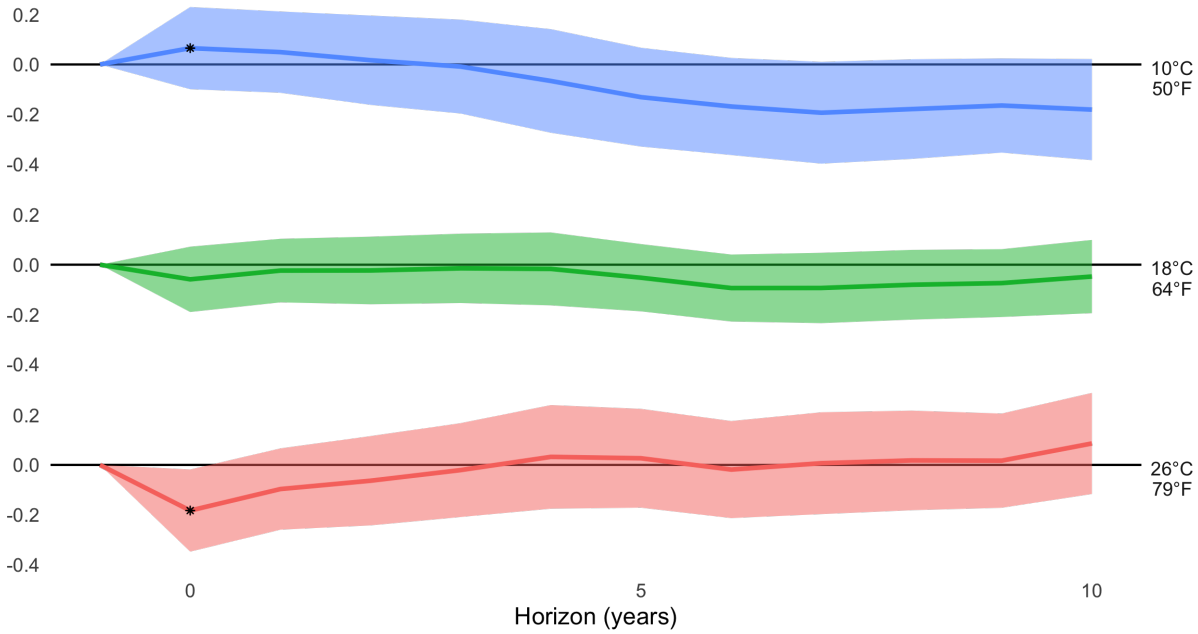
Figure 2.3 plots the marginal effect of an identified temperature shock on democracy scores for regimes in three representative climates, net of the shock's effect on ensuing temperature shocks. Figure 2.3a depicts impacts measured by the widely used Polity IV score while Figure 2.3b report results for scores from the Varieties of Democracy Project which grade country-year regimes according to distinct aspects of democracy.

The results corresponding to the Polity IV score indicate a significant state dependency contemporaneous with an identified temperature but this effect is imprecisely estimate and only large for warm climates. Point estimates ascribe a transitory 0.18 reduction in the polity score to a unit shock in a representative 26°C country. While not exactly rich in intuitive meaning, this is approximately 1.8% of the difference between a borderline democracy and a borderline autocracy going by the group's suggested classification. The results for the V-Dem scores are qualitatively similar but report null effects across the board.

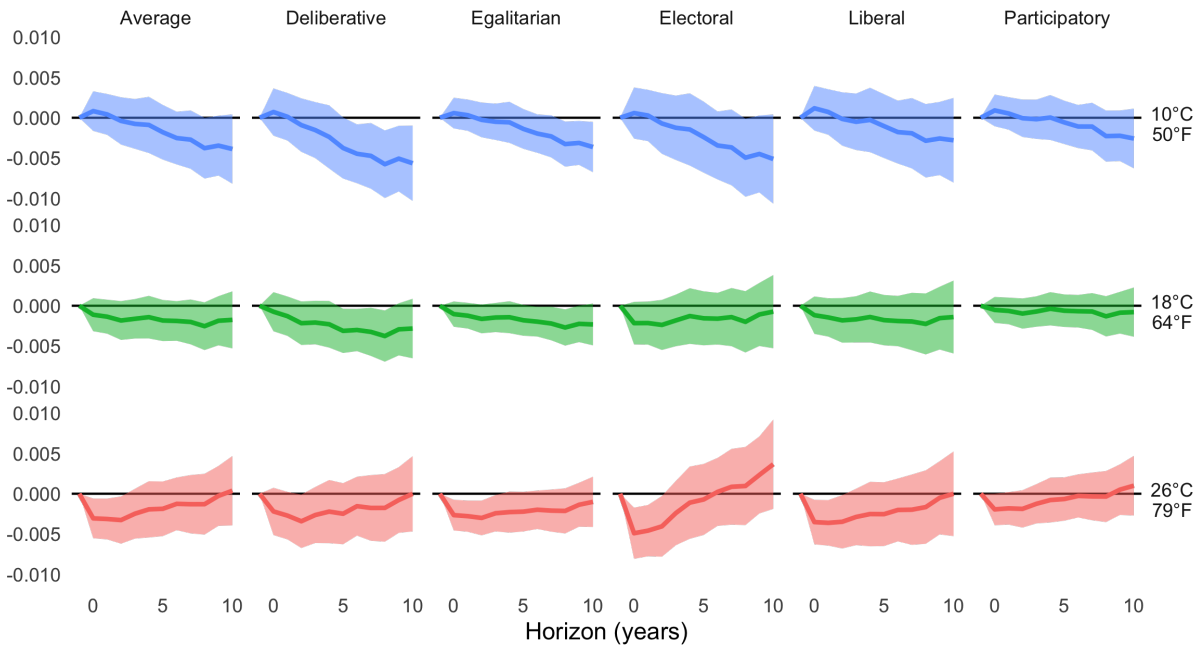
Our final results are summarized in Figure 2.4 which plots estimates of the impact of a unit shock on the propensity for democratic transition and breakdown, split across rows by data source. Figure 2.4a indicates that a transitory temperature shock substantially and significantly increases the propensity of temperate autocracies to democratize by between 0.030 and 0.075 percentage depending on the data source. Compared to an unconditional transition probability of approximately 0.03[2], these represent proportional increases to the probability of a 10°C autocracy to democratically transition of 96% by the ANRR measure, 108% by the CGV measure, and 252% by the Polity IV measure. Referring to Table B.2, the relative responsiveness of the Polity IV score may be explained by its relatively limited focus on "competitiveness of political participation and constraints

Figure 2.3: Intensive margin: Effect of a 1°C shock on democracy scores

(a) Polity IV democracy score (-10 to +10)



(b) V-Dem Scores (0 to 1)



on executive authority". Indeed, the ANRR definition of democracy is strictly more selective since the Polity IV indicator is a necessary but insufficient qualifying condition.

For 26°C autocracies, the propensity to democratize instead decreases as a result of a unit temperature shock by between 0.032-0.048 percentage points corresponding to proportional decreases between 134-236%. Now, this is clearly nonsensical since probabilities cannot take on negative values; this is partially a well-known consequence of linear probability models but more importantly, is worth emphasizing that these impacts are expressed in relation to a 1°C shock, which is much larger in magnitude than shocks we would expect to observe in a typical year in any country. Referring back to the time series of shocks illustrated in Figure 1.1, the median magnitude of a shock varies by the rate of climate change which will in turn vary across geographies and time. For the warmest 33% of country-years in our data, the median shock was approximately 0.19°C in absolute value in the 1970s and then increased to between 0.30-0.37° in the ensuing decades. Thus, effects of a 1°C shock in settings above 25°C can be understood as responses to a shock 3-5 times larger than is typically observed. For the coldest 33% of country-years, the median magnitude of a shock was approximately 0.35°C in the 1970s and 1980s, 0.51-0.53°C in the 1990s and 2000s, and 0.59°C in the 2010s. Their effect sizes then can be interpreted as the response to a shock just 2-3 times larger in magnitude than the median. By the assumption of linearity inherent in the local projections model specified in Equation (2.4), we can scale down these estimated effects accordingly to infer the impact of a typical shock on the probability of democratic transition.

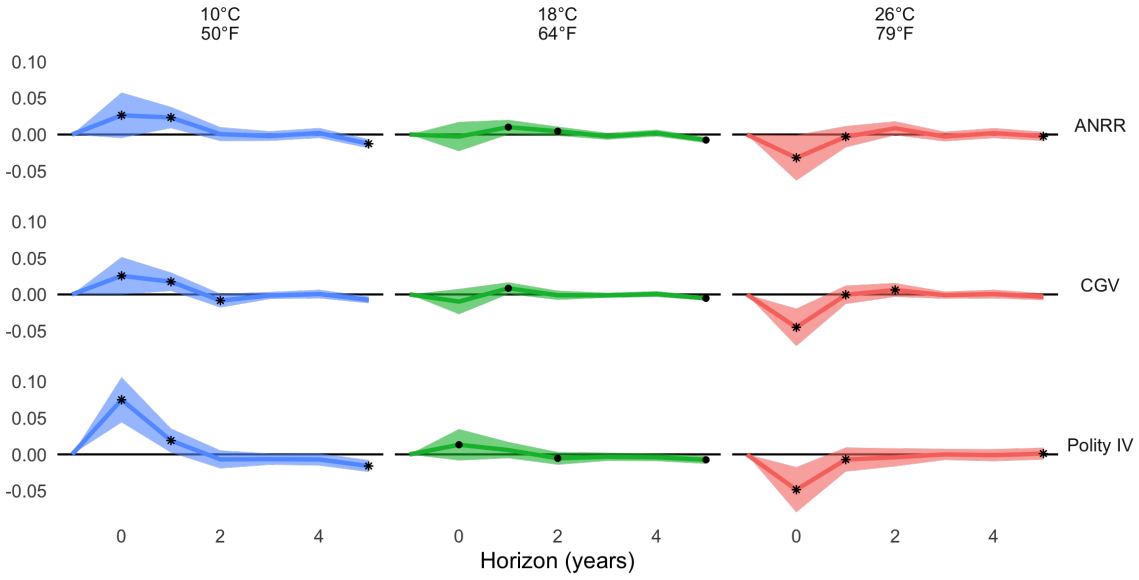
Figure 2.4b reports analogous results for the propensity of democracies to break down. These too exhibit large state dependencies though impacts to cool democracies are found to be null, suggesting they tend to be stable regimes. Impacts to hot countries in comparison are substantial, significant, and persistent, ascribing an increase of 0.032, 0.011 and 0.040 percentage points of democratic reversal in the year of the shock compared to unconditional transition probabilities of between 0.033-0.048. These correspond to proportional increases of 33.0%, 91.8%, and 91.0% for

[2]The unconditional probability to democratize in any given period is surprisingly consistent across climates: 0.027 for autocracies cooler than 17°C compared to 0.024 for those warmer than 25°C by the ANRR measure, 0.024 vs. 0.019 by the CGV measure, and 0.030 vs. 0.032 by the Polity IV measure.

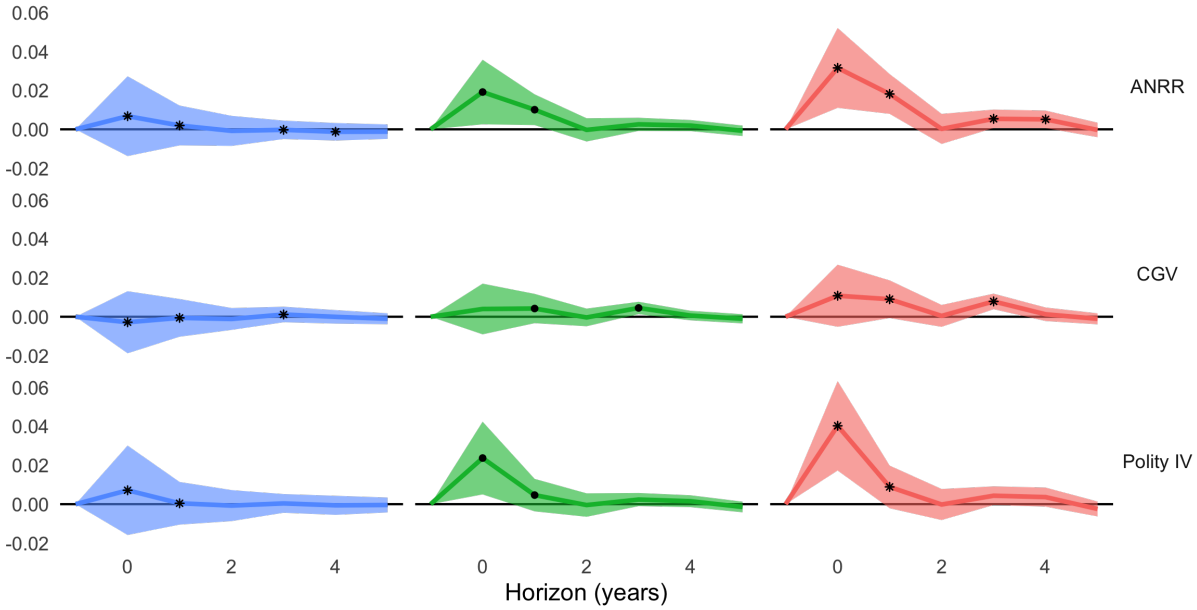
the ANRR, CGV, and Polity IV measures respectively.

Figure 2.4: Extensive margin: Effect of a 1°C shock on political transitions

(a) Change in propensity for democratic transition among autocracies



(b) Change in propensity for democratic reversal among democracies



2.5 Conclusion

In all, the three sets of complementary results presented throughout Section 2.4 provided internally consistent evidence supporting a "states of exception" theory in which adverse climate impacts both diminish latent public demand for democracy and the democratic character of its political institutions. This is contrasted against a competing "mercury uprising" theory with precedent outside climate contexts in which adversely impacted citizens recognize they have less to lose in contesting government suppression and act accordingly to manifest democratic change; to the extent that positive temperature shocks correspond to diminished opportunity costs of such political activity, the theory would imply exactly opposite institutional pressures to the ones recovered at each stage of our analysis.

Still, the present analysis is unsatisfying as a theory of environmental political economy absent consideration of the other side of the "market": the supply of democracy. Aidt and Franck (2015), Rahman et al. (2022), and Wood and Wright (2016) consider cases in which ruling elites anticipate opportunities for political contestation or to consolidate emergency powers, acting to either pre-emptively suppress dissident activity or placate potential challengers. Indeed, our partial equilibrium framework is largely agnostic about the process by which the increased latent demand for democracy we attribute to shocks translates into macro-scale political transformations. Complementary analysis looking at the outcomes such as the incidence of social unrest, electoral participation, and state violence would enrich the work documented thus far.

Further, understanding inequality as both a consequence of climate change (Chapter 1) and a moderator of regime turnover (Acemoglu et al., 2015) raises the question of how inequality in economic outcomes may relate to these inequalities in political representation. That climate change is expected to segment winners from losers within the same populace surely complicates equilibrium outcomes in contexts where the character of political institutions become malleable.

Chapter 3: Evidence of a drought effect on intimate-partner violence

with Tanushree Goyal

3.1 Introduction

The World Health Organization estimates that 27% of women worldwide will experience physical or sexual violence by an intimate partner in their lifetime, making it more prevalent than other public health priorities such as diabetes, depression, or breast cancer (Miller & McCaw, 2019). Including expressions of emotional abuse, decades of research have established associations between intimate-partner violence (IPV) with severe medical and psychiatric conditions (Sugg, 2015). Accordingly, the elimination of gendered violence and harmful gendered practices was identified as a priority target in the United Nations' 2030 Agenda for Sustainable Development with the incidence of intimate-partner violence and child marriage specifically identified as the measure of progress towards this goal (United Nations, 2015).

A developing interdisciplinary literature has drawn attention to "economic abuse"—in which a person's economic dependence on a partner is instrumentalized against them–as a manifestation of and contributor to the prevalence of IPV (Stylianou, 2018). For example, Adams et al. (2024) find that women's labor market outcomes precipitously diminish immediately upon cohabiting with an eventually-abusive partner. The authors interpret this result through a model in which the abuser intentionally creates economic dependence to disincentivize exiting the relationship. Aizer (2010) provides evidence that pay parity reduces physical violence against women.

In this chapter, I explore how explore how shocks to this economic dependence relate to the occurrence of violence in a previously never-violent relationship. Here, I combine high-resolution weather data with comprehensive duration data on the incidence of intimate-partner violence to

estimate how drought-induced negative income shocks may affect these health outcomes. I also replicate a recently published study relating these weather shocks to the risk of child marriage using updated methods and improved data.

Rural India presents an ideal first setting in which to study the malleability of gendered cultural norms and the economics thereof. In addition to the wide availability of research products and infrastructure, this focus allows us to hone in on a region where gender inequities have proven remarkably resilient and where credit constraints and agricultural dependencies are well known to exacerbate the effects of economic shocks. The latter points are vital for our approach as these vulnerabilities combined with the sheer size and diversity of the country's people, cultures, and geography enables us to follow in an established tradition of using random drought shocks in developing countries as natural sources of measurable and localized identifying variation (M. Burke et al., 2014; Corno et al., 2020). In addition, a country marked by variation in gendered and highly economic cultural practices such as dowry payment and patrilocality present opportunities to investigate channels through which gendered effects may operate. Finally, the common cultural practices of virilocality—in which married couples cohabit with the groom's family—and multigeneration households present opportunities for follow-up studies on the influence of socialization and kinship structures.

My findings suggest that negative income shocks substantially reduce the hazard into marital violence. Specifically, a negative precipitation shock of a magnitude one might ordinarily expect once every 4-5 years on average reduces the risk of a never-violent relationship becoming violent by 0.037 percentage points, representing a 70% reduction in the baseline hazard of this transition during the formative years of a marriage.

Updating the main result of Corno et al. (2020), I find corroborating evidence that these weather shocks reduce the hazard into early marriage among girls in India between the ages of 12-18. I find that these effects are milder but more persistent than characterized in the original study. In addition, I find that this result is reproduced in Sub-Saharan African countries, contradicting the headline result of the original study which had reported opposite-signed effects in the two settings

and attributed the difference to consumption smoothing behaviors oppositely incentivized by the cultural practices of dowry vs. bride price payments.

3.1.1 Related litrature

An active area of research intersecting the economics of gender and development interrogates why substantial differences in gendered outcomes cannot be satisfyingly explained by their level of development. Though plenty of evidence corroborates the intuition that economic growth and conventional indicators of women's empowerment should co-move and positively impact one another, surveys of this literature have concluded that "the interrelationships are probably too weak to be self-sustaining." (Duflo, 2012). Prior work has shown this disconnect between economic and gendered progress is reproduced within countries: the correlation between per capita income and many of these measures are weak and inconsistent in sign across Indian states (Jayachandran, 2021). Indeed, much academic attention has rightly concentrated on puzzles such as the labor-force participation rate of women in South Asia, which is markedly lower than in regions of comparable or lower levels of development and have remained so despite decades of sustained economic growth (World Bank Group, 2022).

Explaining the incidence of IPV presents its own specific challenges. In its many psychological and physical manifestations, the pervasiveness of gender-based violence is well understood an acute measure of patriarchal culture. Yet it still presents additional complexities. For one, it does not neatly co-move with economic development: recent surveys by YouGov and the anti-IPV organization Frontline100 reported levels of tolerance for and experiences of IPV in high-income countries of the EU at comparable levels to those self-reported by respondents in rural India, our setting of focus. Nor does IPV incidence necessarily co-move intuitively with other conventional indicators of gender equality. Separate studies have claimed that otherwise culturally conservative countries and subnational regions within countries may report lower IPV incidence if they have proportionally higher Muslim populations; they ascribe the result to Islam's cultural aversion to alcohol (Kovacs, 2017).

Two studies suggest that early-life exposure to differing family structures lead to measurable changes in gender norms. Fernández et al. (2004) finds that war-induced shocks to female labor-force participation, by normalizing working mothers, led male children of such working mothers to seek out working partners in their adulthood. Unterhofer and Wrohlich (2004) find that the takeup of paternity leave induce positive effects on attitudes towards gender norms both in the offspring of the fathers and to the untreated network of these fathers.

3.2 Data

3.2.1 Weather

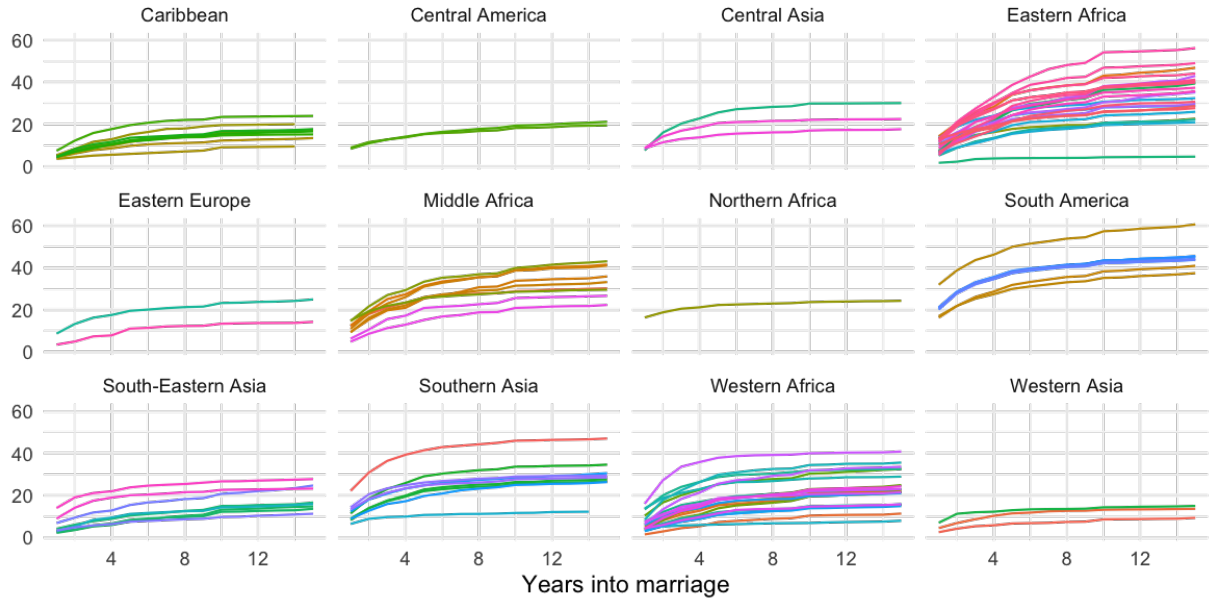
Precipitation data comes from CHIRPS (Funk et al., 2015), a $0.05° \times 0.05°$ gridded product spanning spanning the latitudes from 50°S to 50°N. Derived from a combination of infrared satellite sensors, rain gauge data, and climate reanalysis models, CHIRPS is optimized for monitoring drought conditions in data-scarce regions; this fits our purposes well given our need to identify and differentiate between the historical precipitation experiences of thousands of rural locations (see Figure C.1a).

We make use of this extensive history in constructing the weather shocks which serve as the explanatory variable of interest in all our analyses. Because a given level of precipitation would be differently remarkable across the extremely diverse terrains of sub-Saharan Africa and the Indian subcontinent, an absolute level of precipitation may constitute flooding in one geography but relative dryness in another. For this reason, we follow M. Burke et al. (2014), Corno et al. (2020), and McGavock and Novak (2023) among others in defining 'droughts' relatively: for a given set of coordinates, we extract nearby precipitation observations from every month from 1981 onwards to fit a gamma distribution specific to the location. Then for any event of interest which we can locate in time and space, we can relate the weather observed in the preceding months or years to this local distribution and assign it a locally defined precipitation quantile.

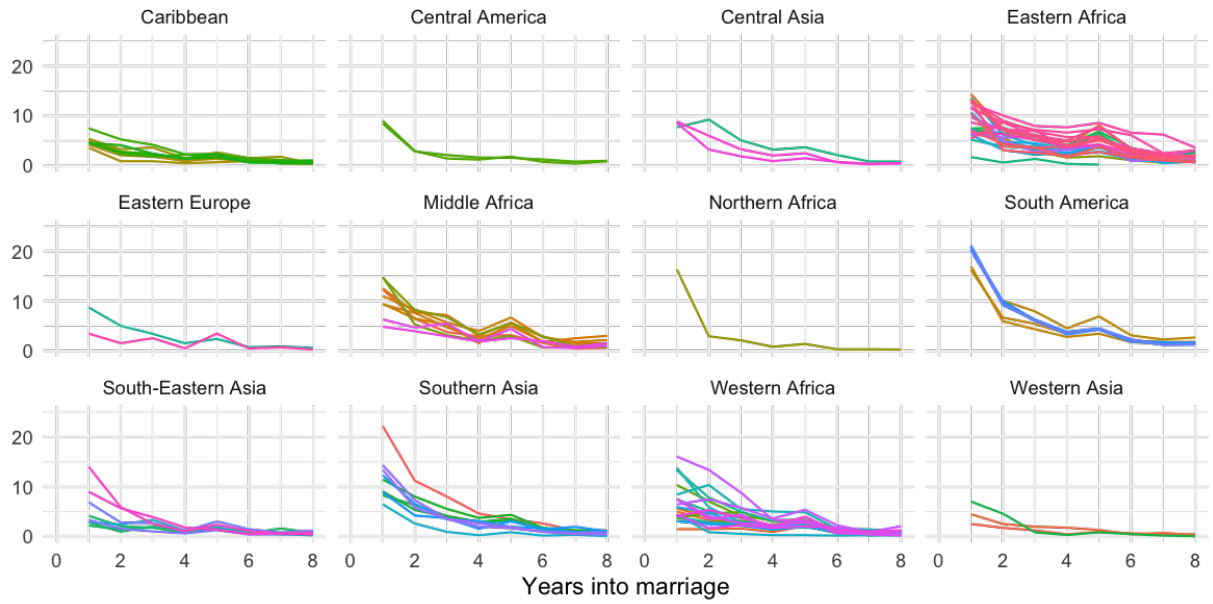
In our main specifications, we use the 15th percentile in the local rainfall distribution as the threshold for qualifying as a relative 'drought'. While this construction is crude and arbitrary in

Figure 3.1: Incidence of self-reported domestic violence

(a) Cumulative percentage of marriages reporting violence



(b) Annual hazard into first-time violence



some respects, it has the advantage of inducing similarly frequent shocks at all locations and can be easily scaled to be made more stringent or less restrictive by using different threshold percentiles to define the shock. We will demonstrate the sensitivity of our results to these alternative thresholds, effectively trading off magnitude of impact for sample size.

3.2.2 Domestic violence

The Demographic and Health Surveys (DHS) are implemented as a series of nationally representative cross-sections. Typically, all women between the ages of 15 and 49 years are selected from each household visited to undergo a detailed interview on topics relevant to public health and impact evaluation categorized into modules. Responses are anonymized but jittered GPS coordinates are provided at the village level; we use these to assign histories of weather exposure to a given respondent identifier relative to available dates such as the month-year of interview, birth, or cohabitation.

DHS surveys have now been conducted in more than 90 low- and middle-income countries but the waves administered in India are by far the most comprehensive with the number of respondents in a given round comparable to those in all of Sub-Saharan Africa; Figure C.1a plots more than 28,000 unique cluster locations provided in its 2016 wave.

While each generation of surveys adheres to a standard format to enable comparisons across settings, some implementations will differ in their sampling methodology or incorporate additional modules on other public health topics. Of particular interest to the present analysis is the development and refinement of an optional domestic violence module, which is administered to one selected woman per household. Inspired by the Conflict Tactics Scale (Straus, 1979; Straus et al., 1996), these respondents are asked whether they have been subject to specific acts of violence (e.g., kicking, striking with a weapon) and if so, their relationship to the perpetrator, if any. The incidence of domestic violence is quantitatively operationalized through discrete affirmative or negative responses to these questions (see Chapter C for summary statistics for variables relevant to domestic violence).

Both the appeal and limitation of this measure is in its simplicity. Tabulating binary answers to unambiguous questions is plausibly less susceptible to social desirability bias than earlier alternatives but also removes contextual information and subjectivities that may meaningfully qualify a notoriously complex social phenomenon (Smith et al., 1995, 1996). Results of this data collection cannot differentiate between chronic battery nor does it consider the meanings and dangers which victims assign to these events.

To investigate the hazard into domestic violence, we make particular use of a single uniquely informative question[1]. After being asked about experiences with a variety of physically violent acts, respondents who answered any question affirmatively were then asked in which specific year of their marriage they first experienced any of these acts. Figure C.2 depicts the distribution of answers to this question for hundreds of survey waves in 12 global regions. Other than mild evidence of bunching at marriage-years divisible by 5, the distributions are consistent across the first nine years and remarkably well-distributed exhibiting approximately exponential decreases from a peak in the first year of marriage. Figure 3.1 depicts the implied cumulative distribution of marital violence over time and Figure 3.1 charts the implied annual hazard into first-time violence: conditional on never previously having been violent, the probability of a marriage becoming violent for the first time is clearly largest in the early years of a marriage. For this reason, we specify the first four years as the baseline risk period for our analysis.

By placing this potentially critical event in time, we can infer more about its context by relating it to the timing of other events. For example, respondent and family-member month-years give us information on the ages of the respondent, perpetrator, and household kinship structure at the time of the event. The sample of respondents for our main results come from the two India survey waves which both provide GPS coordinates and which implemented a version of the domestic-violence module which included this now-defunct timing variable.

[1]Versions of this variable which appeared in the earliest domestic violence surveys restricted focus to sexual violence before it was broadened to consider any of the physical acts enumerated in Chapter C. This question has now been discontinued, having been phased out of DHS survey questionnaires beginning with the launch of DHS-8 in 2018.

3.2.3 Early marriage

We also draw from the DHS surveys to make inferences about the effect of these shocks on the timing of marriage, updating the main result of Corno et al. (2020) using much more recent and higher-quality data.

The original analysis relied on the 1998-1999 survey wave for Indian respondents and 1994-2013 data for Sub-Saharan African countries. In comparison, we use Indian data from the 2015-16 and 2019-21 waves and all available sub-Saharan African survey waves for which GPS coordinates were available. Our estimation sample constitutes 6.8-8.2 million respondent-age observations for India and 5.1-6.3 million for sub-Saharan Africa compared to 329,586 and 1,799,037 for the original analysis. In addition, this older generation of Indian DHS surveys purposely only interviewed women who were married at the time of the interview, which would strictly overstate estimates of the annual hazard into marriage.

Critically, this earlier vintage of Indian DHS data also did not come with GPS coordinates, meaning respondents could only be geographically matched to highly aggregated districts roughly 5000 square kilometers in area on average. In contrast, GPS coordinates provided in later waves are guaranteed to lie within 10 kilometers of the respondent's true location. The higher quality weather data described above and these much more precise DHS locations mean that our shocks are far more localized to the true locations of the respondents.

For additional context, we plot baseline annual hazard rates and cumulative rates of marriage in Chapter C.

3.3 Methods

Our main analysis is an empirical test of whether drought shocks impact the probability of a marriage becoming violent for the first time. While our data is sourced from the repeated cross-sections of the DHS program, the required duration data can be constructed by exploiting the dynamic information implied by the timing variable highlighted in the previous section.

Specifically, we construct a pseudo-panel where each observation corresponds to the experiences of a respondent i in their tth year of marriage. With the provided household GPS coordinates, we can then produce a timeline of drought exposure specific to each respondent under the assumption that the respondent did not move to a substantially different geography over this period. Our main explanatory variable, $D_{i\ell t}$ is a binary indicator for whether the volume of precipitation which occurred in the period t since respondent i was married constituted a relative drought in the history of rainfall specific to location ℓ.

We estimate Equation (3.1) to estimate the effect of drought exposure on the incidence into first-time domestic violence, extending duration data methods employed in other studies of hazards into gendered outcomes such as child mortality (Currie & Neidell, 2005), child marriage (Corno et al., 2020), and female genital cutting (McGavock & Novak, 2023).

$$V_{it} = \sum_{p=0}^{P} \beta_p D_{i,\ell,t-p} + \boldsymbol{\gamma}^\top \mathbf{W}_{i,\ell,t} + \lambda_i + \phi_\ell + \eta_t + \varepsilon_{i,\ell,t} \tag{3.1}$$

The outcome variable V_{it} is a binary indicator of whether the respondent i experienced marital violence for the first time in their tth year of marriage (i.e., the 12 months preceding their tth wedding anniversary). All respondents are observed in our dataset for the first year of their marriage $t = 1$. Respondents who experienced domestic violence in this period 'drop out' from our dataset since they are definitionally no longer at risk of first-time violence. We repeat this process for each $t > 0$ until all respondents have experienced violence or up to a horizon $\bar{t}$ defining the end of the risk period of interest which we had determined from Figure 3.1 to be the fourth year of marriage in our main specification though we also present results under alternative values of $\bar{t}$.

Our main coefficients of interest are the set $\{\beta_p\}_{p=0}^{P}$, corresponding to the absolute change in the annual hazard into domestic violence causally attributed to a drought shock and its P lags. These coefficients can then be divided by the unconditional hazard rates depicted in Figure 3.1 over the risk period of analysis to determine the proportional change in the hazard rate attributable to the shocks.

In our main specification, the matrix $\mathbf{W}_{i,\ell,t}$ of controls includes indicators for the survey wave

the respondent data was collected for, the panel marriage-year t the observation pertains to, the village-level 'cluster' whose coordinates are assigned to the respondent in the data, and the calendar year in which the respondent was married. Errors are clustered at the level of the GPS coordinates from which we derive the treatment variable. In the Indian data, a given cluster's coordinates are assigned to 17 different respondent households on average[2]. Given that our drought shock treatment is assigned on the basis of these coordinates and likely highly autocorrelated across space, households assigned the same cluster are likely to exhibit systematically more correlated responses.

We also estimate the equivalent of Equation (3.1) using entry into child or early marriage as the outcome and women's adolescent years from ages 12-18 as the risk period of analysis. We also include lagged shocks to incorporate dynamic effects not considered in the original analysis by Corno et al. One interpretation of this

3.4 Results

3.4.1 Hazard into intimate-partner violence

Our main results are summarized in Section 3.4.1, representing the effect of a drought on the annual probability of a non-violent marriage becoming violent. Estimated effects across specifications are significantly negative across the different sets of fixed effects considered, suggesting that the experience of a drought in the early years of marriage actually reduces the hazard into marital violence.

Estimates are most pronounced for the first lag of the drought shock, implying that new wives are approximately 0.04 percentage points less likely to experience first-time violence with the occurrence of a drought in the first four years of marriage. For our preferred specification, Model (3), this estimate is 0.037 [0.016, 0.058]. Compared to an unconditional hazard over this period of 0.053 representing, this represents a substantial 70% reduction in the hazard into marital violence.

[2]While this would ordinarily correspond to hundreds of female respondents, the domestic violence module of the DHS is unique in only being administered to one randomly selected woman in each household.

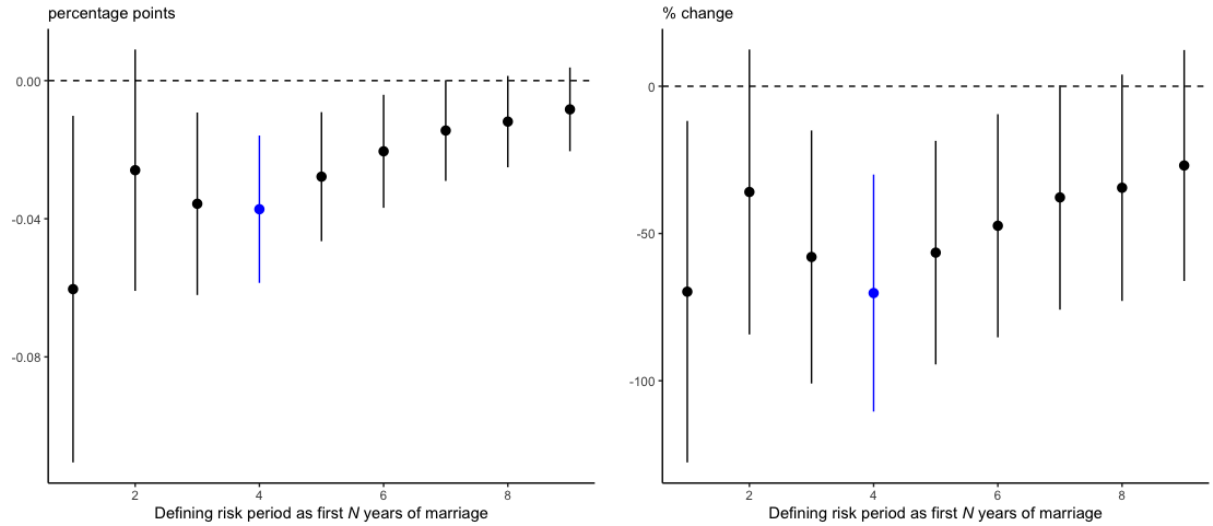
Table 3.1: Effect of drought on the hazard into marital violence

Dependent Variable:	Violent			
Model:	(1)	(2)	(3)	(4)
Variables				
Drought	-0.0004***	-0.0002*	-0.0001	-0.0001
	(0.0001)	(0.0001)	(0.0001)	(0.0001)
Drought, L1	-0.0006***	-0.0004***	-0.0004***	-0.0004***
	(0.0001)	(0.0001)	(0.0001)	(0.0001)
Drought, L2	-0.0002·	2.3×10^{-5}	-6.26×10^{-5}	-5.48×10^{-5}
	(0.0001)	(0.0001)	(0.0001)	(0.0001)
Fixed-effects				
Survey wave (2)	Yes	Yes	Yes	Yes
Panel year of marriage (4)	Yes	Yes	Yes	Yes
Cluster (56,073)		Yes	Yes	Yes
Year married (38)			Yes	Yes
Age at marriage (51)			Yes	Yes
Region (ADM1)-Marriage cohort (291)				Yes
Fit statistics				
Dependent variable mean	0.00538	0.00538	0.00538	0.00538
Observations	3,539,064	3,539,064	3,539,064	3,539,064
Adjusted R^2	0.00068	0.02577	0.02600	0.02607
Within R^2	1.85×10^{-5}	5.34×10^{-6}	3.93×10^{-6}	3.65×10^{-6}

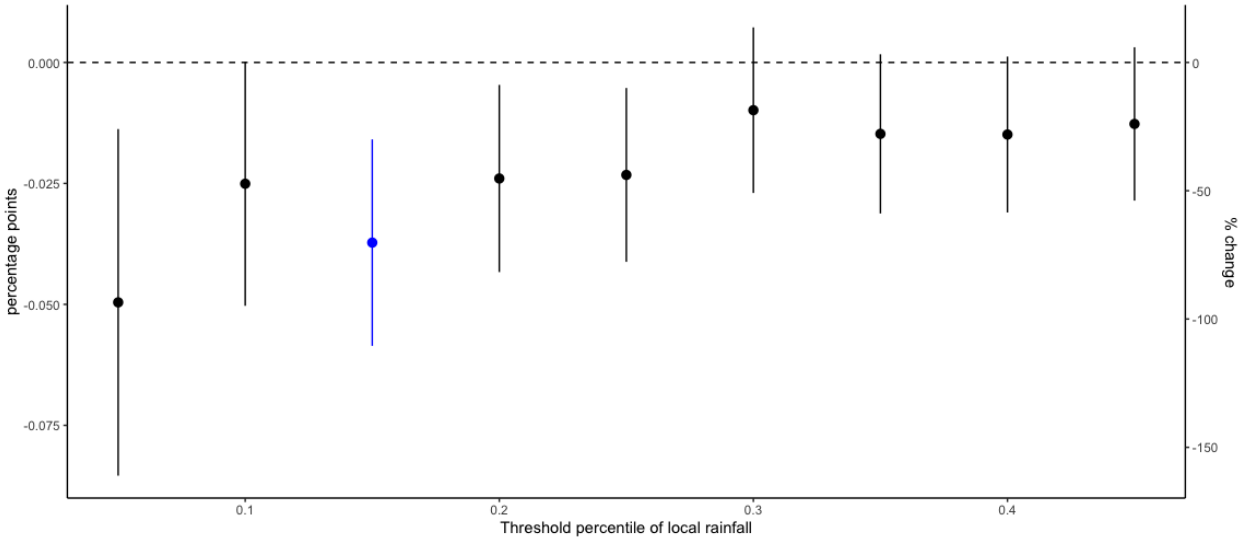
Clustered (Cluster) standard-errors in parentheses
*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, .: 0.1*

Figure 3.2: Estimated effect of drought on hazard into marital violence, sensitivity analysis

(a) Varying reference period of risk



(b) Varying threshold intensity of drought



Sensitivity of coefficient estimates to different definitions of drought and period of ris. Preferred specification in blue, defining droughts as years falling below the 15th percentile in the local distribution of precipitation and defining the reference period of risk as the first four years of marriage

Section 3.4.1 depicts the sensitivity of these estimates to different choices about the relevant risk period and threshold for defining a drought with our preferred specification highlighted in blue and error bands corresponding to a 95% confidence interval. In Figure 3.2, defining the risk period over a sufficiently long period nullifies the drought effect suggesting that environmental conditions (and plausibly the income channels they operate through) can influence the propensity for a marriage only in its earliest formative years. One interpretation of this pattern is that for marriages which have remained non-violent for sufficiently long, breaks from this established norm are more difficult to influence.

Figure 3.2 shows that, unsurprisingly, the less stringent the definition of a drought (i.e., as the threshold percentile increases), the smaller the estimated effect it has on the hazard into marital violence. On the other hand, for sufficiently stringent definitions of drought, the frequency of the 'treatment' variable diminishes, reducing the precision of these estimates. So while years with precipitation less than the 5th percentile in the local historical distribution lead to larger point estimates, the 95% confidence interval ranges from an 8.5% reduction in the hazard to its complete elimination.

Chapter C summarizes results varying the number of lags of the drought variable included and estimating effects separately for the two waves of the Indian DHS survey. We find that regressions which omit the lagged shocks return null effects while including additional lags does not provide additional explanatory power nor does it affect the estimate on the first lag of the drought. The drought effect, then, is entirely transitory and operates with a one-period lag, similar to the drought effect on democratic transition documented by Brückner and Ciccone (2011).

3.4.2 Hazard into early marriage

Section 3.4.2 reports the results of an analogous exercise, measuring the effect of droughts on the hazard into child marriage for women between the ages of 12-18. This represents an update and extension of the main results from Corno et al. (2020) on much newer and improved data[3].

[3]The original analysis used India's 1998-99 DHS survey data compared to our use of India's 2015-16 and 2019-21 rounds. We make note of the substantive differences in data quality between these rounds in Section 3.2.

Table 3.2: Effect of drought on the hazard into child marriage

Dep. Variable:	Married					
Model:	(1) India	(2) India	(3) India	(4) SSA	(5) SSA	(6) SSA
Variables						
Drought	-0.0011***	-0.0009***	-0.0012***	-0.0012***	-0.0015***	-0.0011**
	(0.0002)	(0.0002)	(0.0003)	(0.0003)	(0.0003)	(0.0003)
Drought, L1		-0.0010***	-0.0018***		-0.0025***	-0.0025***
		(0.0002)	(0.0003)		(0.0003)	(0.0003)
Drought, L2			0.0009***			-0.0022***
			(0.0003)			(0.0003)
Fixed-effects						
# Cluster	56,076	56,076	56,076	43,883	43,877	43,872
# Panel age	7	7	6	7	7	6
# Birth year	41	41	40	43	42	40
# Country	1	1	1	36	36	36
# Survey wave	2	2	2	106	106	106
Fit statistics						
Dep. variable mean	0.04919	0.04933	0.05724	0.05978	0.05966	0.06878
Observations	8,237,805	8,165,320	6,815,956	6,267,971	6,166,258	5,127,306
Adjusted R^2	0.05647	0.05662	0.05456	0.06722	0.06729	0.06635
Within R^2	3.24×10^{-6}	5.94×10^{-6}	1.29×10^{-5}	3.23×10^{-6}	1.93×10^{-5}	2.42×10^{-5}

Clustered (Cluster) standard-errors in parentheses
*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, .: 0.1*

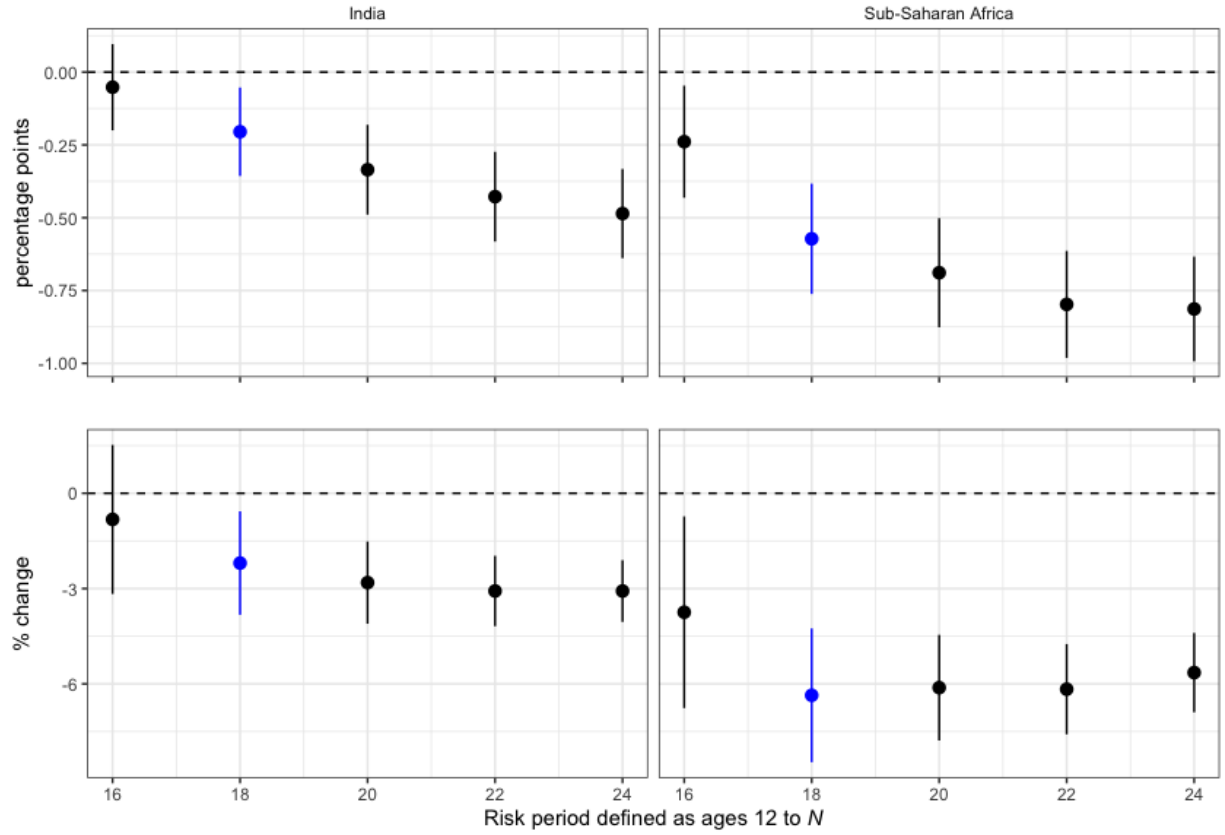
We report new strong evidence that droughts reduce the hazard into child marriage in India, supporting the original finding of Corno et al. Because of differences in data collection methods—for example, the 1999 Indian DHS survey did not interview single women—the older analysis will generally imply larger conditional and unconditional hazard rates. We also incorporate dynamic effects of lagged droughts which were not considered in the original analysis.

Despite these differences, recovered effects in India are on the order of the original results but significantly reduced: in our preferred specification (Column 3), the cumulative effect of droughts experienced between the ages of 12-18 reduces the hazard into child marriage among young women in India by 0.21 percentage points, representing a 2.19% reduction from an unconditional hazard rate of 9.3%. In the original analysis, droughts were found to have reduced this hazard by 0.46-0.47 percentage points, a 4.2-4.3% decline relative to a hazard rate of 10.9 percentage points.

However, our new analysis contradicts the finding of opposite-signed effects in Sub-Saharan Africa, the main result of the original analysis which attributed these opposite dynamics to consumption smoothing induced by opposite customs of marriage payment. Instead, the hazard into child marriage is even more substantialy reduced in the sub-Saharan African sample if anything: a 5.7 percentage-point reduction from a baseline hazard of 9.0%, a reduction of 6.4%.

Figure 3.3 plots the sensitivity of these results to broader definitions of child marriage, finding that a drought event reduces this hazard even more at later ages.

Figure 3.3: Effect of drought on hazard into early marriage, varying definitional risk period



3.5 Conclusion

In the analysis presented in this chapter, we document evidence that droughts during the formative years of a marriage significantly and substantially reduce the hazard into domestic violence in India by 70%. Secondarily, we also find that the same shocks reduce the hazard into child marriage in India and Sub-Saharan Africa, contradicting an earlier published result. These findings recall the finding in McGavock and Novak (2023) that similarly constructed shocks marginally increase the propensity for families in Sub-Saharan Africa to forego the practice of female genital cutting.

Like the authors of that paper note, these results are somewhat challenging in that they do not immediately prescribe responses to the social problems at subject; it would be absurd to use these results to advocate for acute reductions in income and extreme weather events are a means of mitigating these harms.

Our sensitivity analyses suggest timing is a vital consideration in addressing these harmful traditions. The earliest years of a marriage appear to be the most formative in the sense that they are the most malleable. This would imply that efforts to discourage intimate-partner violence would be most effective concentrating on young marriages; beyond the first four years, the hazard into first-time violence appears to be both relatively minimal, as shown in Figure 3.1 and impervious to economic or environmental influence (Figure 3.2).

For some regions, the hazard into child marriage is also high at early ages, as depicted by the density between ages 12-16 in Chapter C. However, Figure 3.3 shows this is precisely the age range where marriage rates are least influenced by external conditions, especially in India where we estimate a relatively null effect of droughts. This would suggest then that the prospects for economic incentives as a means of curtailing or eventually eliminating this particular cultural practice are limited and alternative approaches should be prioritized.

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Appendix A: Supplement to Chapter 1

A.1 Data

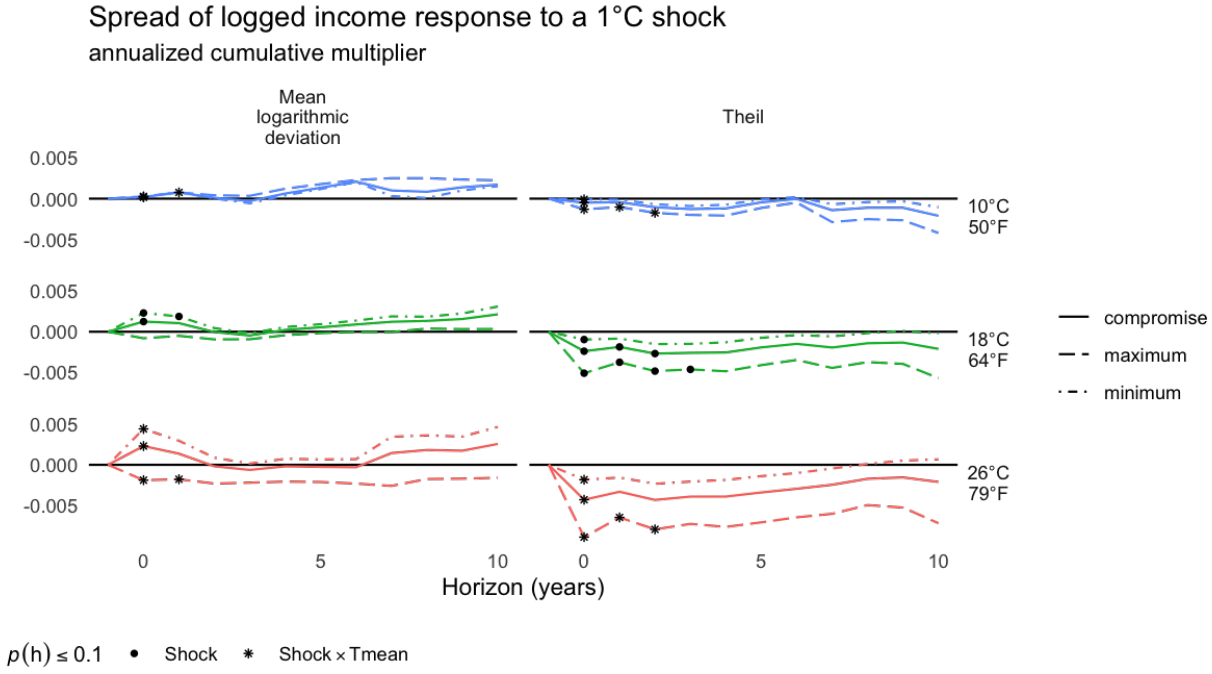
A.1.1 Construction of inequality indices

Data from the World Inequality Database can be extracted at the country-percentile level with variables reporting averages, minimums, maximums, and shares of national income. As grouped data, its use for the construction of inequality indices still requires the modeler to make assumptions about the distribution of household incomes within these disaggregated groups. This becomes increasingly influential for the construction of entropy-based inequality measures such as the mean logarithmic deviation or the Theil index for higher levels of inequality across groups.

For a given inequality measure J, the lower bound J_L is calculated by assuming perfect equality within all groups such that all individuals are assigned their group mean income. The upper bound J_U is calculated by assigning a proportion of the individuals within a group the minimum income of the group and the remainder the maximum income of the group.

Cowell, 2011 describes multiple 'compromise' assumptions between the two extremes, finding that they each approximate an average where the lower bound is given twice the weight of the upper bound. Figure A.1 depicts IRFs of the two inequality measures under each of these three assumptions. For our purposes, we use the compromise construction whenever a choice is necessary.

Figure A.1: Sensitivity of inequality indices to within-quantile inequality



A.2 Methods

Here, we provide additional exposition of our application of the method of local projections for inferring a dynamic social response to the temperature shock described in Section 1.2.3. The supplementary analysis here uses country-level GDP from the Penn World Tables (Feenstra et al., 2015) as the outcome variable of interest. The results documented here may also be compared to other estimates in the prolific literature on global climate impacts estimated by top-down methods, which most commonly use similar country-level aggregates as outcomes.

A.2.1 State-dependent impulse response of GDP

We first estimate Equation (1.3) on a country-year panel dataset using the h-period long difference in the logarithm of GDP as our outcome variable $\Delta_h y_{i,t+h}$. Figure A.2a illustrates the estimated impulse response of GDP up to a projection horizon of 6 years after a unit temperature shock. Col-

ors correspond to three representative temperatures depicted in different colors. The plots in the second row depict the same functions annualized by simplying dividing estimates at each projection horizon h by $h + 1$.

The contemporaneous estimates in the IRFs in the first row of Figure A.2a depict a pattern now standard in the aggregate impacts literature featuring near-null temperature effects in cold climates, mild effects in mid-temperature climates, and the largest negative effects in hot climates. The evolution of these impacts over time is less commonly depicted and here produce a pattern qualitatively similar to those reported by Nath et al. wherein per capita GDP appears to stay persistently depressed over the entire projection horizon. The second row depicts the same estimates but annualized by scaling down estimates by their corresponding horizon $h + 1$ to more clearly depict convergence behavior relative to the contemporaneous effect of the shock.

Global inequality effects of climate change thus are not just dependent on the shape of the impulse response responses across quantiles but also the distribution of warming over space and time as well as the actual distributions of incomes across quantiles. The simulation exercises to be described in Section 1.4 are valuable because they account for four relevant climate inequalities: inequality of temperature effects across an income distribution, inequality in the incidence of temperature effects across different climates, inequality in the global distribution of income, and spatial inequality in the incidence of absolute warming.

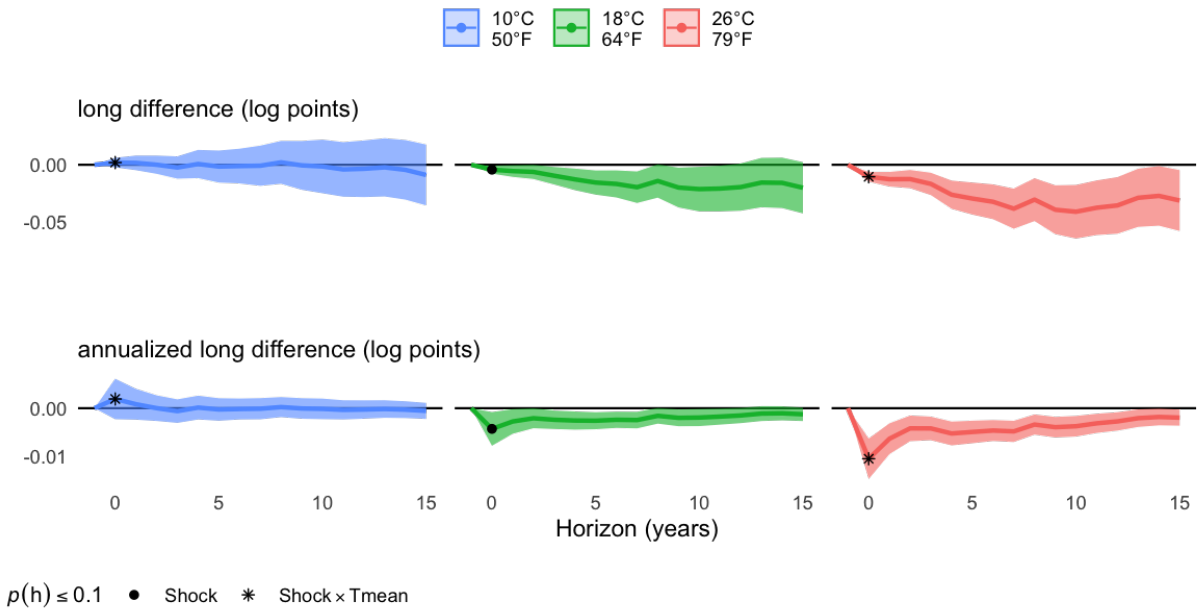
A.2.2 Inferring persistence from local projection estimates

The colored error bands depicted in Figure A.2a reflect pointwise estimation error. This is particularly important to include in this context since the volume of regressions underlying a single LP-IRF precludes convenient summary in a single regression table.[1] One may be tempted to infer from the fact that the hot-country bands in red do not intersect the horizontal axis over the entire projection horizon that these estimates constitute significant evidence to reject a joint null

[1]For example, the results depicted in Figure A.2a derive from 16 different regressions. Later results derive from as many as 91. All graphically depicted errors and significance tests presented in this paper correspond to a significance level of $\alpha = 0.1$.

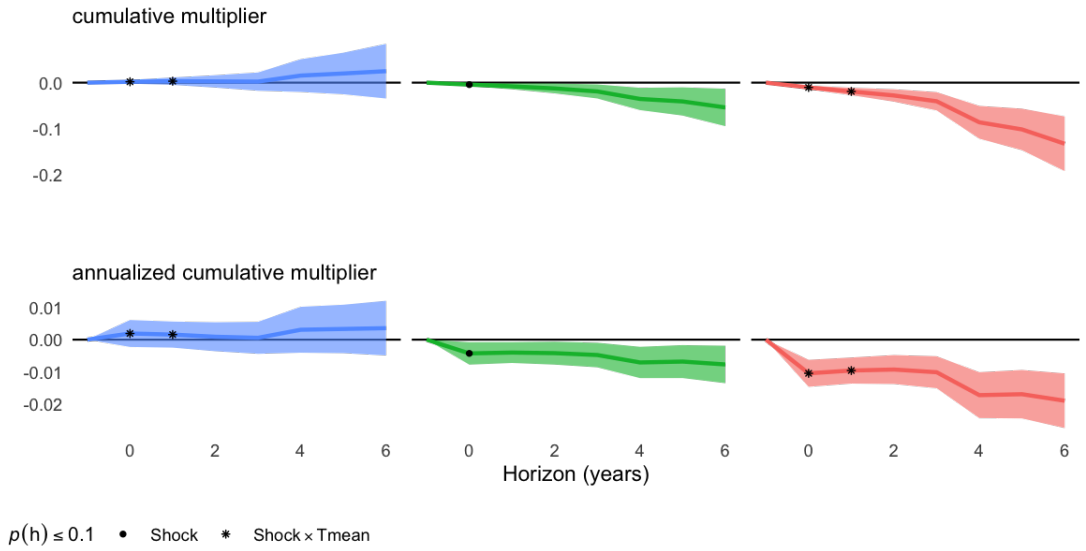
Figure A.2: State-dependent response of GDP to a 1°C shock

(a) State-dependent impulse response of GDP



The plots in the top row correspond to estimates of impulse responses of GDP estimated by the $H+1$ local projections regressions specified in Equation (1.3) for three representative temperatures depicted in different colors. The plots in the bottom row depict the same estimates and error bands annualized by dividing values at each horizon h by $h+1$

(b) State-dependent temperature multiplier of GDP



The cumulative multiplier is interpreted as the change in *cumulative* GDP over h periods attributable to a unit shock in $\widehat{\tau}$, net of its impact on future shocks $\widehat{\tau}$. Annualized multipliers represent the same estimates but divided by the projection horizon for visual convenience.

hypothesis of level effects which persist for fewer than 15 years.

However, the uncertainty depicted by error bands corresponds only to individual hypothesis tests. Since estimates and residuals across an IRF are serially correlated—for example, the green bands straddle the horizontal axis but all estimates are negative— the relevant test of significance is a joint hypothesis test not represented by pointwise error. The significance of the main effect of the impulse response $\mathcal{R}_{\tau\to y}(h)$ amounts to testing the null hypothesis that all main effects up to h are 0:

$$H_0(h) : \beta_{1,0} = \beta_{1,1} = \ldots = \beta_{1,h} = 0$$

Since the number of coefficients being considered in the joint hypothesis increases in h, the coverage of the joint hypothesis test must be adjusted for the inclusion of each additional horizon to compensate for the problem of multiple hypothesis testing. Inverting this joint null hypothesis, Inoue et al. (2024) derives convenient estimators for the implied "significance band" defined to satisfy the following condition:

$$\mathbb{P}\left[\bigcap_{h=0}^{H}\left\{\zeta_{\frac{\alpha}{2(H+1)}}\frac{\sigma_h}{\sqrt{T-h}} < \widehat{\beta}_h < \zeta_{1-\frac{\alpha}{2(H+1)}}\frac{\sigma_h}{\sqrt{T-h}}\right\}\right] \geq 1-\alpha$$

The scaling of the significance level α by $2(H+1)$ is an implementation of the Bonferroni correction. In practice, rejecting the null for the main effect corresponding to horizon $h+1$ is sufficient to reject the null of non-persistence up to horizon h. Graphically, these bands are represented by a region straddling the horizontal axis such that estimates of β_{1h} contained within it are insufficiently large enough to reject the hypothesis of a null effect that the impulse response persists up to period h, analogous to the bands drawn to measure persistence in time series correlograms. The width of these bands increases with the projection horizon because the available estimation sample for a near-balanced panel is reduced by one period per unit for each additional projection.

These significance bands are specific to the main effect and are represented by the gray regions in the middle panels of Figure A.3a. Since ours is a state-dependent model, I derive equivalent significance bands specific to the interaction effects. These are defined relative to the point estimates of the main effects and are depicted by the gray regions in the panels on the left and right. Points outside these regions are considered sufficient to reject the null hypothesis that interaction effects do not persist up to h periods. By the linearity assumption, the null hypothesis of zero interaction effects for a 10° climate is rejected if and only if it is rejected for a 26° climate.

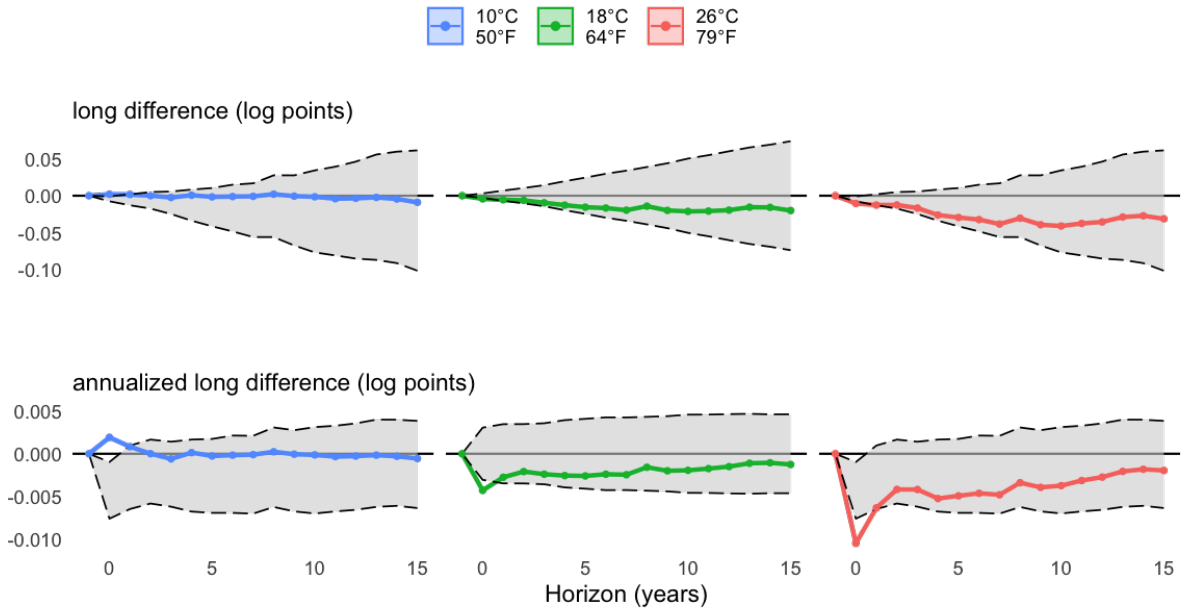
For comparison, Figure A.3b depicts a placebo test where I construct significance bands equivalent to those described above to test the persistence in the effect of temperature on another widely available and potentially non-stationary growth rate: that of population. As I am not aware of any theory or evidence to suggest temperature shocks would meaningfully impact country-level population growth[2], we should expect the tests to return precise null results. Indeed, we find with all point estimates resulting from two separate population datasets easily contained by the corresponding significance bands. Notably, if we were to misuse the error bands as indicators of statistical significance, we would misinterpret the estimated IRFs as evidence to reject the null for large subintervals of the projection horizon for mid-temperature and hot climates. Instead, the entirety of these error bands are easily enveloped by the significance bands.

Since including significance bands, color-coded error bands, and point estimates for multiple IRFs can clutter space, I adopt a convention of omitting the significance bands entirely and instead coloring black the point estimates which are located outside the implicit error bands. In all IRFs beginning with those depicted in Figure A.2a, rounded points correspond to significant main effects while asterisks correspond to significant interaction effects. By this test, these results cannot reject the null hypothesis of one-period level effects.

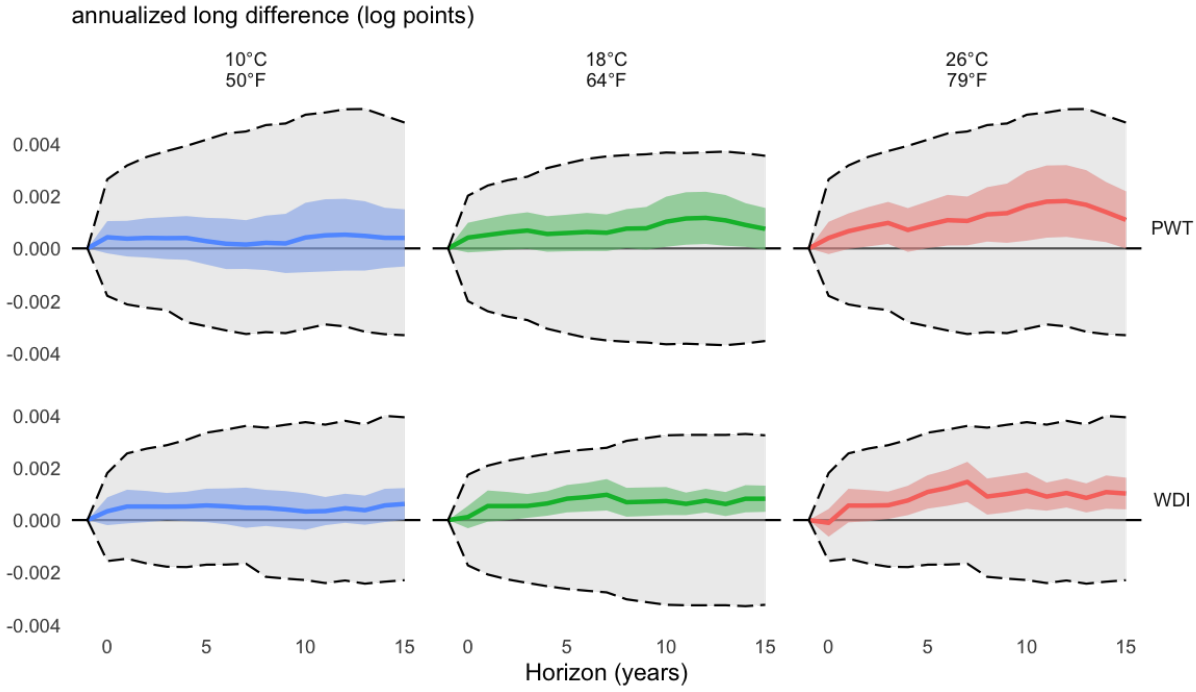
[2]While there is an established literature on climate impacts on mortality rates (A. Barreca et al., 2016; A. I. Barreca & Shimshack, 2012; T. Carleton et al., 2022), absolute magnitudes are miniscule relative to total populations

Figure A.3: Impulse responses to a 1°C shock, including significance bands

(a) GDP impulse response



(b) Placebo test: national population response



A.2.3 The state-dependent temperature multiplier

Another reason to prefer temperature shocks to levels of temperature as treatment variables is that models which use the latter neglect to sufficiently account for autocorrelation in temperature itself when estimating dynamic effects (Bilal & Känzig, 2024; Nath et al., 2024). An analogy can be drawn to the estimation of multipliers in empirical macroeconomics: when estimating the impulse response of, for example, GDP to a fiscal policy shock, one must account for the tendency of stimulus programs to be implemented in stages or to be followed by additional stimulus. Otherwise, the effect sizes implied by simple impulse response functions misattribute cumulative changes in the outcome only to the initial unit shock; multipliers scale these effects at each horizon by the corresponding accumulation of shocks.

Figure 1.3 depicts the impulse response of temperature shocks to a pulse of itself for our three representative climates. Each exhibits very similar persistence patterns across two independent temperature dataset. An application of the persistence test covered in Section A.2.2 implies autocorrelations which persist for up to 10 years though only the first 1-2 periods are qualitatively large (approximately 21% and 9% respectively with insignificant heterogeneity across climates).[3] Cumulative multipliers are estimated in order to scale down the estimated impulse response of GDP $\mathcal{R}_{\tau\to y}(h)$ by the impulse response of the shock to itself $\mathcal{R}_{\tau\to\tau}(h)$ to account for this dynamic treatment schedule.

Cumulative responses are traditionally computed by first constructing *cumulative* versions of the two IRFs: $\mathcal{R}^c_{\tau\to y}(h) := \sum_{j=0}^{h} \mathcal{R}_{\tau\to y}(j)$ for the outcome of interest y and $\mathcal{R}^c_{\tau\to y}(h) := \sum_{j=0}^{h} \mathcal{R}_{\tau\to\tau}(j)$ for the shock $\widehat{\tau}$. Then the cumulative temperature multiplier $m_{\tau\to y}(h)$ would be calculated as the ratio

$$m_{\tau\to y}(h) := \frac{\mathcal{R}^c_{\tau\to y}(h)}{\mathcal{R}^c_{\tau\to\tau}(h)}$$

[3]This contrasts with the finding of NRK who measure shock persistence using levels of temperature to infer that temperature shocks remain 10-20% higher for at least nine years following a unit shock.

However, this approach has a couple of inconvenient downsides. First, as noted by Jordà (2024), computation of standard errors of a ratio of two random variables is complicated and non-standard. Second, it is computationally inefficient in the sense of requiring $2(h+1)$ (or $4(h+1)$ in our state-dependent case) separate local projections to obtain the corresponding multiplier for each projection horizon h.

Instead, we follow the recommendation to use the one-step method described in Ramey (2016) and applied in Ramey and Zubairy (2018). We adapt the method for our state-dependent setting through the procedure summarized in Algorithm 1.

We can use the resulting coefficients to trace a cumulative multiplier function $\mathcal{R}^{c}_{\tau\to y}(h;\overline{T})$ analogous to Equation (1.4) but which accounts for the effect of persistence in the shock. Since estimation of the coefficients follows directly from a system of local projection regressions, standard inference is preserved and the significance testing methods outlined in Section A.2.2 conveniently remain available. Figure A.2b demonstrates an application of the method to estimate temperature multipliers of GDP. The estimated multiplier over horizon h is interpreted as the difference in cumulative GDP over the h periods following an isolated 1°C shock relative to a counterfactual where the shock had never occurred.

Estimated cumulative multipliers are found to be substantial at larger horizons but the significance band test cannot reject a null of non-persistence beyond one period. 90% confidence intervals for the effect of an identified temperature shock on cumulative GDP are $[-0.49, 1.1]$ percent for 10°C climates and and $[-2.7, -1.1]$ percent for 26°C climates.

Algorithm 1: One-step state-dependent cumulative multiplier

for $h = 0$ **to** H **do**

Estimate the IRF $\mathcal{R}_{\tau\to\tau}(h)$ as defined in Equation (1.4), collecting state-dependent shock coefficients $\widehat{\alpha}_{1,h}, \widehat{\alpha}_{2,h}$;

Define $\Delta_h^c y_{i,t+h} := \sum_{j=0}^{h} \Delta_j y_{i,t+j}$;

Define $\widehat{\tau}_{i,t+h}^c := \widehat{\tau}_{it}\left[\sum_{j=0}^{h} \widehat{\alpha}_{1,j} + \widehat{\alpha}_{2,j}\overline{T}_{it}\right]$;

Estimate local projection:

$\Delta_h^c y_{i,t+h} = m_{1,h}\widehat{\tau}_{i,t+h}^c + m_{2,h}\widehat{\tau}_{i,t+h}^c\overline{T}_{it} + \lambda_h\overline{T}_{it} + \mathbf{Z}_{it}\boldsymbol{\gamma}_h + \mu_i + \eta_t + \varepsilon_{i,t+h}$;

end

Output: Multiplier coefficients $\widehat{m}_{1,h}, \widehat{m}_{2,h}, \widehat{\lambda}_h$ for $h \in \{0, ...H\}$

A.3 Supplementary results

Figure A.4: Cumulative state-dependent income responses to a 1°C shock within the top decile

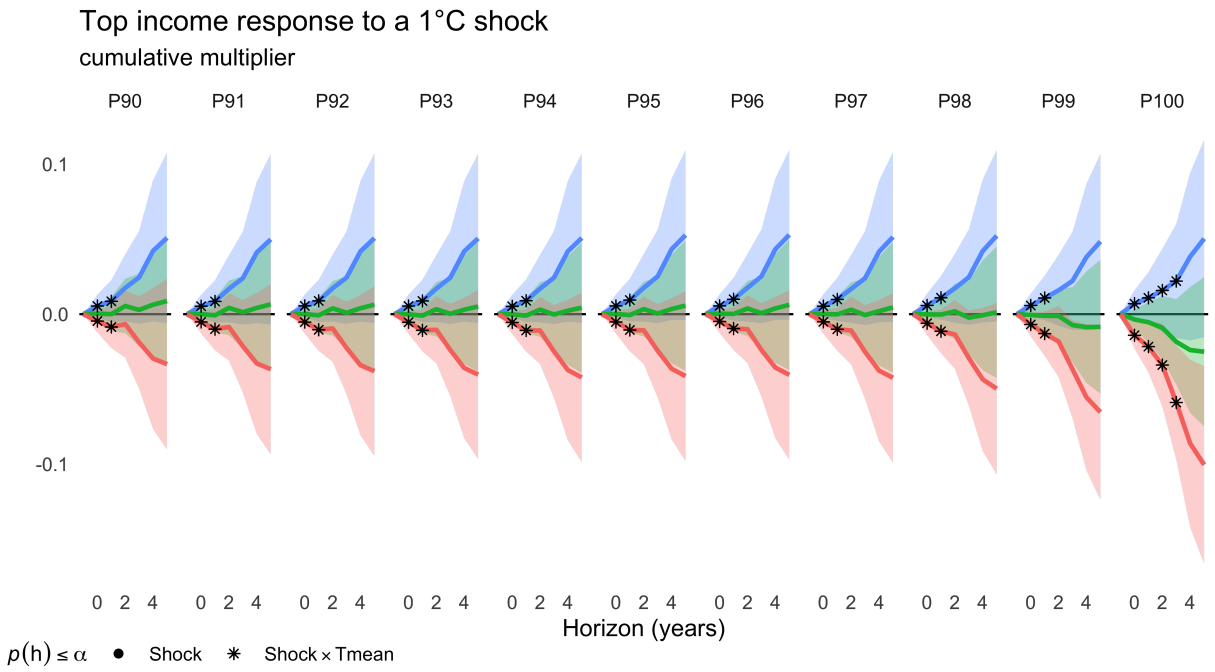
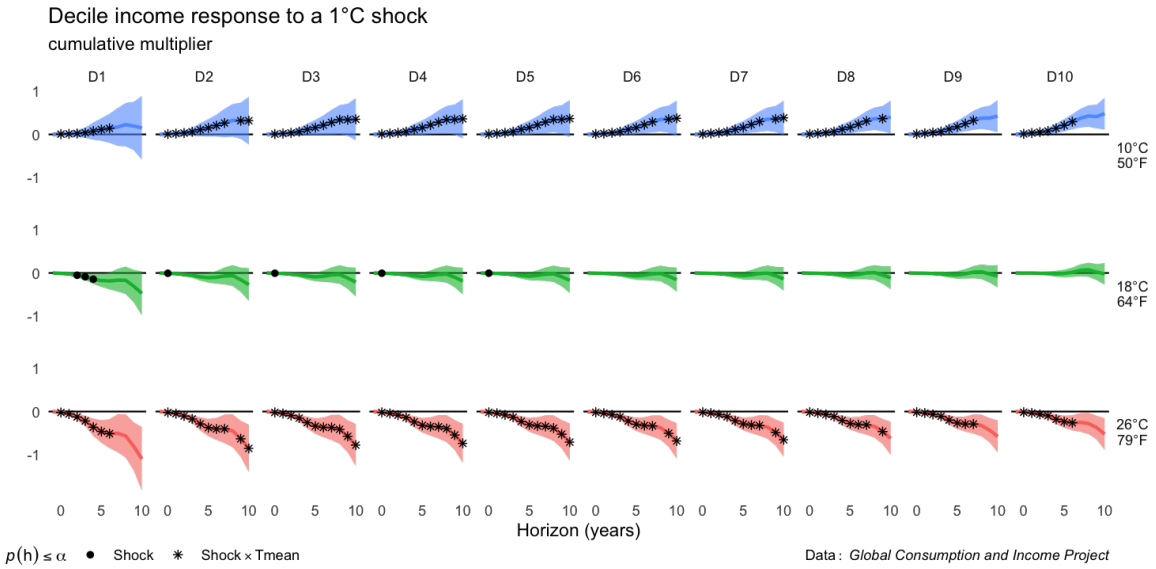
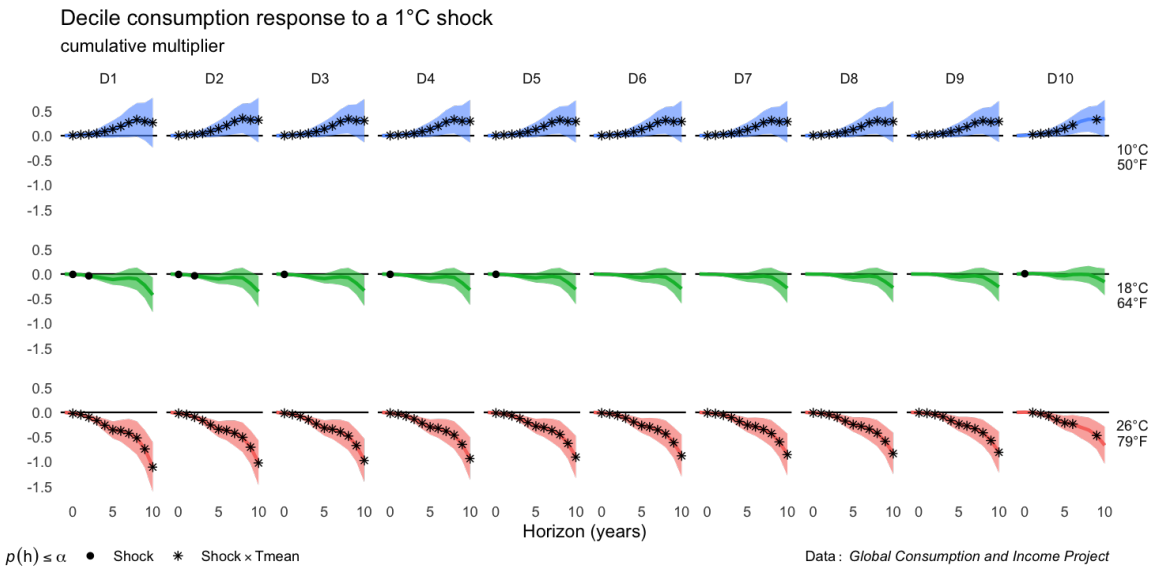


Figure A.5: Temperature multipliers estimated using survey microdata

(a) Impacts on income by decile



(b) Impacts on consumption by decile



A.4 Anthropogenic attribution

To attribute a global inequality effect to historical anthropogenic climate change, we implement a counterfactual analysis which decomposes observed temperature effects into their natural and anthropogenic contributions, adapting methods commonly used in climate attribution science.

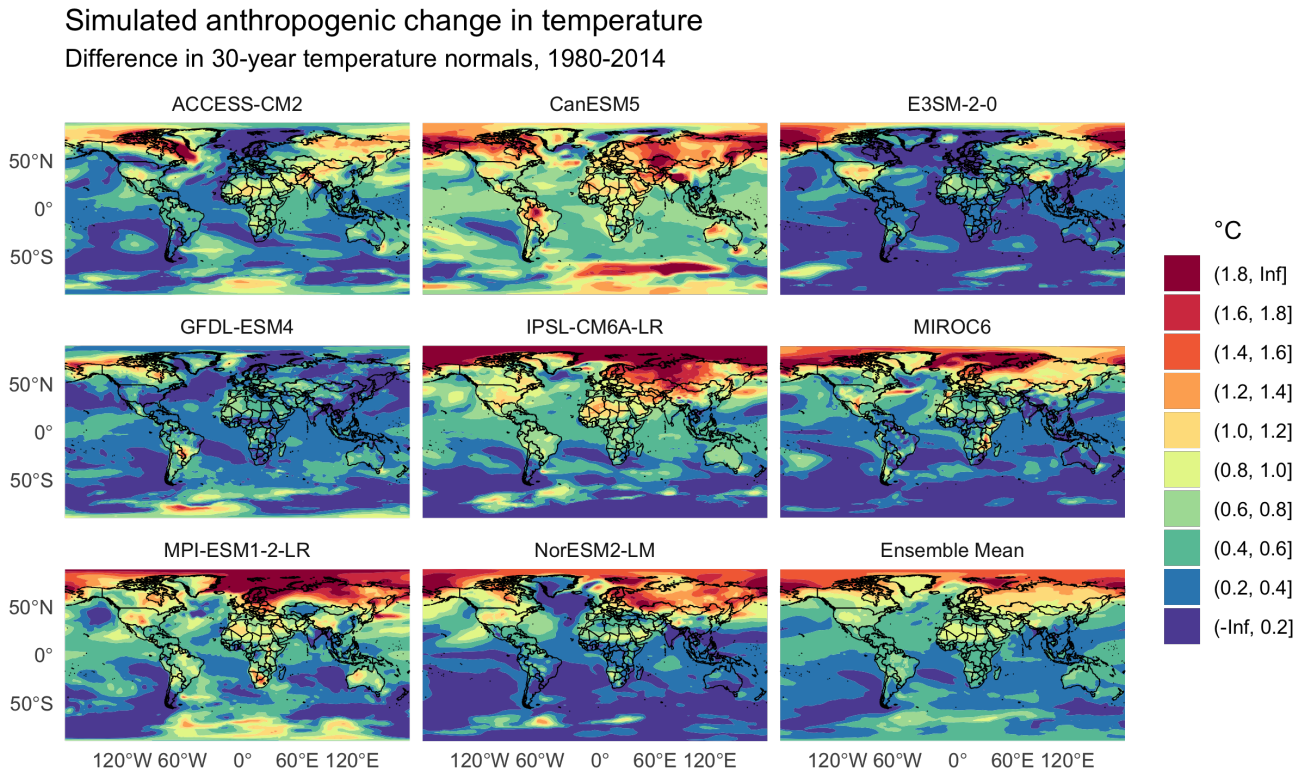
A.4.1 Climate simulation data

Climate simulation data comes from the contributions of climate modeling groups to the World Climate Research Program's Coupled Model Intercomparison Project (CMIP). The CMIP, initiated in 1995, is a framework to coordinate, standardize, and disseminate the results of state-of-the-art simulations of the global climate for the benefit of the international climate research community. Data used in this study comes from CMIP6, the sixth and newest[4] generation of models (Eyring et al., 2016). Data is provided as a collection of globally gridded time series summarizing the joint evolution of several hundred climate variables under various pre-specified calibrations. We will primarily be interested in the models' simulations of near-surface air temperature.

All participating models are "coupled", meaning that they explicitly account for interactions and feedback between distinct components of the global climate system such as the atmosphere, the cryosphere, land surface, and the ocean. Because the manner in which this integration is achieved is idiosyncratic to each model, computationally expensive, and highly sensitive to external calibrations, coordinating the set of "experiments" that participating models run is necessary for comparability and interpretability. For example, one of the four indexes which define an experiment is the realization index which corresponds to the set of geophysical initial conditions at the beginning of the simulation period. By holding the initial conditions fixed, differences in results across models within the same experiment can be more readily attributed to the distinguishing features of the model without being confounded by differences in implementation. Similarly, differences in the same climate model's results across different perturbations to the initial conditions provide a measure of the model's internal variability. The CMIP thus provides a systematic way of decomposing

[4]After a delayed rollout, outputs from most CMIP6 models were made accessible by 2022

Figure A.6: Model-implied spatial distributions of global warming, 1980-2014



Simulated spatial distribution of anthropogenic warming for eight randomly selected climate models. The bottom-right plots the average distribution for all 13 CMIP6 models which ran both historical and historical-natural simulations.

total variation across models and experiments both internal and external to the model. Coordination also enables identification of systematic and idiosyncratic biases in model designs that then inform the next generation of models.

Among the other three indexes which define an experiment, the forcing index is central to our counterfactual analysis. Forcings are factors which affect the net transfer of radiative energy in the climate system, essentially the energy the Earth receives from the sun minus that which it expels back into outer space. Anthropogenic forcings include, for example, industrial greenhouse gas emissions, deforestation, and the transformation of land for arable agriculture. Natural forcings include aerosol expulsions from major volcanic activity and variation in solar irradiance such as those associated with the 11-year solar cycle.[5] Comparing simulations which only differ in their forcings allow one to attribute differences in outcomes to the differences in forcings; we describe next how comparisons of this kind form the basis of both our historical and projection counterfactual analyses.

A.4.2 Construction of historical counterfactuals

Our historical counterfactual analysis uses simulations from a category of experiments called "historical runs". For the CMIP6 generation of models, historical simulations begin at a pre-industrial baseline period in 1850 and run through to 2014. Among the forcing scenarios used in these historical experiments is one simply labeled "historical", which attempts to represent the impacts of influential forcings actually imposed on the climate system over this period, such as the deforestation of the Amazon rainforest and the eruptions of Krakatoa in 1883 and Mt. Pinatubo in 1991. Results from this historical run are required of all participating CMIP6 models since these "hindcasts" can be directly compared to observational data in order to evaluate model accuracy. Models collectively perform remarkably well at reproducing the historical record (Liu et al., 2021; Zelinka et al., 2020; Zhou et al., 2020). For this reason, studies which use CMIP data for the purpose of counterfactual analysis are advised to include the full available "multi-model ensemble" for

[5]Natural forcings are distinct from sources of natural variability such as the ENSO cycle because they are external to coupled climate models.

experiments of interest (Knutti et al., 2010; Tebaldi & Knutti, 2007; van Vuuren et al., 2006).

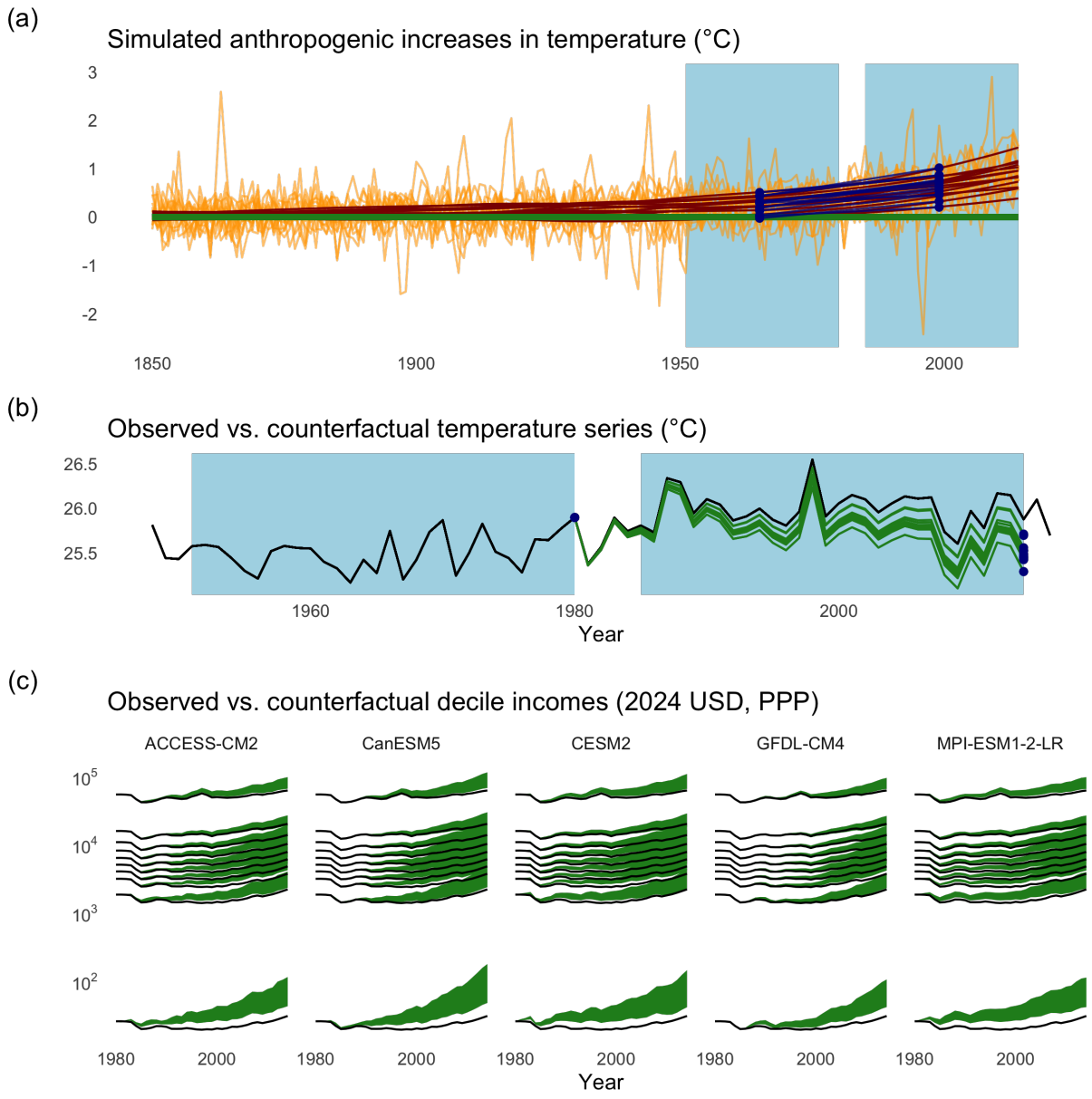
A subset of these models also simulate a set of optional 'historical-natural' experiments which are defined identically except with anthropogenic forcings held fixed at pre-industrial levels. The differences between the two forcings by the same model holding all other variables fixed is then interpreted as the model's simulation of anthropogenic contributions to climate change. Comparisons between historical and historical-natural experiments are commonly used in the subfield of attribution science concerned with quantitatively measuring anthropogenic contributions to the intensity, frequency, or probability of weather events or trends.

For each climate model m and country i, we use an observed series T and the two model-specific historical series $\mathcal{T}^{hist}$ and $\mathcal{T}^{nat}$ to construct a counterfactual series $\widetilde{T}$ where anthropogenic forcings have been held fixed since the first year of the simulation period. The iterative process is summarized in Algorithm 2 and illustrated in Figure A.7 using data from the Philippines as an example. In panel (a), differences between model-specific simulations of national temperature when including all forcings (orange) and when excluding anthropogenic forcings (green horizontal axis). Local regression fits are depicted in red. Light-blue shaded regions correspond to 30-year reference periods used to define temperature normals (dark blue dots) for the start and end of the simulation period 1980-2014.

In panel (b), The linearized difference between normals depicted as dark blue lines in (a) are subtracted from the observed temperature series (black) starting in 1980 to produce counterfactual temperature series (green). Each of these is interpreted as the temperature history that would have been realized had anthropogenic forcings been held fixed at 1980 levels according to a specific model of the climate system.

In panel (c), linearizing the difference between the climate normals defined at either end of the simulation period and subtracting the result from an observed temperature series implements the standard "delta method" for constructing bias-corrected climate counterfactuals. Counterfactual economic series for a random selection of five climate models are depicted in green and are constructed by iteratively applying the decile-level dynamic income responses depicted in Figure 1.4

Figure A.7: Constructing counterfactuals holding anthropogenic forcings fixed at 1980 levels



Demonstration of the delta method for constructing bias-corrected counterfactuals using data for the Philippines as an example.

to the observed and counterfactual temperatures depicted in (b). Their difference is interpreted as the change in income attributable to anthropogenic forcings since 1980. Green bands correspond to 90% confidence intervals generated from 500 bootstrap estimates of the response functions.

Algorithm 2: Constructing temperature counterfactuals holding anthropogenic forcings fixed at 1980 levels

Input: $\{\mathcal{T}^{hist}_{m,i,t}\}$,$\{\mathcal{T}^{nat}_{m,i,t}\}$,$\{T_{it}\}$

for $t^* \in \{1980, 2014\}$ **do**

$\overline{\mathcal{T}}^{hist}_{m,i,t^*} \leftarrow \frac{1}{30}\sum_{j=1}^{30} \mathcal{T}^{hist}_{m,i,t^*-j}$;

$\overline{\mathcal{T}}^{nat}_{m,i,t^*} \leftarrow \frac{1}{30}\sum_{j=1}^{30} \mathcal{T}^{nat}_{m,i,t^*-j}$;

$\overline{\delta}_{m,i,t^*} \leftarrow \overline{\mathcal{T}}^{hist}_{m,i,t^*} - \overline{\mathcal{T}}^{nat}_{m,i,t^*}$;

end

for $t = 1980$ **to** 2014 **do**

$\overline{\delta}_{m,i,t} \leftarrow (t - t_0)\frac{\overline{\delta}_{m,i,2014} - \overline{\delta}_{m,i,1980}}{2014-1980}$;

$\widetilde{T}^{nat}_{m,i,t} \leftarrow T_{it} - \overline{\delta}_{m,i,t}$;

end

Construct states $\{\overline{T}^{nat}_{m,i,t}\}$ by applying Equation (1.1a) to $\{\widetilde{T}^{nat}_{m,i,t}\}$ then subtracting by $\widetilde{T}^{nat}_{m,i,1980}$;

Construct shocks $\{\widehat{\tau}^{nat}_{m,i,t}\}$ by applying Equation (1.1a) and Equation (1.1b) to $\{\mathcal{T}^{nat}_{m,i,t}\}$;

Output: $\{\widetilde{T}^{nat}_{m,i,t}\}$,$\{\overline{T}^{nat}_{m,i,t}\}$,$\{\widehat{\tau}^{nat}_{m,i,t}\}$

Algorithm 3: Constructing economic counterfactuals holding anthropogenic forcings fixed at 1980 levels

Input: $\{\widetilde{T}^{nat}_{m,i,t}\}$,$\{\overline{T}^{nat}_{m,i,t}\}$,$\{\widehat{\tau}^{nat}_{m,i,t}\}$, B bootstrap samples indexed b

Initializing series:

for $b = 1$ **to** B **do**

Specify desired simulation interval: $(t_0, t_1) \leftarrow (1980, 2014)$;

Estimate $\mathcal{R}^{c}_{\tau\rightarrow Y}(h; \overline{T}, b)$ from Equation (1.5);

Specify maximum projection horizon H using persistence test from Section A.2.2;

Define IRFs $\mathcal{R}_{\tau\rightarrow Y}(h; \overline{T}, b) := \mathcal{R}^{c}_{\tau\rightarrow Y}(h; \overline{T}, b) - \mathcal{R}^{c}_{\tau\rightarrow Y}(h-1; \overline{T}, b)$;

Collect IRF coefficients;

$\{\widehat{\beta}_{1,h,b}, \widehat{\beta}_{2,h,b}, \widehat{\lambda}_{h,b}\}^{H}_{h=0}$;

Define $\delta^{\tau}_{m,b,i,t} := \widehat{\tau}^{nat}_{m,i,t} - \widehat{\tau}_{it}$;

Define $\delta^{\tau T}_{m,b,i,t} := \widehat{\tau}^{nat}_{m,i,t} \cdot \overline{T}^{nat} - \widehat{\tau}_{it} \cdot \overline{T}_{it}$;

Define $\delta^{T}_{m,b,i,t} := \overline{T}^{nat}_{m,i,t} - \overline{T}_{it}$;

Initialize $\delta^{Y}_{m,b,i,t} \leftarrow 0$;

Initialize $\widetilde{Y}^{nat}_{m,b,i,t_0} := Y_{i,t_0}$

end

Iteratively constructing series:

for $t = t_0 + 1$ **to** t_1 **do**

for $b = 1$ **to** B **do**

for $h = 0$ **to** H **do**

$\delta^{Y}_{m,b,i,t+h} \leftarrow \delta^{Y}_{m,b,i,t+h} + \widehat{\beta}_{1,h,b} \cdot \delta^{\tau}_{m,b,i,t} + \widehat{\beta}_{2,h,b} \cdot \delta^{\tau T}_{m,b,i,t} + \widehat{\lambda}_{h,b} \cdot \delta^{T}_{m,b,i,t}$;

end

$\widetilde{Y}^{nat}_{m,b,i,t} \leftarrow \left(1 + \log(\frac{Y_{it}}{Y_{i,t-1}}) + \delta^{Y}_{m,b,i,t}\right) \widetilde{Y}^{nat}_{m,b,i,t-1}$

end

end

Output: $\widetilde{N}^{nat}_{m,b,i,t}$

Figure A.8: Anthropogenic growth incidence curves by climate model, 1980-2014

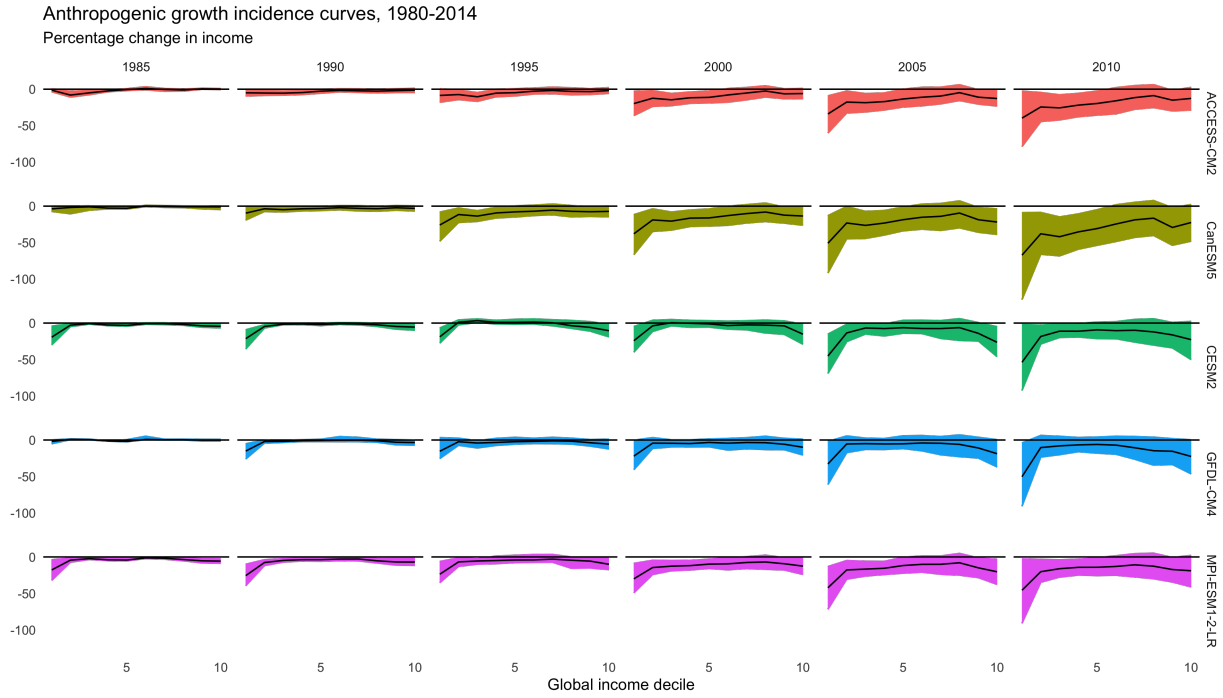
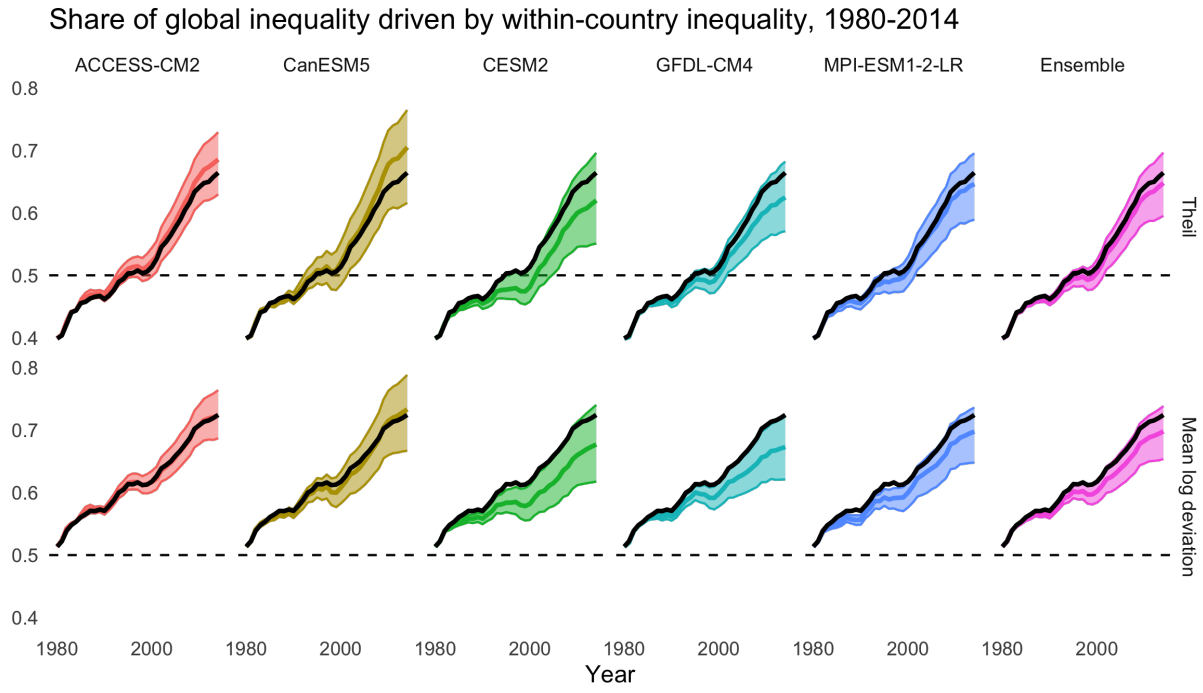


Figure A.9: Anthropogenic influence on within- vs. between-country inequality by climate model



Appendix B: Supplement to Chapter 2

Figure B.1: Distribution of democracy scores

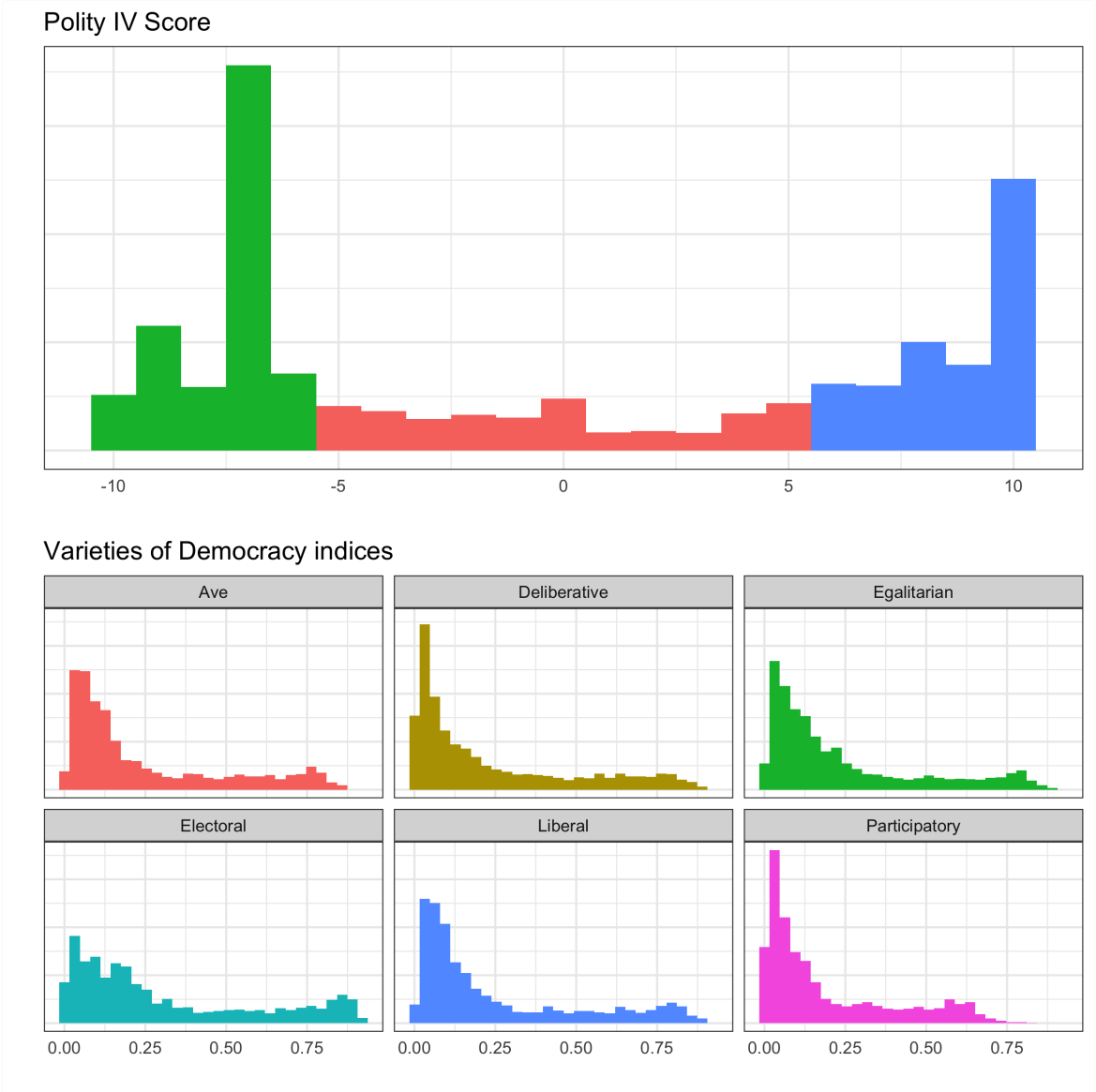
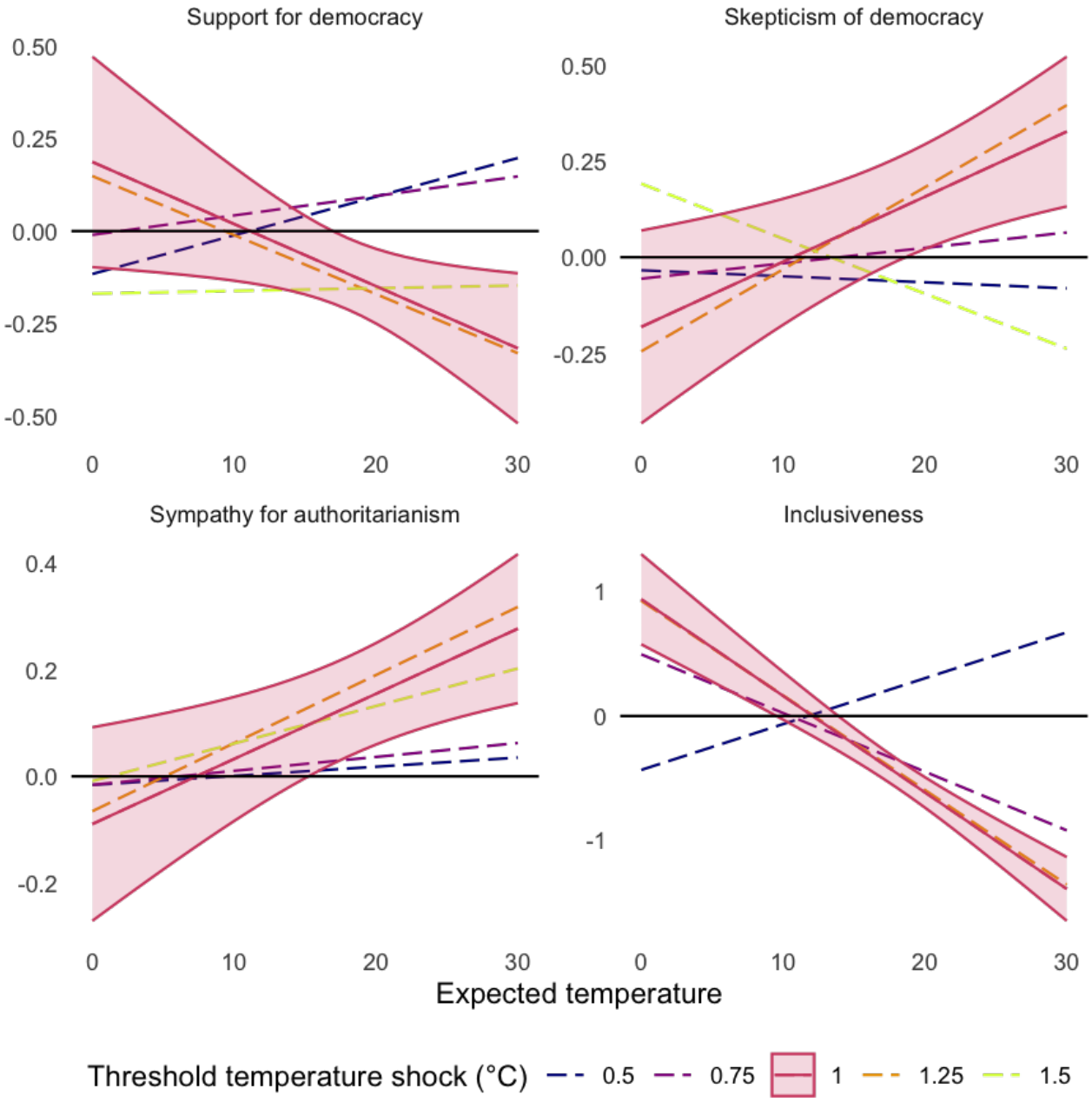


Figure B.2: Effect on political preferences of an additional temperature shock in last three years



Analogous to Figure 2.2 but instead of varying the number of years over which shocks are enumerated, we vary the minimum magnitude to qualify as a temperature shock.

Table B.1: Correlations between democracy indices after de-meaning by country

	Polity	V-Ave	V-Elec	V-Lib	V-Part	V-Delib	V-Egal
Polity	1.000	0.707	0.679	0.706	0.698	0.699	0.697
VDem-Average	0.707	1.000	0.987	0.990	0.988	0.990	0.979
VDem-Electoral	0.679	0.987	1.000	0.973	0.971	0.968	0.952
VDem-Liberal	0.706	0.990	0.973	1.000	0.973	0.979	0.959
VDem-Participatory	0.698	0.988	0.971	0.973	1.000	0.972	0.964
VDem-Deliberative	0.699	0.990	0.968	0.979	0.972	1.000	0.962
VDemx -Egalitarian	0.697	0.979	0.952	0.959	0.964	0.962	1.000

Table B.2: Summary of different binary democracy indicators

Marshall and Gurr, 2020 (Polity IV)

- Country-year regimes scored between -10 and +10 based on competitiveness of political participation and constraints on executive authority
- A country-year is labeled a democracy if scoring at least +5, autocracy if scoring at most -5, anocracy otherwise

Freedom House, 2022 (FH)

- Country-year regimes classified as 'free', 'partially free', or 'not free' based on assessment of political rights and civil liberties
- A country-year is labeled a democracy if free or partially free

---

Cheibub et al., 2010 (CGV)

- Country-year regimes deemed 'democratic' if determined to satisfy four conditions for competitive elections
    1. Chief executive chosen by popular election or elected body
    2. Legislature is popularly elected
    3. More than one party competes in elections
    4. Alternation in power under identical electoral rules

Boix et al., 2012 (BMR)

- Country-year regimes are deemed democratic if it is determined that
    1. Political leaders are chosen through free and fair elections
    2. The electorate satisfies a threshold value of suffrage
- Missing values are assigned to observations during international war or state collapse, unlike other measures that retain these cases
- Long time frame (1800-2007) but in the dataset, only 48 years of data used, the second lowest after FH

Papaioannou and Siourounis, 2008 (PS)

- Focuses only on permanent transitions to democracy, no democratic breakdowns or transitory transitions
- ANRR suggests susceptibility to endogeneity in growth regressions

Acemoglu et al., 2019 (ANRR)

- A country-year regime is deemed 'free' if either:
    1. FH labels it free or partially free and Polity IV assigns a positive score
    2. If FH or Polity IV data is unavailable, CGV or BMR label it as democratic

Table B.3: Impact of additional 1°C shock on support for democracy

Outcome:	Important to live in	Better than alternatives	Index
Model:	(1)	(2)	(3)
Variables			
Shock	-0.0511	-0.0935	-0.1152·
	(0.1639)	(0.0913)	(0.0585)
Shock × $\overline{T}$	-0.0249	-0.0091	-0.0168·
	(0.0228)	(0.0080)	(0.0090)
$\overline{T}$	-2.070·	0.2943	-0.6651·
	(0.9996)	(0.5145)	(0.3382)
Gender	Yes	Yes	Yes
Precip	Yes	Yes	Yes
Fixed effects, # unique values			
Country	76	76	75
Year	11	13	11
Age	85	87	84
Birth year	92	93	92
Survey wave	4	4	4
Observations	160,486	189,456	150,292
Outcome mean	8.4417	3.2526	0.02250
R^2	0.06307	0.09477	0.08924

Clustered (Country & Year) standard-errors in parentheses
*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, .: 0.1*

Table B.4: Impact of additional 1°C shock on skepticism of democracy

Outcome:	Causes disorder	Too bureaucratic	Bad for economy	Index
Model:	(1)	(2)	(3)	(4)
Variables				
Shock	0.0803	0.2320***	0.0030	0.1235
	(0.0801)	(0.0423)	(0.0040)	(0.0815)
Shock × $\overline{T}$	0.0107	0.0268***	0.0059**	0.0169*
	(0.0071)	(0.0027)	(0.0016)	(0.0072)
$\overline{T}$	-0.1319	-0.3056	-0.1672	-0.2732
	(0.3052)	(0.4138)	(0.3569)	(0.3732)
Gender	Yes	Yes	Yes	Yes
Precip	Yes	Yes	Yes	Yes
Fixed effects, # unique values				
Country	77	76	76	76
Year	13	13	13	13
Age	87	87	87	87
Birth year	95	95	95	93
Survey wave	4	4	4	4
Observations	185,838	185,402	181,414	170,798
Outcome mean	2.2753	2.5324	2.2669	-0.00387
R^2	0.08693	0.08300	0.07527	0.10248

Clustered (Country & Year) standard-errors in parentheses
*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, .: 0.1*

Table B.5: Impact of additional 1°C shock on preference for authoritarian governance

Outcome:	Strongman	Army rule	Experts over government	Index
Model:	(1)	(2)	(3)	(4)
Variables				
Shock	0.4612*	0.2957*	0.1370	0.1303*
	(0.1628)	(0.1246)	(0.0831)	(0.0568)
Shock × $\overline{T}$	0.0462**	0.0272*	0.0146·	0.0122*
	(0.0157)	(0.0111)	(0.0082)	(0.0052)
$\overline{T}$	0.4493*	-0.1414	0.1461	0.0832
	(0.1754)	(0.1650)	(0.1719)	(0.1460)
Gender	Yes	Yes	Yes	Yes
Precip	Yes	Yes	Yes	Yes
Fixed effects, # unique values				
Country	98	98	98	98
Year	21	21	21	21
Age	87	87	87	87
Birth year	101	101	101	101
Survey wave	8	8	8	8
Observations	351,770	349,842	342,351	317,655
Outcome mean	2.2227	1.7531	2.6405	-0.03017
R^2	0.16859	0.21701	0.10108	0.18527

Clustered (Country & Year) standard-errors in parentheses
*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, .: 0.1*

Table B.6: Impact of additional 1°C shock on support for inclusive institutions

Outcome:	Free elections	Civil rights	Gender equality	Tax the rich	Income equality	Unemployment aid	Index
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Variables							
Shock	-0.0498	-0.4239$^{\cdot}$	0.2250	-0.1899	19.38*	-0.0151	-0.4582**
	(0.1486)	(0.2244)	(0.1389)	(0.2499)	(5.119)	(0.1883)	(0.0704)
Shock $\times \overline{T}$	-0.0188	-0.0641**	0.0085	-0.0657$^{\cdot}$	2.885*	-0.0400$^{\cdot}$	-0.0775**
	(0.0215)	(0.0188)	(0.0160)	(0.0295)	(0.8057)	(0.0214)	(0.0116)
$\overline{T}$	-2.325$^{\cdot}$	-2.041	-0.1941	-1.115	0.7127	-1.702	-0.3258*
	(1.190)	(1.215)	(0.8698)	(1.078)	(1.789)	(1.323)	(0.0970)
Gender	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Precip	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fixed effects, # unique values							
Country	75	75	75	75	59	75	58
Year	11	11	11	11	6	11	6
Age	85	85	85	85	83	85	83
Birth year	92	92	92	92	87	92	86
Survey wave	4	4	4	4	3	4	3
Observations	157,769	153,650	158,623	156,230	85,972	157,150	77,392
Outcome mean	8.2808	7.6900	8.2046	6.4546	5.9951	7.1533	-0.05107
R^2	0.07563	0.08909	0.10281	0.10524	0.15725	0.10367	0.17156

Clustered (Country & Year) standard-errors in parentheses
*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, .: 0.1*

Appendix C: Supplement to Chapter 3

Figure C.1: Spatial coverage of Indian DHS surveys

(a) Map of unique cluster locations in Indian DHS surveys



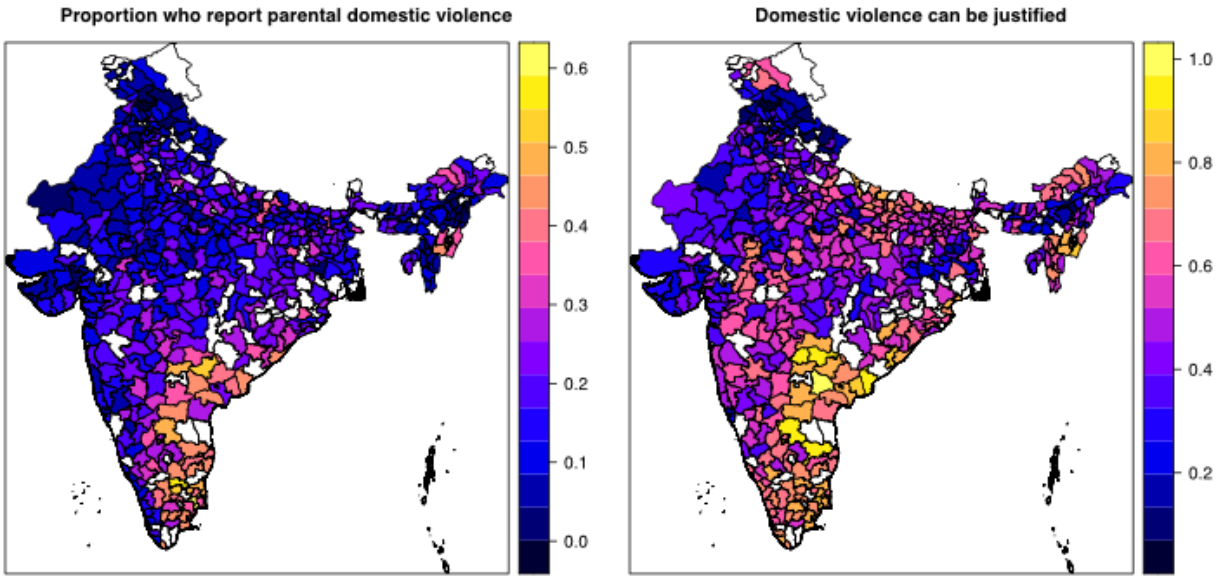
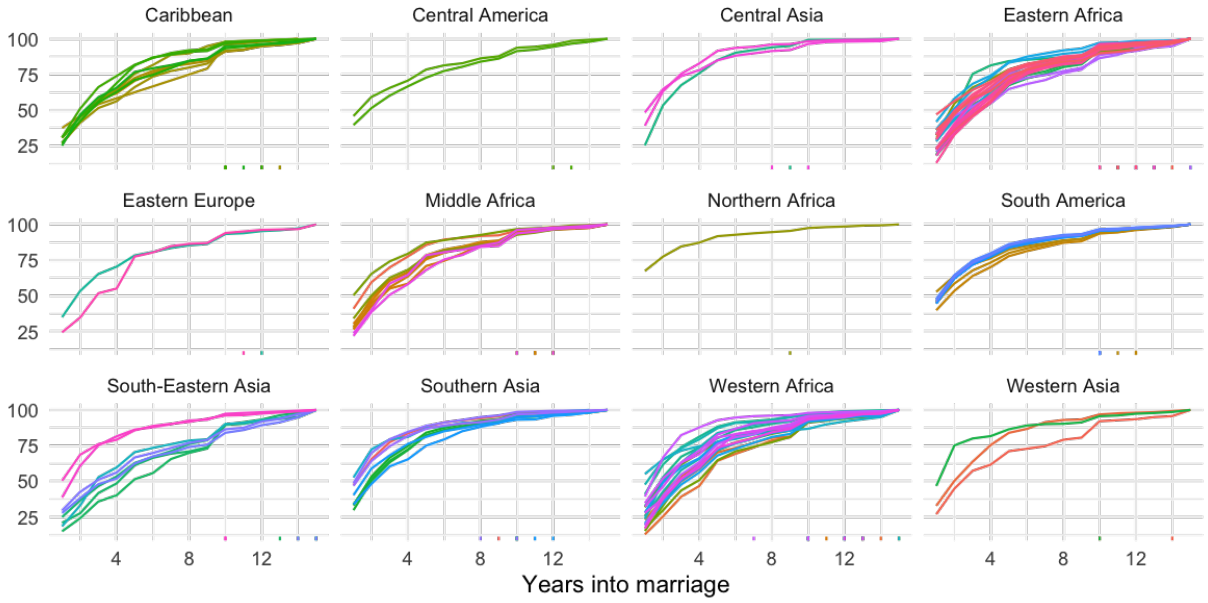


Figure C.2: Incidence of violence within marriages, conditional on ever reporting violence

(a) Timing of first violent event



(b) Reported year of first violent event

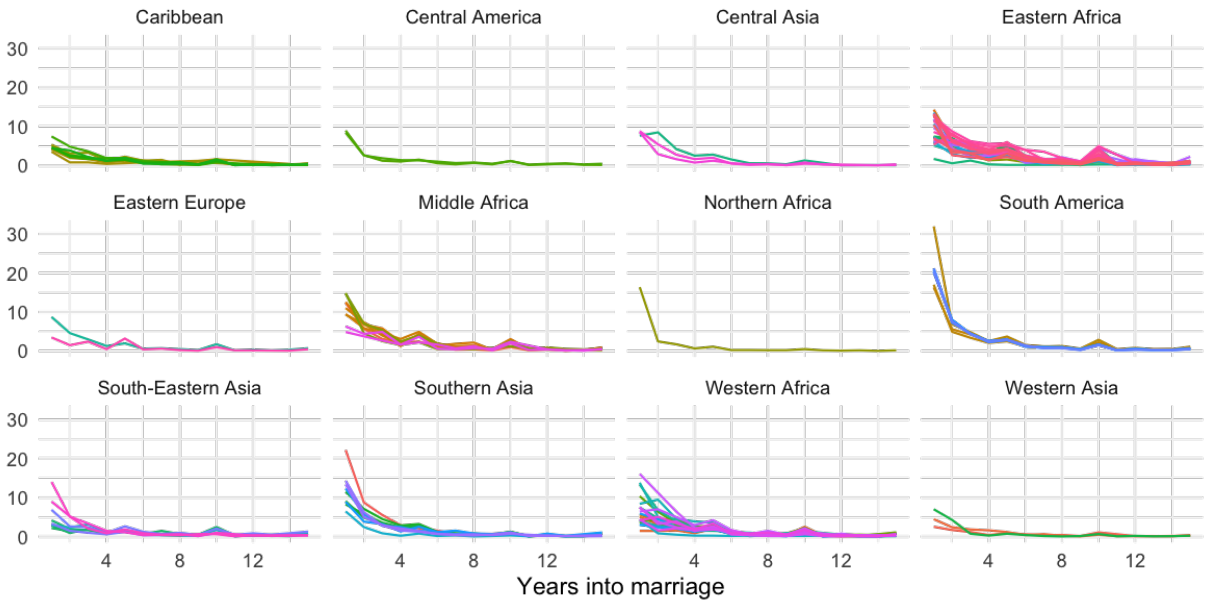


Table C.1: Categorization of relevant questions from DHS domestic violence module

	Sub-Saharan Africa		India	
	Obs	Mean	Obs	Mean
IPV justified if wife:				
Argued with husband	116,599	0.692	86,520	0.719
Burnt food	116,713	0.826	86,678	0.813
Neglected child	116,891	0.657	86,786	0.670
Refused to have sex	116,341	0.716	86,049	0.860
Went out without permission	116,722	0.667	86,718	0.737
Wife justified in refusing sex if:				
Infidelity	97,264	0.357	85,594	0.152
Not in mood	28,271	0.361	85,176	0.175
Partner has an STI	33,861	0.204	84,387	0.141
No birth control	96,691	0.209	0	–
Recently gave birth	10,524	0.214	0	–
Household decisions wife has influence over				
Wife's money	51,924	0.879	25,969	0.855
Husband's money	87,411	0.463	80,850	0.769
Household purchases	101,126	0.556	81,876	0.755
Visits to family/friends	101,121	0.653	81,876	0.787
Contraceptive use	49,250	0.851	57,666	0.921
Health decisions	99,593	0.593	81,876	0.774
Sexual consent	84,654	0.351	0	–
Controlling behavior by husband				
Accuses wife of infidelity	113,060	0.228	86,973	0.085
Restricts contact, family	113,181	0.123	86,998	0.122
Restricts contract, friends	113,171	0.176	86,983	0.172
Exhibits jealousy	112,706	0.520	86,754	0.240
Access to money	41,389	0.209	86,898	0.201
Demands knowledge of whereabouts	113,164	0.405	86,923	0.166
Experience of physical violence, last 12 months				
Pulled hair / Twisted arm	99,827	0.077	87,163	0.129
Kicked	113,896	0.116	87,163	0.098
Punched	114,725	0.108	87,163	0.095
Pushed	115,554	0.129	87,163	0.139
Slapped	114,740	0.249	87,163	0.297
Strangled	115,545	0.035	87,163	0.021
Hit with a weapon	108,377	0.027	87,163	0.013
Experience of emotional violence, last 12 months				
Humiliated	114,306	0.173	87,163	0.107
Insulted	100,316	0.241	87,163	0.082
Threatened	114,288	0.102	87,163	0.057

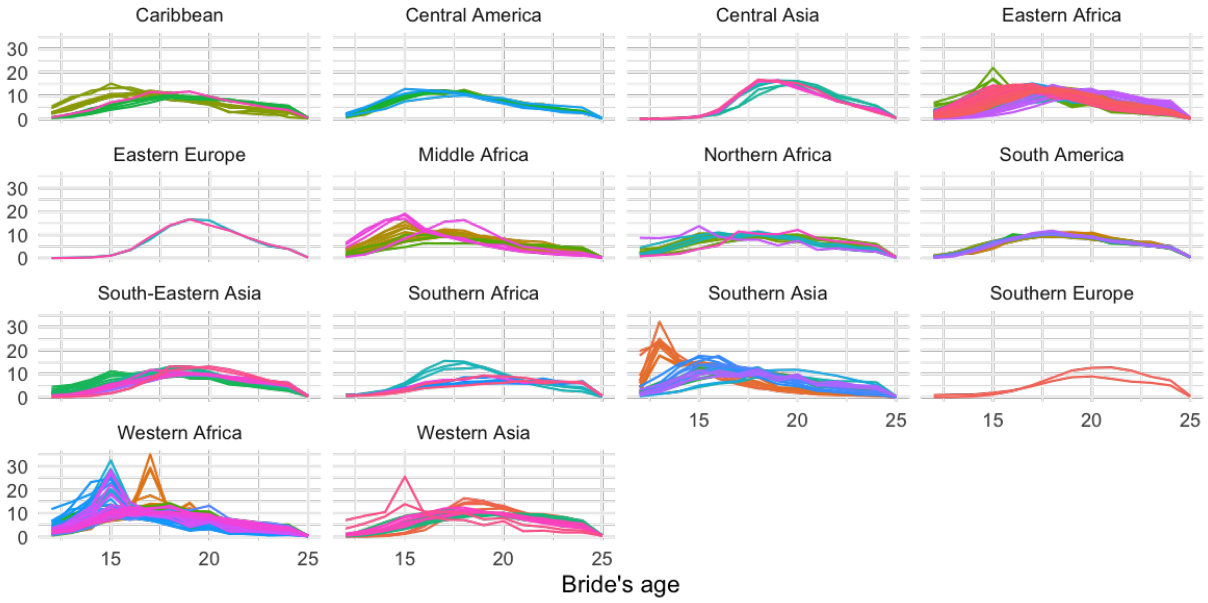
Table C.2: IPV hazard results, varying lags and splitting survey waves

Dependent Variable:	Violent					
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
Drought	-0.0002	-0.0001	-0.0001	-0.0002	-0.0001	-0.0001
	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)
Drought, L1		-0.0004***	-0.0004***		-0.0004***	-0.0004***
		(0.0001)	(0.0001)		(0.0001)	(0.0001)
Drought, L2			-6.26×10^{-5}			$-6.26 \times 10^{-}$
			(0.0001)			(0.0001)
Fixed-effects						
Survey wave (2)	Yes	Yes	Yes	Yes	Yes	Yes
Panel year of marriage (4)	Yes	Yes	Yes	Yes	Yes	Yes
Cluster (56,073)	Yes	Yes	Yes	Yes	Yes	Yes
Year married	Yes	Yes	Yes	Yes	Yes	Yes
Age at marriage	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics						
# Year married	40	39	38	40	39	38
# Age at marriage	52	52	51	52	52	51
Dependent variable mean	0.00538	0.00538	0.00538	0.00538	0.00538	0.00538
Observations	3,550,939	3,546,383	3,539,064	3,550,939	3,546,383	3,539,064
Adjusted R^2	0.02592	0.02593	0.02600	0.02592	0.02593	0.02600
Within R^2	5.38×10^{-7}	3.7×10^{-6}	3.93×10^{-6}	5.38×10^{-7}	3.7×10^{-6}	3.93×10^{-6}

Clustered (Cluster) standard-errors in parentheses
*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, .: 0.1*

Figure C.3: Incidence of early marriage by age

(a) Annual hazard into early marriage



(b) Cumulative percentage of women married

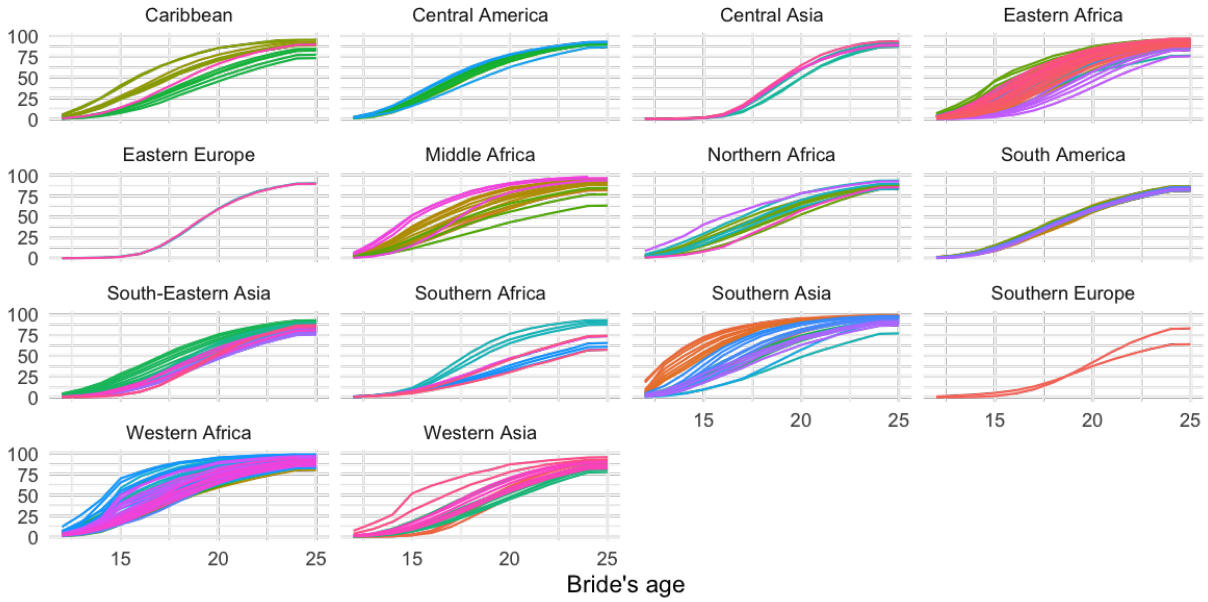
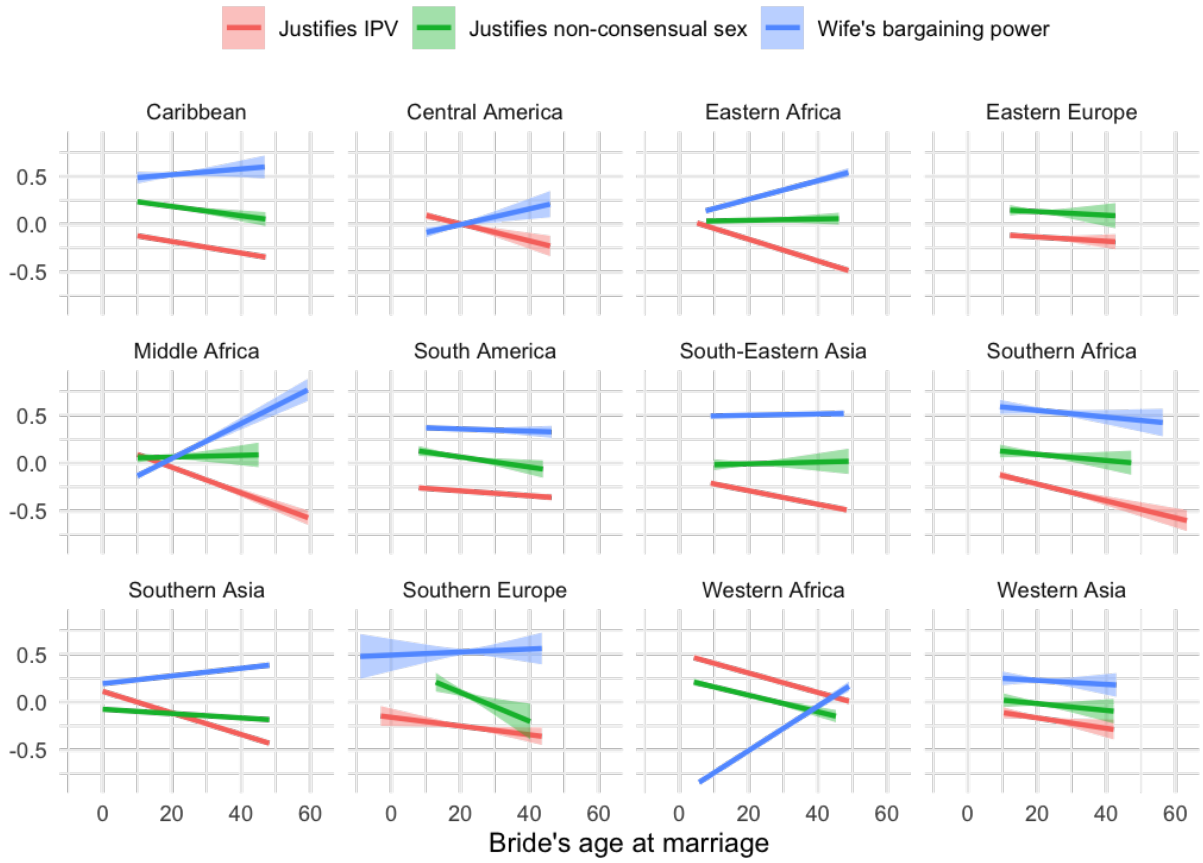


Figure C.4: Correlations, bridal age at marriage and groom preferences for gender roles



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